

Project:

MSSR 2000 ID

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Interface Control Document - ASTERIX  
Implementation

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## CHANGE RECORD

| Version | Change date | Changed pages  | Brief description of change                                                                                                                                                                                                                           | Change Request |
|---------|-------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 01      | 2004-07-16  | all            | Work in of comments and requests from several projects concerning Version 06 of the document "MSR 2000 I ICD ASTERIX Implementation"                                                                                                                  | 502263         |
| 02      | 2004-07-28  | See version 02 | See version 02                                                                                                                                                                                                                                        | 509476         |
| 03      | 2006-06-13  | See version 03 | See version 03                                                                                                                                                                                                                                        | 513883         |
| 04      | 2007-02-02  | See version 04 | This version was finalised for MSSR 2000 I SW Version 7.01<br>See List of Changes in V04.                                                                                                                                                             | 525765         |
| 05      | 2009-09-03  | See version 05 | This version was finalised for MSSR 2000 I SW Version 8.01<br>See List of Changes in V05.                                                                                                                                                             | 537951         |
| 06      | 2011-02-22  | See version 06 | V06 finalised for MSSR 2000 I Version 8.03.<br>Reason for major changes: new functions, bug fix, precise description, obsolete chapters, preparation for customising, trace to System Specification and Adaption Data.<br>See List of Changes in V06. | 570033         |
| 07      | 2013-02-15  | See version 07 | V07 finalised for MSSR 2000 I Version 8.04.<br>Reason for major changes: new functions, bug fix, precise description, new KIV78 functions.<br>See List of Changes in V07.                                                                             | 606071         |
| 08      | 2013-04-19  | See version 08 | V08 finalised for MSSR 2000 I Version 8.06.                                                                                                                                                                                                           | 626504         |
| 09      | 2015-10-20  | See version 09 | V09 finalised for MSSR 2000 ID Version 9.01.02                                                                                                                                                                                                        | 633466         |
| 10      | 2016-06-07  | 102, 104       | Start phase adjustment, access level of Silent Sectors                                                                                                                                                                                                | 646103         |
| 11      | 2017-01-20  | See version 11 | V11 finalised for MSSR 2000 ID Version 9.02.01                                                                                                                                                                                                        | 647814         |
| 12      | 2018-02-27  | See version 12 | Update for MSSR 2000 ID Version 10.01.<br>A detailed list of changes is provided in V12.                                                                                                                                                              | 702996         |
| 13      | 2019-11-11  | See version 13 | Update for MSSR 2000 ID Version 10.01.03.<br>A detailed list of changes is provided below.                                                                                                                                                            | 711355         |
| 14      | 2023-08-11  | See version 14 | Update for MSSR 2000 ID Version 10.04.01.<br>A detailed list of changes is provided below.                                                                                                                                                            | 736463         |
| 15      | 2024-06-20  | See version 15 | Update for MSSR 2000 ID Version 10.05.<br>A detailed list of changes is provided below.                                                                                                                                                               | 739148         |

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**Exported from DOORS module:**

'/MSSR 2000 I/04 ExtIntf/ICD Asterix'

**version:**

baseline '9.12' (created on 20.06.2024)

**from view:**

'50 Export 10.05 civ'

**with filtering set as:**

'(((A\_Configuration includes 10.05.xx) OR (A\_Configuration includes all)) AND (Object Heading &lt; 5)) AND ((A\_civ\_mil == civil) OR (A\_civ\_mil == all))'

**List of changes from document version 14 to version 15. Chapter numbers are from version 15.**

| Chapter (V15) | Change Note                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.2           | added ref /SysArc M23/                                                                                                                                                                  |
| 4.6.7.2       | added "Extension of Primary Subfield", explanation of "Structure of Subfield # 3 Part 2 (MkX info Flight Level)" and "Structure of Subfield # 3 Part 3 (MkX info M3/MC code)" improved. |
| 4.6.8.17      | added reference /SysArc M23/ for transformation from ITEM 40 (Range,Azimuth) to ITEM 42 (X,Y).                                                                                          |
| 4.7.5.1       | added info that the sequence of reports may change                                                                                                                                      |
| 4.11.1.2      | added on Rules for the design of CMS SW                                                                                                                                                 |
| 4.11.8.6      | updated description of "Start Of Sector" > "End Of Sector".                                                                                                                             |
| 4.11.8.10     | added description of "Start Of Sector" > "End Of Sector".                                                                                                                               |
| 4.11.9.2      | improved explanation by adding a "Note"                                                                                                                                                 |
| 4.11.9.22     | updated for enabling/disabling of "Mode 4" and "Supermode" at the same time.                                                                                                            |

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# 1 General Information

## 1.1 Purpose of this Document

This document is part of a set of documents containing the interface overview and the interface description of the MSSR 2000 ID External Interfaces. This document is focused on the description of the EUROCONTROL ASTERIX formats (ASTERIX = All-purpose Structured Eurocontrol surveillance Information Exchange).

### 1.1.1 Brief Description

The MSSR 2000 ID is an IFF and SSR interrogator for long and medium range radar systems, providing Mk XIIA and Mode S functionality.

The MSSR 2000 ID includes all functions to

- send interrogation signals,
- receive replies from transponders,
- process the replies,
- generate target information reports,
- transmit the target information to the succeeding ATC or weapon system processing stages.

## 1.2 Referenced Documents

/EC CAT 253/

Eurocontrol ASTERIX Category 253 Specification,  
Doc. ID: none, Issue 1.0,  
November 1999

Note 1: There is a more recent version of this document (Issue 5.0, November 2010) available. This issue is incompatible with the implementation of the MSSR 2000 ID. The incompatibility lies within the UAP having a different FSPEC scheme, where the enumeration of the message types has changed. The MSSR 2000 ID is therefore based on the noted issue.  
Note 2: As clearly indicated by Eurocontrol within the document Issue 1 and Issue 5: "Until now, no standard exists for ASTERIX category 253. Because the requirements of the different users of ASTERIX category 253 differ too much for a common definition, no attempt for standardisation of ASTERIX category 253 is made"

/EC ASTERIX p1/

EUROCONTROL Standard Document for Surveillance Data Exchange,  
Part 1 All Purpose Structured Eurocontrol Surveillance Information Exchange  
EUROCONTROL-SPEC-0149, Edition: 2.2,  
Edition Date: October 2014, Status: Released Issue

/EC CAT 048/

EUROCONTROL Specification for Surveillance Data Exchange,  
ASTERIX Part 4 Cat. 048, Monoradar Target Reports,  
EUROCONTROL-SPEC-0149-4, Edition 1.24,  
Edition Date : March 2019, Status : Released Edition

/EC CAT 048 REF 1.7/

Eurocontrol Standard Document for Surveillance Data Exchange,  
Appendix A to Part 4: Coding rules for "Reserved Expansion Field",  
Edition 1.7,  
Edition Date: 01/09/2014, Status: Released Issue

/EC CAT 034/

Eurocontrol Standard Document for Surveillance Data Exchange,  
Part 2b: Transmission of Monoradar Service Messages (=Category 034),  
SUR.ET1.ST05.2000-STD-02b-01, Edition 1.27,  
Edition Date: May 2007, Status: Released Issue

/EC CAT 017/

POEMS Document for ASTERIX Category 017  
Part 5: Mode S Surveillance Coordination Function Messages,  
SUR.E2.ST03.3111-SPC-02-00, Edition: 1.3,  
Edition Date: January 2009; Status: Released Issue

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## /EC CAT 018/

EUROCONTROL Specification for Surveillance Data Exchange,  
 ASTERIX Part 06 Cat. 018, Mode S Datalink Function Messages,  
 EUROCONTROL-SPEC-0149-06, Edition: 1.7,  
 Edition Date: November 2015; Status: Released

## /EC GDLP ICD/

EATCHIP GDLP/Local User ICD for POEMS,  
 SUR.ET2.ST03.3112-SPC-02-00, Edition: 1.7,  
 Edition Date, 17 March 1999, Status: Proposed Issue

## /EC CAT 021/

Eurocontrol Standard Document for Surveillance Data Exchange,  
 Part 12: Category 021, ADS-B Reports  
 SUR.ET1.ST05.2000-STD-12-01, Edition: 2.1,  
 Edition Date: May 2011; Status: Released Issue

## /EC CAT 007/

EUROCONTROL Specification for Surveillance Data Exchange - ASTERIX Part 21 Category  
 007 - Transmission of Directed  
 Interrogation Messages (EUROCONTROL-SPEC-0149-21), Edition 1.7, Edition Date:  
 24/10/2016, Status: Released Edition

## /EC CAT 007 REF/

Eurocontrol Standard Document for Surveillance Data Exchange,  
 ASTERIX Part 21 Category 007, Appendix A, Coding rules for "Reserved Expansion Field",  
 SUR.ET1.ST05.2000-STD-21-02, Edition 1.4,  
 Edition Date: November 2012,  
 Status: Released Issue

## /ECMS/

**European Mode S** Station Functional Specification  
 SUR/MODES/EMS/SPE-01,  
 Edition 3.11, 09 May 2005

## /ISO 7498/

ISO 7498, Open System Interconnection Model  
 ISO/IEC 7498, Part 1, The Basic Model,  
 ISO/IEC 7498, Part 3, Naming and Addressing

## /RFC 5905/

Internet Engineering Task Force, Request for Comments 5905:  
 "Network Time Protocol Version 4: Protocol and Algorithms Specification"

## /RFC 793/

Transmission Control Protocol, Darpa Internet Program, Protocol Specification

## /RFC 768/

User Datagram Protocol

## /RFC 791/

Internet Protocol, Darpa Internet Program, Protocol Specification

## /RFC 1112/

Host Extensions for IP Multicasting

## /RFC 826/

Ethernet Address Resolution Protocol

## /IEEE 802.3/

IEEE Standard for Information Technology: Telecommunications and information exchange  
 between systems--Local and metropolitan area networks--Specific requirements--Part 3:  
 Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and  
 Physical Layer Specifications

## /MSSR SW Arch/

MSSR 2000 I  
 SW Architecture Interrogator Unit  
 Document-No.: 20393102 EDO 000

## /Parameter/

MSSR 2000 I  
 Interface Description Document Adaptation Data  
 Doc No: 30032012 EDO 000

|                                                |            |                         |          |          |
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/MSSR ICD EI/  
MSSR 2000 I  
Interface Control Document - External Interfaces,  
Doc No: 30005092 EDO 000

/MSSR ICD AI - Annex/  
MSSR 2000 ID  
Annex to Interface Control Document - Asterix Implementation  
Document No: 31323082 EDO 000

/ICD ACA/  
ICD MSSR-ACA Interface  
Doc. ID: 30874893 EDO 000

/SysArc M23/  
System Architecture MSSR 2000 ID M23  
Doc No: 20393098 EDO 000 Version 08

### 1.3 List of Definitions and Acronyms

See document:

MSSR 2000 I Definitions and Acronyms  
Document-No.: 30004181 EDO 000

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## 2 Logical Interface Description

### 2.1 General

The MSSR 2000 ID System exists in several variants.

For the different power variants, there is nearly no difference relating to the ASTERIX Interface. But in case of a Dual Redundant (DR) variant the ASTERIX Interface is applied to each chain of the system, because each chain consists of a complete interrogator. One chain acts as Master and the other as Slave. In this Document 'Master' stands for 'Master chain' and 'Slave' stands for 'Slave chain'.

DR specific features are explained within this document.

### 2.2 LAN Interface

#### 2.2.1 Layer Model

According to the ISO/OSI Layer Model, the following protocols are used for communication:

Application Layer 6 and Presentation 7:

ASTERIX and NTP are used.

Session Layer 5:

BSD Sockets (Version 4.4) are used.

Transport Layer 4:

TCP/IP Protocol as described in /RFC 793/.

UDP/IP Protocol is as described in /RFC 768/.

Network Layer 3:

IP-Layer compliant with /RFC 791/.

The MSSR 2000 ID supports IPv4 and is ready for IPv6.

Host Extensions for IP multicasting according to /RFC 1112/.

Note: The multicast is in accordance to RFC 1112 "Host Extensions for IP Multicasting".

The Ethernet Address Resolution Protocol compliant with /RFC 826/ (ARP is used to translate IPv4 addresses (Layer 3) into Ethernet MAC addresses (Layer 2)).

Data Link Layer 2 and Physical Layer 1:

10Base-T or 100Base-TX Ethernet according to IEEE 802.3.

Auto sensing capability to determine the transfer rate 10 or 100 Mbits/s.

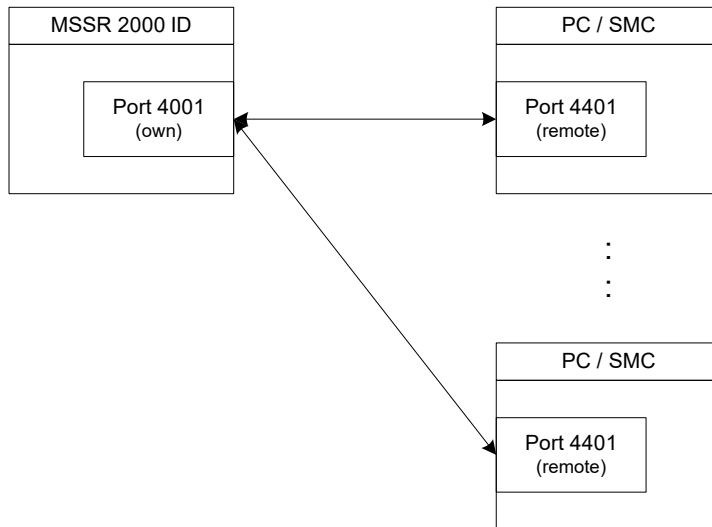
|                                                |            |                  |          |          |
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## 2.2.2 Logical Structure of the UDP/IP Interface

Example for UDP Connection Structure:

*Figure 2-1. UDP Connection Structure for MSSR 2000 ID*



The MSSR 2000 ID provides the following service at its own UDP ports:

- CAT 007 (Master, only unicast)
- CAT 021 (Master; uni-, multi- or broadcast)
- CAT 034 (Master and Slave; uni-, multi- or broadcast)
- CAT 048 (Master and Slave for processed plots and tracks; raw plot only sent by Master; uni-, multi- or broadcast)
- CAT 253 (Master and Slave, only unicast)

**Note:**

In case of DR system, the relevant chain is mentioned in brackets.

Note: in case of a DR system configuration the output of CAT 048 and CAT 034 data reports from the slave chain can be disabled.

The own Port is configurable (parameter LAN\_UDP\_ASTERIX, e.g. own port = 4001).

Optional it is possible to configure a list up to 16 remote host addresses and ports for uni- and multicast.

Data from a remote client is only accepted if the remote port and address (filtered with netmask) fits to the configured (parameter LAN\_UDP\_ASTERIX, e.g. remote port = 4401).

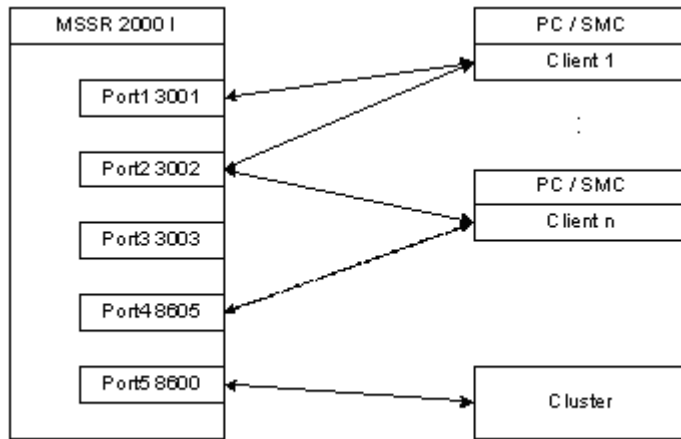
The transmission to an UDP/IP multicast group shall be activated by setting a valid multicast address as UDP address (parameter LAN\_UDP\_ASTERIX, address). A valid multicast address must be in the range of 224.X.X.X - 239.X.X.X.

|                                                |            |                         |          |          |
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### 2.2.3 Logical Structure of the TCP/IP Interface

Example for TCP Connection Structure:

*Figure 2-2. TCP Connection Structure for MSSR 2000 I*



The MSSR 2000 ID provides the following services at TCP Port 1, Port 2 and Port 3 (if enabled):

- CAT 007 (Master)
- CAT 018 (Master) at an average rate of 150 messages/second.
- CAT 021 (Master)
- CAT 034 (Master and Slave)
- CAT 048 (Master and Slave for processed plots and tracks; raw plot only sent by Master)
- CAT 253 (Master and Slave)

Note, in case of a DR system configuration the output of CAT 048 and CAT 034 data reports from the slave chain can be disabled.

Port 1 is configurable (e.g. 3001) and

Port 2 has automatically the port number from Port 1 plus one.

Port 3 has automatically the port number from Port 1 plus two.

The system supports as multi client server up to 64 clients at TCP Port 1, Port 2 and Port 3.

That would result in  $3 * 64 = 192$  clients in total that is full supported by the new data processor 3 boards.

Note:

The total number of clients is limited to 150 on older (slower) PC-104 modules (data processor 2 boards).

The overall target load is limited to 3000 targets/second for all ports. Therefore it is possible to support 16 clients on each port under max. load conditions.

The following service shall be provided (only Master) at TCP Port 4 and Port 5, if SCF is enabled:

- Port 4: CAT 017 data recording for up to 64 clients.
- Port 4 is used for integration and test to supervise the communication between the nodes. Messages exchanged with other nodes are available for recording (the port is configurable in the scf.conf file e.g. 8605).
- Port 5: CAT 017 SCN data for up to 6 clients (5 nodes + cluster controller) at an average rate of 150 messages/second.
- Port 5 is used for communication between the cluster nodes. Port number and the IP address is defined in the cluster map. It is recommended by EUROCONTROL to use Port 8600 for SCN service.

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### 3 The Network Time Protocol of the MSSR 2000 ID

#### 3.1 Related Standards

The MSSR 2000 ID implementation of NTP is in accordance with /RFC 5905/.

#### 3.2 Mode of Operation of MSSR 2000 ID for NTP

MSSR 2000 ID works in Client Mode (see /RFC 5905/, chapter 2 "Modes of Operation").

The IP addresses of up to two NTP Server per chain are configurable (NTP\_ADDRESS and NTP\_2\_ADDRESS).

In case of time synchronisation by NTP, each chain shall synchronise to a NTP server.

If the connections to NTP fails, the chain synchronises automatically to the other chain, independent of the Master - Slave status.

NTP authentication is not implemented in MSSR 2000 ID.

The MSSR 2000 ID uses the iburst - option for fast time synchronisation after start up.

The following parameters for the MSSR 2000 ID are used:

maxpoll: the maximum poll intervals for NTP messages [s]

minpoll: the minimum poll intervals for NTP messages [s]

|                                                |            |                         |          |          |
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## 4 ASTERIX Message Format

### 4.1 General

ASTERIX is emerging as the international standard for surveillance data. The format has proven to be an extremely powerful and flexible basis for a wide range of applications. Today a comprehensive set of ASTERIX systems and tools support the market.

ASTERIX stands for "All-purpose Structured Eurocontrol suRveillance Information Exchange". It was defined in the late eighties to overcome the deficiencies of the existing variety of national and vendor-specific formats and protocols.

Serving as an intermediate format, which can convey all the information present in any of the existing surveillance data formats, ASTERIX contributes strongly to the standardisation within the surveillance domain.

Based on a flexible structural base, ASTERIX is open to extensions resulting from developments in radar and radar processing technology, as well as with respect to different applications using the format.

ASTERIX is an ideal device to substitute yesterday's outdated and inflexible solutions. It is a main prerequisite contributing to harmonisation of the world's skies.

MSSR 2000 ID will use standard ASTERIX message formats for the communication with external devices.

For MSSR 2000 ID the following categories of ASTERIX are used:

- CAT 034 Transmission of Monoradar Service Messages
- CAT 048 Transmission of Monoradar Target Reports
- CAT 253 Transmission of Remote Station Monitoring and Control Information
- CAT 007 Transmission and Reception of Directed Interrogation Messages
- CAT 017 Transmission of Mode S Surveillance Coordination Function Messages
- CAT 018 Mode S Datalink Function Messages
- CAT 021 ADS-B Messages

Note:

It is recommended to use CAT 007 for Directed Interrogation but for compatibility reason the previous implementation via CAT 048M is still supported but not updated with the new functionalities.

#### Communication Perspective:

In ISO/OSI terminology, ASTERIX is a presentation layer standard (Layer 6) and does not prescribe a specific link or transport protocol to support it. This separation into layers has the invaluable advantage that now ASTERIX can be used over a wide range of underlying transport links, including LAN and WAN protocols of the ISO/OSI and TCP/IP stacks. This way ASTERIX also has the potential to be integrated in a straight-forward manner into the emerging ATN (Aeronautical Telecommunications Network).

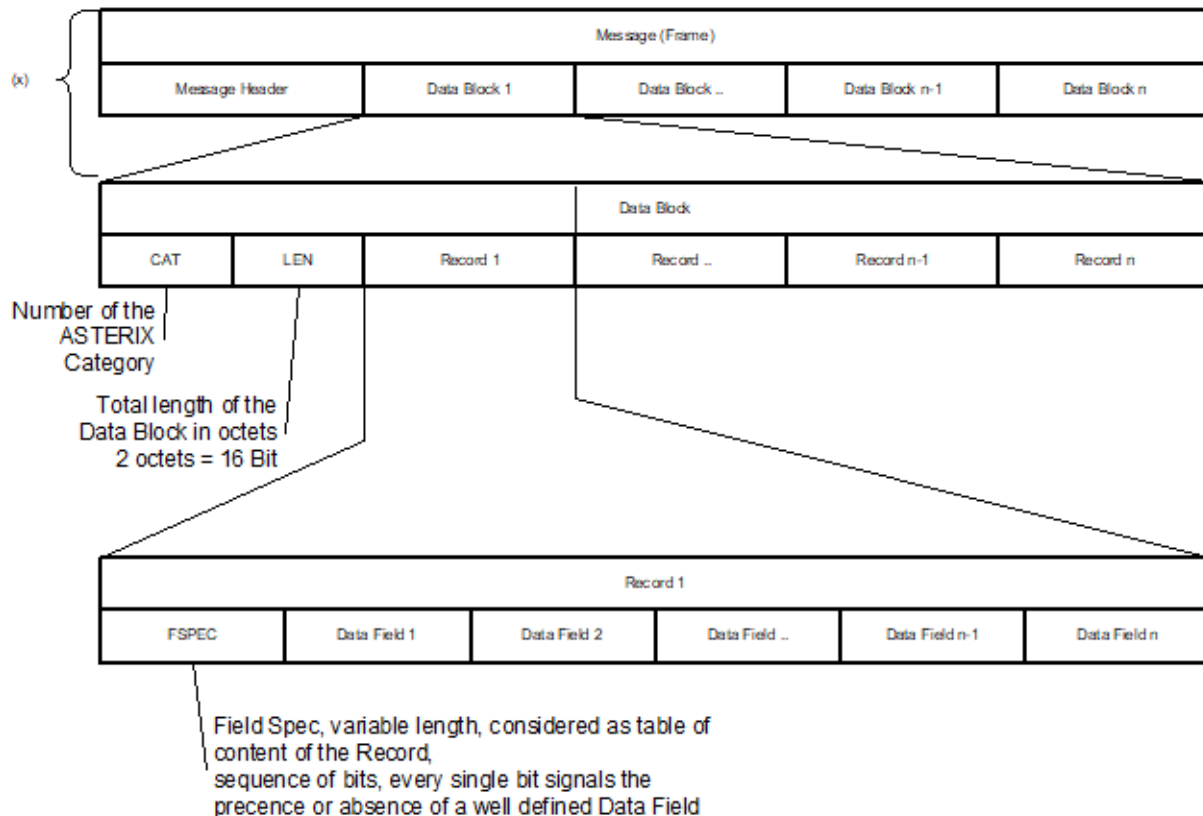
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## 4.2 Structure of ASTERIX Data Blocks

This chapter shall give a brief introduction to the structure of the ASTERIX Messages for a better understanding of the following chapters.

ASTERIX Messages are sent as Data Blocks. Ref. /EC ASTERIX p1/ shows the principle structure of an ASTERIX Data Block.

Figure 4-1. Principal Structure of ASTERIX Data Blocks



(x): The structure of this layer depends on the type of data link which is used to transfer the ASTERIX messages. This layer is not part of the ASTERIX standard.

Octet: ASTERIX uses the term “octet” for a set of 8 bit. An octet is not exactly the same as a byte because of the different bit numbering scheme.

The format and maximum length of a message depends on the selected interface type (X.21 or LAN) and the selected protocol.

A message can contain one or more data blocks, which can be of different type.

Each Data Block contains one or more records. These records are all off the same category.

NOTE: If TCP is used to transport ASTERIX Data Blocks, the receiver of the TCP stream may not assume, that each TCP packet payload starts with the first byte of an ASTERIX Data Block, or ends with the last byte of an ASTERIX Data Block. Instead, ASTERIX Data Blocks may be fragmented over multiple (typically two) TCP packets. The receiver of the TCP stream is responsible to re-assemble fragmented ASTERIX Data Blocks.

The Length Indicator (LEN) is a two octet field indicating the total length of the Data Block in octets, including the CAT and LEN fields.

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Maximum length of a Data Block:  $2^{16} = 65536$  octets. This is the theoretical value. Restrictions may occur by shorter messages given by the used type of communication lines (ISO/OSI Level 0 - 5).

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## 4.3 User Application Profiles

### 4.3.1 General

The contents of a record is defined by the UAP (User Application Profile).

The UAP defines the data fields, transferred via a record of the dedicated category and the order of the data fields within the record. The contained data fields and the order cannot be changed but unused data fields can be omitted.

The parameter FSPEC (Field Specification) specifies, which data fields are transferred (the pursuant bit is set to "1") or omitted (the pursuant bit is set to "0").

The contents and meaning of the data fields is described in the respective ASTERIX documentation.

The personalisation of the UAPs is described within the relevant project documents.

#### Column Description of the following UAP's:

**FRN:**

Indicates the FRN (Field Reference Number) associated with each Data Item used in the UAP.

**Data Item:**

Reference to the Data Item.

**Data Item Description:**

Description of the Data Item.

**Length:**

Gives the format and the length of each item, a stand-alone figure indicates the octet count of a fixed-length Data Item, 1+ indicates a variable-length Data Item comprising a first part of one-octet followed by n-octets extent as necessary, 1+1+ indicates a variable-length Data Item as for 1+, but with a major number of n-octets extent as necessary.

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#### 4.3.2 UAP for CAT 034

For Category 034, the UAP as described in the following table shall be used:

*Table 4-1. Standard UAP for Category 034*

| FRN | Data Item    | Data Item Description           | Length  |
|-----|--------------|---------------------------------|---------|
| 1   | I034/010     | Data Source Identifier          | 2       |
| 2   | I034/000     | Message Type                    | 1       |
| 3   | I034/030     | Time-of-Day                     | 3       |
| 4   | I034/020     | Sector Number                   | 1       |
| 5   | I034/041     | Antenna Rotation Period         | 2       |
| 6   | I034/050     | System Configuration and Status | 1+      |
| 7   | I034/060     | System Processing Mode          | 1+      |
| FX  | n.a.         | Field Extension Indicator       | n.a.    |
| 8   | I034/070     | Message Count Values            | (1+2*N) |
| 9   | I034/100     | Generic Polar Window            | 8       |
| 10  | I034/110     | Data Filter                     | 1       |
| 11  | I034/120     | 3D-Position of Data Source      | 8       |
| 12  | I034/090     | Collimation Error               | 2       |
| 13  | RE-Data Item | Reserved Expansion Field        | 1+1+    |
| 14  | SP-Data Item | Special Purpose Field           | 1+1+    |
| FX  | n.a.         | Field Extension Indicator       | n.a.    |

|                                                |                     |                                             |                |                  |
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### 4.3.3 UAP for CAT 048

For Category 048, the UAP as described in the following table shall be used:

*Table 4-2. Standard UAP for Category 048*

| FRN | Data Item    | Data Item Description                              | Length |
|-----|--------------|----------------------------------------------------|--------|
| 1   | I048/010     | Data Source Identifier                             | 2      |
| 2   | I048/140     | Time-of-Day                                        | 3      |
| 3   | I048/020     | Target Report Descriptor                           | 1+     |
| 4   | I048/040     | Measured Position in Slant Polar Co-ordinates      | 4      |
| 5   | I048/070     | Mode-3/A Code in Octal Representation              | 2      |
| 6   | I048/090     | Flight Level in Binary Representation              | 2      |
| 7   | I048/130     | Radar Plot Characteristics                         | 1+1+   |
| FX  | n.a.         | Field Extension Indicator                          | n.a.   |
| 8   | I048/220     | Aircraft Address                                   | 3      |
| 9   | I048/240     | Aircraft Identification                            | 6      |
| 10  | I048/250     | Mode S MB Data                                     | 1+8*n  |
| 11  | I048/161     | Track Number                                       | 2      |
| 12  | I048/042     | Calculated Position in Cartesian Co-ordinates      | 4      |
| 13  | I048/200     | Calculated Track Velocity in Polar Representation  | 4      |
| 14  | I048/170     | Track Status                                       | 1+     |
| FX  | n.a.         | Field Extension Indicator                          | n.a.   |
| 15  | I048/210     | Track Quality                                      | 4      |
| 16  | I048/030     | Warning/Error Conditions                           | 1+     |
| 17  | I048/080     | Mode-3/A Code Confidence Indicator                 | 2      |
| 18  | I048/100     | Mode-C Code and Confidence Indicator               | 4      |
| 19  | I048/110     | Height Measured by 3D Radar                        | 2      |
| 20  | I048/120     | Radial Doppler Speed                               | 1+     |
| 21  | I048/230     | Communications / ACAS Capability and Flight Status | 2      |
| FX  | n.a.         | Field Extension Indicator                          | n.a.   |
| 22  | I048/260     | ACAS Resolution Advisory Report                    | 7      |
| 23  | I048/055     | Mode-1 Code in Octal Representation                | 1      |
| 24  | I048/050     | Mode-2 Code in Octal Representation                | 2      |
| 25  | I048/065     | Mode-1 Code Confidence Indicator                   | 1      |
| 26  | I048/060     | Mode-2 Code Confidence Indicator                   | 2      |
| 27  | SP-Data Item | Special Purpose Field                              | 1+1+   |
| 28  | RE-Data Item | Reserved Expansion Field                           | 1+1+   |
| FX  | n.a.         | Field Extension Indicator                          | n.a.   |

|                                                |                     |                                             |                |                  |
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#### 4.3.4 UAP for CAT 253

For Category 253, the UAP as described in the following table shall be used:

*Table 4-3. UAP for Extended Category 253*

| FRN | Data Item | Data Item Description                 | Length |
|-----|-----------|---------------------------------------|--------|
| 1   | I253/010  | Data Source Identifier                | 2      |
| 2   | I253/015  | Local Data Source Identifier          | 1      |
| 3   | I253/020  | Data Destination Identifier           | 1+2*n  |
| 4   | I253/025  | Data Destination and Local Identifier | 4      |
| 5   | I253/030  | Source Application Identifier         | 2      |
| 6   | I253/035  | Data Origin Identifier                | 3      |
| 7   | I253/040  | Message Type                          | 1      |
| FX  | n.a.      | Field Extension Indicator             | n.a.   |
| 8   | I253/050  | Message Sequence ID                   | 1+2*n  |
| 9   | I253/060  | Blocking Information                  | 2      |
| 10  | I253/070  | Time of Day                           | 3      |
| 11  | I253/080  | Application Data Structure            | 4      |
| 12  | I253/090  | Application Data                      | 1+16*n |
| 13  | I253/100  | Transparent Application Data 1        | 1+n    |
| 14  | I253/110  | Transparent Application Data 2        | 1+n    |
| FX  | n.a.      | Field Extension Indicator             | n.a.   |

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#### 4.3.5 UAP for CAT 007

The Category 007 is derived from the Category 048, but additional Data Items provided to support Directed Interrogation (see reference /EC CAT 007/).

For Category 007, the UAP as described in the following table shall be used for messages from the MSSR 2000 ID to an external device (Downlink):

*Table 4-4. Standard UAP for Category 007 "Downlink"*

| FRN | Data Item | Data Item Description                              | Length |
|-----|-----------|----------------------------------------------------|--------|
| 1   | I007/010  | Data Source Identifier                             | 2      |
| 2   | I007/025  | Data Destination Identifier                        | 2      |
| 3   | I007/410  | Directed Interrogation Message Type                | 1      |
| 4   | I007/140  | Time-of-Day                                        | 3      |
| 5   | I007/400  | Directed Interrogation Request Number              | 2      |
| 6   | I007/020  | Target Report Descriptor                           | 1+     |
| 7   | I007/040  | Measured Position in Slant Polar Co-ordinates      | 4      |
| FX  | n.a.      | Field Extension Indicator                          | n.a.   |
| 8   | I007/070  | Mode-3/A Code in Octal Representation              | 2      |
| 9   | I007/090  | Flight Level in Binary Representation              | 2      |
| 10  | I007/130  | Radar Plot Characteristics                         | 1+1+   |
| 11  | I007/220  | Aircraft Address                                   | 3      |
| 12  | I007/240  | Aircraft Identification                            | 6      |
| 13  | I007/250  | Mode S MB Data                                     | 1+8*n  |
| 14  | I007/161  | Track Number                                       | 2      |
| FX  | n.a.      | Field Extension Indicator                          | n.a.   |
| 15  | I007/042  | Calculated Position in Cartesian Co-ordinates      | 4      |
| 16  | I007/200  | Calculated Track Velocity in Polar Representation  | 4      |
| 17  | I007/170  | Track Status                                       | 1+     |
| 18  | I007/210  | Track Quality                                      | 4      |
| 19  | I007/030  | Warning/Error Conditions                           | 1+     |
| 20  | I007/080  | Mode-3/A Code Confidence Indicator                 | 2      |
| 21  | I007/100  | Mode-C Code and Confidence Indicator               | 4      |
| FX  | n.a.      | Field Extension Indicator                          | n.a.   |
| 22  | I007/110  | Height Measured by 3D Radar                        | 2      |
| 23  | I007/120  | Radial Doppler Speed                               | 1+     |
| 24  | I007/230  | Communications / ACAS Capability and Flight Status | 2      |
| 25  | I007/260  | ACAS Resolution Advisory Report                    | 7      |
| 26  | I007/055  | Mode-1 Code in Octal Representation                | 1      |
| 27  | I007/050  | Mode-2 Code in Octal Representation                | 2      |
| 28  | I007/065  | Mode-1 Code Confidence Indicator                   | 1      |
| FX  | n.a.      | Field Extension Indicator                          | n.a.   |
| 29  | I007/060  | Mode-2 Code Confidence Indicator                   | 2      |
| 30  | I007/450  | Directed Interrogation Result                      | 1+     |

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|    |          |                                  |      |
|----|----------|----------------------------------|------|
| 31 | I007/085 | Mode 5, extended Mode 1, X-Pulse | 1+   |
| 32 | -        | Unused                           | -    |
| 33 | -        | Unused                           | -    |
| 34 | SPF      | Special Purpose Field            | 1+1+ |
| 35 | REF      | Reserved Expansion Field         | 1+1+ |
| FX | n.a.     | Field Extension Indicator        | n.a. |

For Category 007, the UAP as described in the following table shall be used for messages to the MSSR 2000 ID from an external device (Uplink):

*Table 4-5. Standard UAP for Category 007 "Uplink"*

| FRN | Data Item | Data Item Description                             | Length |
|-----|-----------|---------------------------------------------------|--------|
| 1   | I007/010  | Data Source Identifier                            | 2      |
| 2   | I007/025  | Data Destination Identifier                       | 2      |
| 3   | I007/410  | Directed Interrogation Message Type               | 1      |
| 4   | I007/140  | Time-of-Day                                       | 3      |
| 5   | I007/400  | Directed Interrogation Request Number             | 2      |
| 6   | I007/040  | Measured Position in Slant Polar Co-ordinates     | 4      |
| 7   | I007/220  | Aircraft Address                                  | 3      |
| FX  | n.a.      | Field Extension Indicator                         | n.a.   |
| 8   | I007/161  | Track Number                                      | 2      |
| 9   | I007/042  | Calculated Position in Cartesian Co-ordinates     | 4      |
| 10  | I007/200  | Calculated Track Velocity in Polar Representation | 4      |
| 11  | I007/415  | Required Interrogation Modes                      | 1+     |
| 12  | I007/420  | Directed Interrogation Window                     | 8      |
| 13  | I007/440  | Directed Interrogation BDS Register Request       | 1+     |
| 14  | -         | Unused                                            | -      |
| FX  | n.a.      | Field Extension Indicator                         | n.a.   |
| 15  | -         | Unused                                            | -      |
| 16  | -         | Unused                                            | -      |
| 17  | -         | Unused                                            | -      |
| 18  | -         | Unused                                            | -      |
| 19  | -         | Unused                                            | -      |
| 20  | SPF       | Special Purpose Field                             | 1+1+   |
| 21  | REF       | Reserved Expansion Field (Unused)                 | -      |
| FX  | n.a.      | Field Extension Indicator                         | n.a.   |

|                                                |                     |                                             |                |                  |
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#### 4.3.6 UAP for CAT 017

For Category 017, the UAP as described in the following table shall be used:

*Table 4-6. Standard UAP for Category 017*

| FRN | Data Item | Data Item Description                       | Length |
|-----|-----------|---------------------------------------------|--------|
| 1   | I017/010  | Data Source Identifier                      | 2      |
| 2   | I017/012  | Data Destination Identifier                 | 2      |
| 3   | I017/000  | Message Type                                | 1      |
| 4   | I017/350  | Cluster station / node list                 | 1+2*n  |
| 5   | I017/220  | Aircraft Address                            | 3      |
| 6   | I017/221  | Duplicate address reference number          | 2      |
| 7   | I017/140  | Time of Day                                 | 3      |
| FX  | n.a.      | Field Extension Indicator                   | n.a.   |
| 8   | I017/045  | Aircraft position in WGS 84 co-ordinates    | 6      |
| 9   | I017/070  | Mode 3/A code in octal representation       | 2      |
| 9   | I017/050  | Flight Level in Binary Representation       | 2      |
| 10  | I017/200  | Track velocity in polar co-ordinates        | 4      |
| 11  | I017/230  | Transponder capability                      | 1      |
| 12  | I017/240  | Track status                                | 1      |
| 13  | I017/210  | Mode-S address list                         | 2+3*n  |
| FX  | n.a.      | Field Extension Indicator                   | n.a.   |
| 15  | I017/360  | Cluster controller command state            | 1      |
| 16  | -         | Spare                                       | -      |
| 17  | -         | Spare                                       | -      |
| 18  | -         | Spare                                       | -      |
| 19  | -         | Spare                                       | -      |
| 20  | -         | Spare                                       | -      |
| 21  | -         | Reserved for Special Purpose Indicator (SP) | -      |
| FX  | n.a.      | Field Extension Indicator                   | n.a.   |

|                                                |                     |                                             |                |                  |
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#### 4.3.7 UAP for CAT 018

For Category 018, the UAP as described in the following table shall be used:

*Table 4-7. Standard UAP for Category 018*

| FRN | Data Item | Data Item Description     | Length  |
|-----|-----------|---------------------------|---------|
| 1   | I018/036  | D_Source                  | 2       |
| 2   | I018/037  | D_Destination             | 2       |
| 3   | I018/000  | D_Message_type            | 1       |
| 4   | I018/001  | D_Result                  | 1       |
| 5   | I018/005  | D_Mode_S_address          | 3       |
| 6   | I018/016  | D_Packet_number           | 4       |
| 7   | I018/017  | D_Packet_number_list      | (1+4*N) |
| FX  | n.a.      | Field Extension Indicator | n.a.    |
| 8   | I018/018  | D_Packet_properties       | 1       |
| 9   | I018/019  | D_Mode_S_packet           | 3+      |
| 10  | I018/028  | D_GICB_periodicity        | 2       |
| 11  | I018/030  | D_GICB_properties         | 2       |
| 12  | I018/025  | D_GICB_number             | 4       |
| 13  | I018/027  | D_BDS_code                | 1       |
| 14  | I018/029  | D_GICB_extracted          | 7       |
| FX  | n.a.      | Field Extension Indicator | n.a.    |
| 15  | I018/002  | D_Time                    | 3       |
| 16  | I018/006  | D_Mode_S_address_list     | 1+3*N   |
| 17  | I018/007  | D_Data_link_command       | 1       |
| 18  | I018/008  | D_data_link_status        | 1+      |
| 19  | I018/009  | D_Report_request          | 1+      |
| 20  | I018/010  | D_COM                     | 1       |
| 21  | I018/011  | D_ECA                     | 7       |
| FX  | n.a.      | Field Extension Indicator | n.a.    |
| 22  | I018/014  | D_Polar_position          | 4       |
| 23  | I018/015  | D_Cartesian_position      | 4       |
| 24  | I018/020  | D_Broadcast_number        | 4       |
| 25  | I018/021  | D_Broadcast_properties    | 6       |
| 26  | I018/022  | D_Broadcast_prefix        | 4       |
| 27  | I018/023  | D_Broadcast               | 7       |
| 28  | I018/004  | D_II_Code                 | 1       |
| FX  | n.a.      | Field Extension Indicator | n.a.    |
| 29  | I018/031  | D_Identity                | 6       |
| 30  | I018/032  | D_Mode_A                  | 2       |
| 31  | I018/033  | D_Hight                   | 2       |
| 32  | I018/034  | D_Speed                   | 2       |

|                                                |                     |                                             |                |                  |
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|----|----------|---------------------------|------|
| 33 | I018/035 | D_Heading                 | 2    |
| 34 | I018/012 | D_CQF                     | 1    |
| 35 | I018/013 | D_CQF_method              | 1    |
| FX | n.a.     | Field Extension Indicator | n.a. |

#### 4.3.8 UAP for CAT 021

For Category 021, the UAP as described in the following table shall be used:

*Table 4-8. Standard UAP for Category 021*

| FRN | Data Item | Information                                          | Length |
|-----|-----------|------------------------------------------------------|--------|
| 1   | I021/010  | Data Source Identification                           | 2      |
| 2   | I021/040  | Target Report Descriptor                             | 2+     |
| 3   | I021/161  | Track Number                                         | 2      |
| 4   | I021/015  | Service Identification                               | 1      |
| 5   | I021/071  | Time of Applicability for Position                   | 3      |
| 6   | I021/130  | Position in WGS-84 co-ordinates                      | 6      |
| 7   | I021/131  | Position in WGS-84 co-ordinates high res.            | 8      |
| FX  | -         | Field extension indicator                            | -      |
| 8   | I021/072  | Time of Applicability for Velocity                   | 3      |
| 9   | I021/150  | Air Speed                                            | 2      |
| 10  | I021/151  | True Air Speed                                       | 2      |
| 11  | I021/080  | Target Address                                       | 3      |
| 12  | I021/073  | Time of Message Reception of Position                | 3      |
| 13  | I021/074  | Time of Message Reception of Position-High Precision | 4      |
| 14  | I021/075  | Time of Message Reception of Velocity                | 3      |
| FX  | -         | Field extension indicator                            | -      |
| 15  | I021/076  | Time of Message Reception of Velocity-High Precision | 4      |
| 16  | I021/140  | Geometric Height                                     | 2      |
| 17  | I021/090  | Quality Indicators                                   | 1+     |
| 18  | I021/210  | MOPS Version                                         | 1      |
| 19  | I021/070  | Mode 3/A Code                                        | 2      |
| 20  | I021/230  | Roll Angle                                           | 2      |
| 21  | I021/145  | Flight Level                                         | 2      |
| FX  | -         | Field extension indicator                            | -      |
| 22  | I021/152  | Magnetic Heading                                     | 2      |
| 23  | I021/200  | Target Status                                        | 1      |
| 24  | I021/155  | Barometric Vertical Rate                             | 2      |
| 25  | I021/157  | Geometric Vertical Rate                              | 2      |
| 26  | I021/160  | Ground Vector                                        | 4      |
| 27  | I021/165  | Track Angle Rate                                     | 2      |
| 28  | I021/077  | Time of Message Transmission                         | 3      |
| FX  | -         | Field extension indicator                            | -      |

|                                                |                     |                                             |                |                  |
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|    |          |                                          |       |
|----|----------|------------------------------------------|-------|
| 29 | I021/170 | Target Identification                    | 6     |
| 30 | I021/020 | Emitter Category                         | 1     |
| 31 | I021/220 | Met Information                          | 1+    |
| 32 | I021/146 | Intermediate State Selected Altitude     | 2     |
| 33 | I021/148 | Final State Selected Altitude            | 2     |
| 34 | I021/110 | Trajectory Intent                        | 1+    |
| 35 | I021/016 | Service Management                       | 1     |
| FX | -        | Field extension indicator                | -     |
| 36 | I021/008 | Aircraft Operational Status              | 1     |
| 37 | I021/271 | Surface Capabilities and Characteristics | 1+    |
| 38 | I021/132 | Message Amplitude                        | 1     |
| 39 | I021/250 | Mode S MB Data                           | 1+N*8 |
| 40 | I021/260 | ACAS Resolution Advisory Report          | 7     |
| 41 | I021/400 | Receiver ID                              | 1     |
| 42 | I021/295 | Data Ages                                | 1+    |
| FX | -        | Field extension indicator                | -     |
| 43 | -        | Not Used                                 | -     |
| 44 | -        | Not Used                                 | -     |
| 45 | -        | Not Used                                 | -     |
| 46 | -        | Not Used                                 | -     |
| 47 | -        | Not Used                                 | -     |
| 48 | RE       | Reserved Expansion Field                 | 1+    |
| 49 | SP       | Special Purpose Field                    | 1+    |
| FX | -        | Field extension indicator                | -     |

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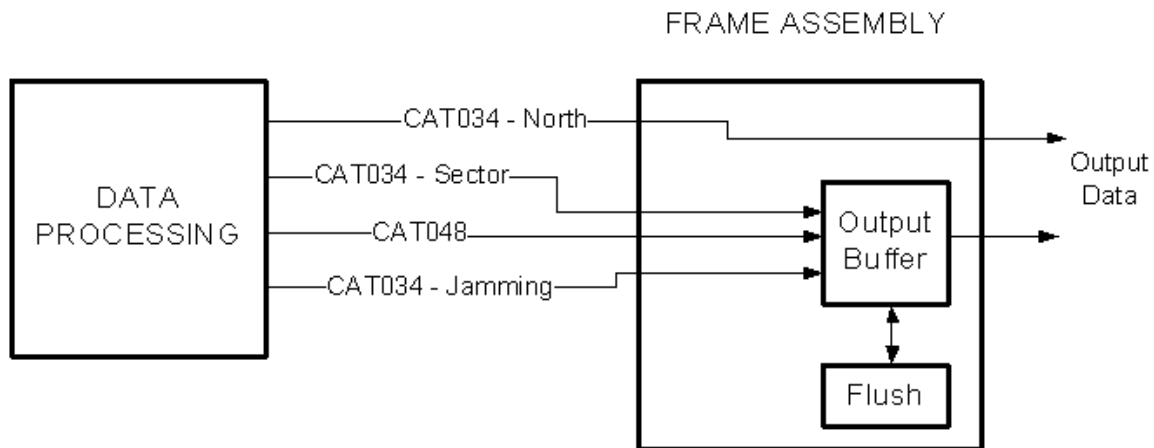
|                                                |                     |                                             |                |                  |
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## 4.4 Data Handling

### 4.4.1 Message Handling at the Output Interfaces

The timely order and the handling of the ASTERIX Messages is in accordance with the ASTERIX standards.

Figure 4-2. ASTERIX Message Output Handling



The ASTERIX Messages output handling is as follows:

- The Signal Processing of the MSSR 2000 ID generate mode reports from the received replies.
- The Data Processing of the MSSR 2000 ID generate the CAT 048 messages from the mode reports.
- The maximum length of a CAT 048 message (Asterix data block) is set to 512 Byte.
- The Output Buffer Size is set to 512 Byte.
- The Messages are held in the Output Buffer until:
  - The Output Buffer is full
  - A Sector Crossing Message has to be transferred
  - All plots and tracks from a sector have been processed
  - The antenna stopped
- If one of the above criteria appears, the Messages are transmitted via the selected interface.

An example at north crossing illustrates the order of sending messages:

- North marker message is sent without buffering to the output
- Sector crossing message (sector number 0) is sent to output buffer and the output buffer is flushed:
  - Output Jamming strobes messages, generated in sector 31
  - Output Raw Plots from sector 30 (if the output delay is accordant configured)
- The sector crossing message starts the processing of Plots and Tracks from sector number 28, if the sector processing is ready the output buffer is flushed again:
  - Output processed Plots and Tracks from sector number 28 (if the delay is configured for 45°)

|                                                |                     |                                             |                |                  |
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#### 4.4.2 Co-ordinates and Moving Platform

##### Definitions for Co-ordinates and Position Data:

If not otherwise specified, all position data shall be provided in co-ordinates, with the origin at radar head and with a reference to local geographical north.

If not otherwise specified, all azimuth or sector related data is north oriented.

##### Special case Moving Platform (if ACP/ARP pulses not north orientated):

Additional to the sector messages generated directly from the antenna azimuth encoder, north orientated sector messages, using heading, pitch and roll values are generated. The north orientated sector messages are used to trigger the internal data processing. The MSSR 2000 ID 'thinks' north orientated.

But it is possible to send Processed Plot and the corresponding north marker and sector crossing messages in the local co-ordinate system (bow orientated).

Note: The ASTERIX standard was not designed for moving platforms, therefore not north orientated data is not conform to the standard /EC CAT 048/.

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## 4.5 Service Messages (CAT 034)

### 4.5.1 General

Mono Radar Service Messages are generated from a radar station (Secondary or Primary), and sent to one or more Surveillance Data Processing systems.

Monoradar service messages are transferred via Data Category 034. The Category 034 service message types supported by the MSSR 2000 ID are described in the following subsections.

Dual redundant system:

CAT 034 delivered from Master and Slave chain.

If the internal connection to the Master chain is interrupted the Slave set the NOGO Bit in I034/050, because it is no more possible to send valid target data.

### 4.5.2 North Marker Message

The north marker indicates the north crossing of the antenna.

Processing delay:

This event is processed without any further delay within the MSSR 2000 ID and immediately sent to the output interfaces. This means that an interrupt from the antenna azimuth encoder forces the generation of this message, but due to internal data processing a target load dependent time (e. g. 10ms up to 80ms) delays the output at LAN port.

Configuration parameter for output enables:

AST\_TX\_NORTH (additional, the sector messages should be present at output port)

Output in state:

NOGO, BITE, STANDBY, TX, CALIBRATION

Moving Platform (if ACP/ARP pulses not north orientated):

The MSSR 2000 ID calculates from the given heading, pitch and roll values a north marker message, additional to the bow orientated 'north' marker message.

If the sector messages at the output Port is north orientated, then the north marker message is send at 'north', and if the sector messages at the output Port is bow orientated, then the north marker message is send at 'bow'.

The North Marker Message is composed of the following data items:

*Table 4-9. North Marker Message*

| Data Item | Description                  | System Units | Information                                                                                |
|-----------|------------------------------|--------------|--------------------------------------------------------------------------------------------|
| I034/010  | Data Source Identifier       | n.a.         | From system configuration.                                                                 |
| I034/000  | Message Type                 | n.a.         | 001 fixed                                                                                  |
| I034/030  | Time of Day                  | 1/128 s      | Time of north crossing (*)                                                                 |
| I034/020  | ---                          | ---          | ---                                                                                        |
| I034/041  | Antenna Rotation Period      | 1/128 s      | Measured from the last north marker<br>Always sent by MSSR 2000 (optional in /EC CAT 034/) |
| I034/050  | System Configuration &Status | n.a.         | Status information determined by system controller, see chapter for Data Item I034/050.    |
| I034/060  | System Processing Mode       | n.a.         | Not used by MSSR 2000 ID (optional in /EC CAT 034/)                                        |
| I034/070  | Message Count Values         | n.a.         | Message Counter, see chapter                                                               |

|                                                |                     |                                             |                |                  |
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|          |                        |                                                                                  |                                                                                                                                                            |
|----------|------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                        |                                                                                  | for Data Item I034/070.                                                                                                                                    |
| I034/100 | ---                    | ---                                                                              | ---                                                                                                                                                        |
| I034/110 | ---                    | ---                                                                              | ---                                                                                                                                                        |
| I034/120 | 3D -position of source | Height: 1 m<br>Latitude: $180^\circ/(2^{23})$<br>Longitude: $180^\circ/(2^{23})$ | see chapter for Data Item I034/120                                                                                                                         |
| I034/090 | Collimation Error      | Range: $1/128$ NM<br>Azimuth: $360^\circ/(2^{14})$                               | Not used by MSSR 2000 ID (optional in /EC CAT 034/)                                                                                                        |
| I034/SPF | Special Purpose Field  | ---                                                                              | SPF Subfields in this message:<br>#1 and #2 (Speed / Heading)<br>#4 (Cluster Info)<br>See Subfield Encoding Rules in Section "I034 Special Purpose Field". |

(\*)Timestamp from system time which is derived from a time source. If no time source found (GPS receiver not connected or NTP server not active), the internal system time is used.

#### 4.5.2.1 System Configuration & Status (I034/050)

This Item and the subfields of this Item sent if information is present.

The MSSR 2000 ID uses the Subfield #1: Common Part (COM)

The MSSR 2000 ID uses the Subfield #6: Mode S Sensor (MDS)

As Subfield #5: "Specific Status for SSR Sensor (SSR)" does not contain additional information or the information is not appropriate, it is not present in the case of MSSR 2000 ID.

The other subfields are not present.

##### Subfield #1: Common Part (COM):

|         |                                                                                                                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NOGO:   | Operational Release Status of the System<br>= 0: System is released for operational use = TRANSMIT of the MSSR 2000 ID;<br>= 1: Operational use of System is inhibited = all other States of the MSSR 2000 ID.<br>In case of a dual redundant system, the NOGO bit is additional set if the MSSR 2000 ID is in REDUNDANT-MAINTENANCE Mode. |
| RDPC:   | Radar Data Processing Chain,<br>= 0: Chain 1 in case of a dual redundant system or single interrogator system<br>= 1: Chain 2 in case of a dual redundant system                                                                                                                                                                           |
| RDPR:   | Radar Data Processing Chain Reset / Restart<br>= 0: default = for MSSR 2000 ID, this value is always = 0<br>= 1: a reset / restart (not used for the MSSR 2000 ID)                                                                                                                                                                         |
| OVL RDP | Radar Data Processor Overload Indicator<br>= 0: default, no processing overload detected within the MSSR 2000 ID<br>= 1: a processing overload situation within the MSSR 2000 ID has been detected.                                                                                                                                        |
| OVL XMT | Transmission Subsystem Overload Status<br>= 0: default, no communication overload detected within the MSSR 2000 ID<br>= 1: a communication overload situation within the MSSR 2000 ID has been detected.                                                                                                                                   |
| MSC     | Monitoring System Connected Status<br>= 0: at least one Monitoring System (SMC) is connected via CAT 253<br>= 1: no Monitoring System is connected                                                                                                                                                                                         |
| TSV     | Time Source Validity<br>= 0: valid (synchronised to NTP or GPS and accurate)<br>= 1: invalid (not accurate)                                                                                                                                                                                                                                |

##### Subfield #6: Mode S Sensor (MDS):

|      |                                    |
|------|------------------------------------|
| ANT: | Selected Antenna<br>= 0: Antenna 1 |
|------|------------------------------------|

|                                                |                     |                                             |                |                  |
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- = 1: Antenna 2  
Not used for MSSR 2000 ID, always set to '0'.
- CH-A/B: Channel A/B selection status for surveillance  
= 00: No channel selected (single chain system without switch over unit)  
= 01: Channel A only selected (Chain 1 is the master chain)  
= 10: Channel B only selected (Chain 2 is the master chain)  
= 11: Illegal combination
- OVL SUR: Overload condition  
= 0: No overload  
= 1: Overload  
Not used for MSSR 2000 ID, always set to '0'.
- MSC Monitoring System connection Status  
= 0: Monitoring System connected  
= 1: Monitoring System disconnected  
Not used for MSSR 2000 ID, always set to '0'.
- SCF Channel A/B selection status for Surveillance Co-ordinated Function  
= 0: Channel A in use  
= 1: Channel B in use  
Not used for MSSR 2000 ID, always set to '0'.
- DLF Channel A/B selection status for Data Link Function  
= 0: Channel A in use  
= 1: Channel B in use  
Not used for MSSR 2000 ID, always set to '0'.
- OVL SCF: Overload in Surveillance Co-ordinated Function  
= 0: No overload  
= 1: Overload  
Not used for MSSR 2000 ID, always set to '0'.
- OVL DLF: Overload in Data Link Function  
= 0: No overload  
= 1: Overload  
Not used for MSSR 2000 ID, always set to '0'.

#### 4.5.2.2 Message Count Values (I034/070)

This Item is always sent by the MSSR 2000 ID (optional in /EC CAT 034/) but only the following types are valid:

- Type:
- = 0: no detection (number of misses, coasted track updates)
  - = 2: single SSR target report (non Mode S)
  - = 4: single All-Call target report (Mode S)
  - = 5: single Roll-Call target report (Mode S)

All other values are not used for MSSR 2000 ID.

Note:

- 1) The types 2, 4 and 5 are the numbers of all reports (Raw Plots, Processed Plots and Tracks).
- 2) For each target the best quality will be count where Roll Call targets are considered to have the highest quality, All Call target are considered to have the second highest quality and SSR targets are considered to have the least quality.

|                                                |            |                  |          |          |
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#### 4.5.2.3 3D-Position of Data Source (I034/120)

The 3D-Position of Data Source is always sent by the MSSR 2000 ID in the North Marker Message.

The value of this data item is either from a configuration value or from a connected navigation or GPS system:

- a) As default the **configured** position (Latitude/Longitude/Altitude) is used.
- b) If a **navigation** interface is configured and navigation data received, the latest received navigation data will be transmitted within data item I034/120. If the navigation interface is down, the configured position is used as fallback.

Note:

The configured position is automatically updated with the current received position in steps of 1 NM.

- c) If a **GPS** is configured and GPS data received, the latest received position will be transmitted within data item I034/120. If the GPS interface is down, the configured position is used as fallback.

Note:

The configured position is automatically updated with the current received position in steps of 1 NM.

The reference point to compute the 3D-Position of the Data Source is set by the TRACK\_POS\_REFERENCE parameter, and is either equal to the radar-head position or the SIRP.

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### 4.5.3 Sector Crossing Messages

The sector crossing indicates that the antenna crosses a certain relative azimuth. In case of MSSR 2000 ID, 32 sectors of equidistant azimuth angles ( $0^\circ$ ,  $11.25^\circ$ ,  $22.50^\circ$ , ...) are used.

Sector Crossing Messages are generated when the antenna crosses the respective azimuths. This means that an interrupt from the antenna azimuth encoder forces the generation of a sector crossing message, but due to internal data processing a target load dependent time (e. g. 10ms up to 100ms) delays the output at LAN port.

Configuration parameter for output enable:

AST\_TX\_SECTOR for north orientated (default) or

AST\_TX\_SCTRAD for bow orientated (it is not allowed to enable both at the same port)

Output in state:

NOGO, BITE, STANDBY, TX, CALIBRATION

Moving Platform (if ACP/ARP pulses not north orientated):

The MSSR 2000 ID calculates from the given heading, pitch and roll values 'virtual' north orientated sector crossing messages, which are used to trigger the tracking process. But it is possible to switch the 'old' bow orientated or the 'virtual' north orientated sector messages to output.

The Sector Crossing Message is composed of the following data items:

*Table 4-10. Sector Crossing Message*

| Data Item | Description                   | System Units                  | Information                                                                                                      |
|-----------|-------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| I034/010  | Data Source Identifier        | n.a.                          | from system configuration.                                                                                       |
| I034/000  | Message Type                  | n.a.                          | 002 fixed                                                                                                        |
| I034/030  | Time of Day                   | 1/128 s                       | time of sector crossing<br>GPS synchronised (see a.)                                                             |
| I034/020  | Sector Number                 | $360^\circ/(2^{\text{exp}8})$ | sector azimuth position                                                                                          |
| I034/041  | ---                           | ---                           | ---                                                                                                              |
| I034/050  | System Configuration & Status | n.a.                          | Status information determined<br>by system controller, see<br>chapter for Data Item I034/050.<br>(optional item) |
| I034/060  | ---                           | ---                           | ---                                                                                                              |
| I034/070  | ---                           | ---                           | ---                                                                                                              |
| I034/100  | ---                           | ---                           | ---                                                                                                              |
| I034/110  | ---                           | ---                           | ---                                                                                                              |
| I034/120  | ---                           | ---                           | ---                                                                                                              |
| I034/090  | ---                           | ---                           | ---                                                                                                              |

a.) Timestamp from system time which is derived from a time source. If no time source found (GPS receiver not connected or NTP server not active), the internal system time is used.

|                                                |                     |                                             |                |                  |
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#### 4.5.4 Jamming Strobe Messages

Optionally, the MSSR 2000 ID can output a third type of radar service messages, called "Jamming Strobes". By default, this output is deactivated.

Jamming strobes are generated from different sources:

- From the Mode Jamming information from the digital signal processing. This information is computed from the number of MK XA replies per second. If the number of Mk XA replies exceeds a threshold, the sector is declared a jammed sector.
- From the rail jamming information from the digital signal processing. This information is computed from the Mode 4 replies. If the Mode 4 reply processing detects a rail jammer the sector is declared a jammed sector.
- From the CW - jamming information from the digital signal processing. This is detected if a signal exceeds length and amplitudes limit. To detect a sector the signal must end within 360°

Note:

The Jamming Strobe Message will not be transmitted until the end of the jammed sector is detected. If one of jamming information is detected a Jamming Strobe Message is generated and transmitted, but the type of source is not within the data content.

Processing delay:

After generation, the message is sent immediately to the output buffer, but due to internal data processing, a target load dependent time (e. g. 10ms up to 100ms) delays the output at LAN port.

Configuration parameter for output enables:

AST\_TX\_JAMMING

Output in state:

TX

Moving Platform (if ACP/ARP pulses not north orientated):

Jamming Strobe Messages are not north corrected!

The Jamming Strobe Message comprises the following information:

*Table 4-11. Jamming Strobe Message*

| Data Item | Description            | System Units                                       | Information                                                                       |
|-----------|------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------|
| I034/010  | Data Source Identifier | n.a.                                               | from system configuration.                                                        |
| I034/000  | Message Type           | n.a.                                               | 004 fixed                                                                         |
| I034/030  | Time of Day            | 1/128 s                                            | Estimated time of detection of jamming                                            |
| I034/020  | ---                    | ---                                                | ---                                                                               |
| I034/041  | ---                    | ---                                                | ---                                                                               |
| I034/050  | ---                    | ---                                                | ---                                                                               |
| I034/060  | ---                    | ---                                                | ---                                                                               |
| I034/070  | ---                    | ---                                                | ---                                                                               |
| I034/100  | Generic Polar Window   | Range: 1/256NM<br>Azimuth: 360°/(2 <sup>16</sup> ) | The jammed sector range.<br>The range window is set to 0 to 256 NM                |
| I034/110  | ---                    | ---                                                | ---                                                                               |
| I034/120  | ---                    | ---                                                | ---                                                                               |
| I034/090  | ---                    | ---                                                | ---                                                                               |
| I034/SPF  | Special Purpose Field  | ---                                                | SPF Subfields in this message:<br>#3 (Jammer Type)<br>See Subfield Encoding Rules |

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|  |  |  | in Section "I034 Special Purpose Field". |
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#### 4.5.5 I034 Special Purpose Field

Several MSSR 2000 ID specific data subfields can be transmitted in the I034 Special Purpose Field:

- Heading and ground speed for moving platforms
- Jammer type information in case of detected RF jamming attempts
- Cluster information

The subfields are described in detail in the following sections. Note that the I034/SPF is not contained in all I034 messages, and I034 messages containing the SPF do not necessarily contain all subfields. Please see the encoding rules for each subfield as defined in the following sections.

Note: implementation specific SPF subfields are in accordance with the ASTERIX standards and do not violate the interface compatibility.

The following structure shows the I034 Special Purpose Field as it is implemented within MSSR 2000 ID.

##### 4.5.5.1 Data Item I034/SPF, Special Purpose Length Indicator

**Definition:** CAT 034 Special Purpose Length Indicator

**Format:** One-octet fixed length Data Item

Structure of Length Indicator:

| Octet No. 1 |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| LEN         |   |   |   |   |   |   |   |

Bit 8/1(LEN) Length of SPF in octets, including the Length Indicator itself.

Encoding Rule:

This data item shall be present in every SPF

##### 4.5.5.2 Data Item I034/SPF, Special Purpose Data Field

**Definition:** CAT 034 Special Purpose Data Field.

**Format:** Compound Data Item, comprising a primary subfield of one octet, followed by up to four defined subfields.

Structure of Primary Subfield:

| Octet No. 1 |   |   |    |     |       |    |   |
|-------------|---|---|----|-----|-------|----|---|
| 8           | 7 | 6 | 5  | 4   | 3     | 2  | 1 |
| GS          | H | J | CI | res | Spare | FX |   |

Bit 8 (GS) Subfield #1: Ground Speed

= 0 Absence of Subfield #1

= 1 Presence of Subfield #1

Bit 7 (H) Subfield #2: Heading

= 0 Absence of Subfield #2

= 1 Presence of Subfield #2

Bit 6 (J) Subfield #3: Jammer Type

= 0 Absence of Subfield #3

= 1 Presence of Subfield #3

Bit 5 (CI) Subfield #4: Cluster Info

= 0 Absence of Subfield #4

= 1 Presence of Subfield #4

Bit 4 (reserved) Reserved bit set to 0

Bit 3/2(Spare) Spare bits set to 0

Bit 1 (FX) Field Extension Indicator (for future use, now set to 0)

Structure of Subfield # 1 (Ground Speed):

| Octet No. 1 |    |    |    |    |    |    |   | Octet No. 2 |   |   |   |   |   |   |   |
|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

|                                                |                     |                                             |                |                  |
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|                            |     |
|----------------------------|-----|
| GROUNDSPPEED (max. 2 NM/s) | LSB |
|----------------------------|-----|

Bit 1 (LSB) =  $(2^{-14})$  NM/s = approx. 0.22 kt

This Subfield shall be used if a Groundspeed from an external navigation system is available. No calculation, except adjusting to the required resolution, is applied to the received Groundspeed information.

Structure of Subfield # 2 (Heading):

|             |    |    |    |    |    |    |   |             |   |   |   |   |   |   |     |
|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|-----|
| Octet No. 1 |    |    |    |    |    |    |   | Octet No. 2 |   |   |   |   |   |   |     |
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1   |
| HEADING     |    |    |    |    |    |    |   |             |   |   |   |   |   |   | LSB |

Bit 1 (LSB) =  $360^\circ / 2^{16}$  = approx. 0.0055°

This Subfield shall be used if a Heading from an external navigation system is available. No calculation, except adjusting to the required resolution, is applied to the received heading information.

Encoding Rules for Subfield #1 and #2:

Subfields #1 and #2 are used only in the North Marker Message, under the following conditions:

- The system must be configured to receive navigation data from a navigation system and must be configured as a moving platform in system parameter NAV\_MOVING\_PLATFORM.
- The respective subfield must be activated in system parameter AST\_CAT034\_SPF\_CONTROL.
- If a navigation system is configured but navigation data is not received or not valid, the Subfields #1 and #2 are not transmitted.

Structure of Subfield # 3 (Jammer Type):

|             |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|
| Octet No. 1 |   |   |   |   |   |   |   |
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Jammer Type |   |   |   |   |   |   |   |

Jammer Type:

- 0: not used
- 1: CW Jammed Sector
- 2: Noise Jammed Sector
- 3-10: not used
- 11: Mode 4 Reply Jammer
- 12: Mode 4 Pulse Jammer
- 13: Mode 4 Rail Jammer
- 14- 20: not used
- 21: MkXA Reply Jammer
- 22: MkXA Pulse Jammer
- 23: MkXA Rail Jammer
- 24 - 30: not used
- 31: Mode 5 Preamble Symbol Jammer
- 32: Mode 5 Reply Repeater Jammer
- 33: Mode 5 Report Repeater Jammer (future use)
- 34 - 255: not used

Encoding Rules:

- Subfield #3 is used only in the Jamming Strobe Message.

Structure of Subfield # 4 (Cluster Info):

|             |   |   |   |   |   |   |   |             |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|-------------|---|---|---|---|---|---|---|
| Octet No. 1 |   |   |   |   |   |   |   | Octet No. 2 |   |   |   |   |   |   |   |
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| SI Code     |   |   |   |   |   |   |   | II Code     |   |   |   |   |   |   |   |

Note:

|                                                |                     |                                             |                |                  |
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SI = 0 indicates, that the II Code is used.

SI > 0 means that the SI code is used, regardless of the II code value.

The SI and II code fields indicate the currently used II / SI code coming from the cluster map.

If multiple sectors with different II/SI codes are configured, this subfield indicates the II/SI code which is valid at zero degree azimuth.

|                                                                   |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |
|-------------------------------------------------------------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| Octet No. 3                                                       |    |    |    |    |    |    |   | Octet No. 4 |   |   |   |   |   |   |   |
| 16                                                                | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Clustermap Version Major (Range: 0000h - 03E7h = 0 - 999 decimal) |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |
| Octet No. 5                                                       |    |    |    |    |    |    |   | Octet No. 6 |   |   |   |   |   |   |   |
| 16                                                                | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Clustermap Version Minor (Range: 0000h - 03E7h = 0 - 999 decimal) |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |

Example:

Clustermap Version 12.34 = 000Ch 0022h

Encoding Rules:

- Subfield #4 is used only in the North Marker Message, and only if activated in system parameter AST\_CAT034\_SPF\_CONTROL.

#### 4.5.6 Specific Content of the Data Items

##### 4.5.6.1 Content of Data Item I034/030, Time of Day

In a North Crossing Message the Item I034/030 represents the time where the antenna has crossed north. Some correction may apply to compensate for the internal processing delays.

In a Sector Crossing Message the Item I034/030 represents the time where the antenna has crossed the sector boundary. Some correction may apply to compensate for the internal processing delays.

|                                                |            |                  |          |          |
|------------------------------------------------|------------|------------------|----------|----------|
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## 4.6 Target Reports (CAT 048)

### 4.6.1 General

Target Messages are generated from a radar station (Secondary or Primary), in our case by the MSSR 2000 ID, and sent to one or more Surveillance Data Processing systems.

Note: The term “if available” in the following tables means, the system will provide this information when the system has gathered this information and the aircraft has delivered the information.

The term “if relevant” in the following tables means, the system will provide this information when the Data Items contain relevant information. If all bits within such a Data Item are = 0 the information is not relevant.

Dual redundant system:

- Raw Plots generated only from the Master chain.
- Processed Plots generated from the Master and the Slave.
- Extended Plots generated from the Master and the Slave.
- Tracks generated from the Master and the Slave.

### 4.6.2 Raw Plots

Raw Plots are generated without processing in the tracking stage. They are intended to present the 'measured' data and for the quickest possible reporting to an adjacent system. They are not intended to achieve the standardised ASTERIX CAT 048 conformity. The output of Raw Plots is per default disabled.

Important Note:

It is strongly recommended not to use the Raw Plot Messages for operational use. Some data items are used in a way which is not compliant with the CAT 048 documentation. Differences are stated here in this document.

Processing delay:

A Plot is processed with a configurable azimuth delay (PA\_OUTPUT\_DELAY\_MP, PA\_OUTPUT\_DELAY\_SW, PA\_OUTPUT\_DELAY\_M5\_SW), which is necessary to associate the mode reports given from the digital signal processing to a plot. The plot information is sent to the tracking stage for further processing and directly to the output buffer as a Raw Plot.

Example:

The default value for the output delay is 12.7°. The output takes place between this value and next sector border (22.5°). At the antenna turn time of 5 sec and low target load, the target delay can be estimated by:  $5s \cdot (22.5^\circ / 360^\circ) + 10ms = 322.5ms$

Configuration parameter for output enable:

AST\_TX\_RAW\_PLOT

Output in state:

TX, CALIBRATION

Moving Platform (if ACP/ARP pulses not north orientated):

Raw Plots not north corrected.

**Table 4-12. Radar Target Messages: Raw Plots**

| Data Item | Description                                  | System Units                                      | Information                 |
|-----------|----------------------------------------------|---------------------------------------------------|-----------------------------|
| I048/010  | Data Source Identifier                       | n.a.                                              | from system configuration.  |
| I048/140  | Time-of-Day                                  | 1/128 s                                           | estimated time of detection |
| I048/020  | Target Report Descriptor                     | n.a.                                              | content is target dependent |
| I048/040  | Measured Position in Slant Polar Coordinates | Range: 1/256NM<br>Azimuth: $360^\circ / (2^{16})$ | actual measurement          |

|                                                |                     |                                             |                |                  |
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|          |                                                    |        |                                                                    |
|----------|----------------------------------------------------|--------|--------------------------------------------------------------------|
| I048/070 | Mode-3/A Code (octal)                              | n.a.   | if available                                                       |
| I048/090 | Flight Level in binary representation              | 1/4 FL | if available                                                       |
| I048/130 | Radar Plot Characteristics                         | n.a.   | determined by reply and plot processing                            |
| I048/220 | Aircraft Address                                   | n.a.   | If available                                                       |
| I048/240 | Aircraft Identification                            | n.a.   | If available                                                       |
| I048/250 | Mode S MB Data                                     | n.a.   | If available                                                       |
| I048/161 | ---                                                | ---    | ---                                                                |
| I048/042 | ---                                                | ---    | ---                                                                |
| I048/200 | ---                                                | ---    | ---                                                                |
| I048/170 | ---                                                | ---    | ---                                                                |
| I048/210 | ---                                                | ---    | ---                                                                |
| I048/030 | Warning/Error Conditions                           | n.a.   | not used                                                           |
| I048/080 | Mode-3/A Code Confidence                           | n.a.   | If relevant                                                        |
| I048/100 | Mode-C Code and Confidence                         | n.a.   | Measured value, always available regardless of decode able or not. |
| I048/110 | ---                                                | ---    | ---                                                                |
| I048/120 | ---                                                | ---    | ---                                                                |
| I048/230 | Communications / ACAS Capability and Flight Status | n.a.   | If available                                                       |
| I048/260 | ACAS Resolution Advisory                           | n.a.   | If available                                                       |
| I048/055 | Mode-1 Code (octal)                                | n.a.   | If available                                                       |
| I048/050 | Mode-2 Code (octal)                                | n.a.   | If available                                                       |
| I048/065 | Mode-1 Code Confidence                             | n.a.   | If relevant                                                        |
| I048/060 | Mode-2 Code Confidence                             | n.a.   | If relevant                                                        |
| I048/SPF | Special Purpose Field                              | ---    | MSSR 2000 ID specific extension if necessary                       |
| I048/REF | Reserved Expansion Field                           | ---    | if available<br>see /EC CAT 048 REF 1.7/,<br>/EC CAT 048 REF 1.3/  |

If the parameter ANTENNA\_NOT\_TURNING is set, the item I048/040 contains the OBI value in the 16 bit field "theta" instead the azimuth value.

|                                                |                     |                                             |                |                  |
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### 4.6.3 Processed Plots

High quality plots as they are generated by the tracking stage.

Processing delay:

The Plot is processed with a configurable azimuth delay (SU\_UPD\_DELAY), which is necessary to check the association with a track. Then the Plot is sent to the output buffer.

Example:

The default value for the output delay is 45°. The output takes place between this value and next sector border (45°). At the antenna turn time of 5 sec and low target load, the target delay can be estimated by:  $5s \cdot (45^\circ / 360^\circ) + 10ms = 635ms$

Initial generation of Processed Plot (All-Call):

First scan of detection. Even if there is no internal track, the plot can associate with. This behaviour is against the specification in reference /ECMS/ but necessary to provide data output following Directed Interrogations.

Configuration parameter for output enable:

AST\_TX\_PLOT

AST\_TX\_NO\_ALLCALL (disable output of Mode S All-Call targets)

AST\_TX\_CHECK\_MAP (Output only targets in surveillance area)

Output in state:

TX

Moving Platform (if ACP/ARP pulses not north orientated):

Processed Plots are north corrected, but if the parameter SU\_PROC\_PLOT\_REF\_SYS = 1, then the plot position is transformed back to ship co-ordinate system before sent to output buffer.

The following conditions must be fulfilled to compensation platform movement:

- valid navigation data (heading, pitch, roll, latitude, longitude) from the platform
- correct high information from the target

Radar Target Messages: Processed Plots (Actual / Test / FFM):

**Table 4-13. Radar Target Messages: Processed Plots (Actual / Test / FFM)**

| Data Item | Description                                  | System Units                                      | Information                                                                                    |
|-----------|----------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------|
| I048/010  | Data Source Identifier                       | n.a.                                              | from system configuration.                                                                     |
| I048/140  | Time-of-Day                                  | 1/128 s                                           | estimated time of detection                                                                    |
| I048/020  | Target Report Descriptor                     | n.a.                                              | content is target dependent<br>SIM, TST is set for internal test targets<br>RAB is set for FFM |
| I048/040  | Measured Position in Slant Polar Coordinates | Range: 1/256NM<br>Azimuth: $360^\circ / (2^{16})$ | actual measurement                                                                             |
| I048/070  | Mode-3/A Code (octal)                        | n.a.                                              | if available or smoothed when configured                                                       |
| I048/090  | Flight Level in binary representation        | 1/4 FL                                            | if available                                                                                   |
| I048/130  | Radar Plot Characteristics                   | n.a.                                              | determined by reply and plot processing                                                        |
| I048/220  | Aircraft Address                             | n.a.                                              | present only for Mode S targets                                                                |
| I048/240  | Aircraft Identification                      | n.a.                                              | present only for Mode S targets after first Aircraft Identification extraction when configured |
| I048/250  | Mode S MB Data                               | n.a.                                              | present only for Mode S targets if                                                             |

|                                                |                     |                                             |                |                  |
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|          |                                                    |      |                                                                   |
|----------|----------------------------------------------------|------|-------------------------------------------------------------------|
|          |                                                    |      | extracted in current scan                                         |
| I048/161 | Track Number                                       | ---  | present only if the TRACK_ID_IN_PPLOT_ENABLE parameter is enabled |
| I048/042 | ---                                                | ---  | ---                                                               |
| I048/200 | ---                                                | ---  | ---                                                               |
| I048/170 | ---                                                | ---  | ---                                                               |
| I048/210 | ---                                                | ---  | ---                                                               |
| I048/030 | Warning/Error Conditions                           | n.a. | if indicated                                                      |
| I048/080 | Mode-3/A Code Confidence                           | n.a. | If relevant or smoothed when configured                           |
| I048/100 | Mode-C Code and Confidence                         | n.a. | If relevant                                                       |
| I048/110 | ---                                                | ---  | ---                                                               |
| I048/120 | ---                                                | ---  | ---                                                               |
| I048/230 | Communications / ACAS Capability and Flight Status | n.a. | present only for Mode S targets                                   |
| I048/260 | ACAS Resolution Advisory                           | n.a. | if available                                                      |
| I048/055 | Mode-1 Code (octal)                                | n.a. | if available or smoothed when configured                          |
| I048/050 | Mode-2 Code (octal)                                | n.a. | if available or smoothed when configured                          |
| I048/065 | Mode-1 Code Confidence                             | n.a. | If relevant                                                       |
| I048/060 | Mode-2 Code Confidence                             | n.a. | If relevant                                                       |
| I048/SPF | Special Purpose Field                              | ---  | MSSR 2000 ID specific extension if necessary                      |
| I048/REF | Reserved Expansion Field                           | ---  | if available<br>see /EC CAT 048 REF 1.7/, /EC CAT 048 REF 1.3/    |

|                                                |                     |                                             |                |                  |
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#### 4.6.4 Extended Plot Message and Track Creation/Update Message

A Track Creation/Update Message is generated with every scan for a maintained track. In ASTERIX creation of a track is implicit by sending the first update message.

An Extended Plot Message is generated in the first scan of target detection to report Plot information with a preliminary Track number at the Track output port.

Processing delay:

The Track messages are processed with the same delay like the Processed Plot messages (see chapter above).

Initial generation of Tracks (All-Call):

Extended Plot in the first scan of target detection at Track Port.

Track Creation in the second scan of target detection.

Track Update in the scan after Track Creation.

Configuration parameter for output enables:

AST\_TX\_TRACK (together with the Track Creation/Update Message)

AST\_TX\_NO\_ALLCALL (disable output of Mode S All-Call targets)

AST\_TX\_CHECK\_MAP (Output only targets in surveillance area)

EXTENDED\_PLOT\_ENA (Enable or disable the output of extended plots )

TRACK\_POS\_REFERENCE (Item 42 - calculated position - is referenced to either the radar head, or the SIRP)

Output in state:

TX

Moving Platform (if ACP/ARP pulses not north orientated):

Tracks are north corrected.

The following conditions must be fulfilled to compensate platform movement:

- valid navigation data (heading, pitch, roll, latitude, longitude, timestamp) from the platform
- correct height information from the target

Track Creation/Update Message:

*Table 4-14. Extended Plot Message and Track Creation/Update Message*

| Data Item | Description                                  | System Units                                       | Information                                                                                       |
|-----------|----------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------|
| I048/010  | Data Source Identifier                       | n.a.                                               | from system configuration.                                                                        |
| I048/140  | Time-of-Day                                  | 1/128 s                                            | estimated time of detection or predicted time in case no plot contributes to the track (coasting) |
| I048/020  | Target Report Descriptor                     | n.a.                                               | content is target dependent                                                                       |
| I048/040  | Measured Position in Slant Polar Coordinates | Range: 1/256NM<br>Azimuth: 360°/(2 <sup>16</sup> ) | actual measurement                                                                                |
| I048/070  | Mode-3/A Code (Octal)                        | n.a.                                               | if available or smoothed                                                                          |
| I048/090  | Flight Level                                 | 1/4 FL                                             | FL as measured from the supporting Plot, if it is fitting into the 3D motion model of the track   |
| I048/130  | Radar Plot Characteristics                   | n.a.                                               | determined by reply and plot processing                                                           |
| I048/220  | Aircraft Address                             | n.a.                                               | present only for Mode S targets                                                                   |
| I048/240  | Aircraft Identification                      | n.a.                                               | present only for Mode S                                                                           |

|                                                |                     |                                             |                |                  |
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|          |                                                    |                                                                                                        |                                                                              |
|----------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
|          |                                                    |                                                                                                        | targets after first Aircraft Identification extraction                       |
| I048/250 | Mode S MB Data                                     | n.a.                                                                                                   | present only for Mode S targets if extracted in current scan                 |
| I048/161 | Track Number                                       | 0-4095                                                                                                 | internal track number, a preliminary number is included in the Extended Plot |
| I048/042 | Calculated Position in Cartesian Coordinates       | 1/128 NM                                                                                               | computed, included in the Extended Plot                                      |
| I048/200 | Calculated Track Velocity in Polar Representation  | Groundspeed: $2^{-14}$ NM/s = 0,22 kt.<br>Heading: $360^\circ/2^{-16} = 0,0055^\circ$                  | computed by tracker, not included in the Extended Plot                       |
| I048/170 | Track Status                                       | n.a.                                                                                                   | determined by tracker, not included in the Extended Plot                     |
| I048/210 | Track Quality                                      | $\sigma_x$ [1/128 NM],<br>$\sigma_y$ [1/128 NM],<br>$\sigma_v$ [0.22 kt] and<br>$\sigma_h$ [0.08789°]. | determined by tracker, included in the Extended Plot                         |
| I048/030 | Warning/Error Conditions                           | n.a.                                                                                                   | if indicated                                                                 |
| I048/080 | Mode 3/A Code Confidence                           | ---                                                                                                    | If relevant or smoothed                                                      |
| I048/100 | Mode-C Code and Confidence                         | ---                                                                                                    | If relevant                                                                  |
| I048/110 | ---                                                | ---                                                                                                    | ---                                                                          |
| I048/120 | ---                                                | ---                                                                                                    | ---                                                                          |
| I048/230 | Communications / ACAS Capability and Flight Status | n.a.                                                                                                   | present only for Mode S targets                                              |
| I048/260 | ACAS Resolution Advisory                           | n.a.                                                                                                   | if available or smoothed                                                     |
| I048/055 | Mode-1 Code (octal)                                | n.a.                                                                                                   | if available or smoothed                                                     |
| I048/050 | Mode-2 Code (octal)                                | n.a.                                                                                                   | if available or smoothed                                                     |
| I048/065 | Mode-1 Code Confidence                             | n.a.                                                                                                   | If relevant                                                                  |
| I048/060 | Mode-2 Code Confidence                             | n.a.                                                                                                   | If relevant                                                                  |
| I048/SPF | Special Purpose Field                              | ---                                                                                                    | MSSR 2000 ID specific extension if necessary                                 |
| I048/REF | Reserved Expansion Field                           | ---                                                                                                    | if available<br>see /EC CAT 048 REF 1.7/,<br>/EC CAT 048 REF 1.3/            |

|                                                |                     |                                             |                |                  |
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#### 4.6.5 Track Termination Message

A Track Termination Message is sent when the tracking system within MSSR 2000 ID cancels the track. No further position updates are generated for the respective track.  
Behaviour like Track Creation/Update Message (see chapter above)

*Table 4-15. Track Termination Message*

| Data Item | Description              | System Units | Information                 |
|-----------|--------------------------|--------------|-----------------------------|
| I048/010  | Data Source Identifier   | n.a.         | from system configuration.  |
| I048/140  | Time-of-Day              | 1/128 s      | time of transmission        |
| I048/020  | Target Report Descriptor | n.a.         | content is target dependent |
| I048/040  | ---                      | ---          | ---                         |
| I048/070  | ---                      | ---          | ---                         |
| I048/090  | ---                      | ---          | ---                         |
| I048/130  | ---                      | ---          | ---                         |
| I048/220  | ---                      | ---          | ---                         |
| I048/240  | ---                      | ---          | ---                         |
| I048/250  | ---                      | ---          | ---                         |
| I048/161  | Track Number             | 0-4095       | deleted track               |
| I048/042  | ---                      | ---          | ---                         |
| I048/200  | ---                      | ---          | ---                         |
| I048/170  | Track Status             | n.a.         | track termination           |
| I048/210  | ---                      | ---          | ---                         |
| I048/030  | ---                      | ---          | ---                         |
| I048/080  | ---                      | ---          | ---                         |
| I048/100  | ---                      | ---          | ---                         |
| I048/110  | ---                      | ---          | ---                         |
| I048/120  | ---                      | ---          | ---                         |
| I048/230  | ---                      | ---          | ---                         |
| I048/260  | ---                      | ---          | ---                         |
| I048/055  | ---                      | ---          | ---                         |
| I048/050  | ---                      | ---          | ---                         |
| I048/065  | ---                      | ---          | ---                         |
| I048/060  | ---                      | ---          | ---                         |
| I048/SPF  | ---                      | ---          | ---                         |
| I048/REF  | ---                      | ---          | ---                         |

**Note:**

The Track Termination Message is sent for Extended Plot and Tracks if the internal Track number has been deleted.

|                                                |                     |                                             |                |                  |
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#### 4.6.6 Discrimination between different types of CAT 048 Messages

To discriminate between Raw Plot, Processed Plot, Extended Plot, Track Creation/Update and Track Termination Messages, the following criteria can be used:

*Table 4-16. Discrimination between different types of CAT 048 Messages*

| Message                                                | SP-Field, RP-Bit      | Track Status, I048/170    |
|--------------------------------------------------------|-----------------------|---------------------------|
| Raw Plot                                               | present, RP = 1       | not present               |
| Processed Plot                                         | not present or RP = 0 | not present               |
| Extended Plot                                          | not present or RP = 0 | not present               |
| Track Creation (if output of tentative tracks enabled) | not present or RP = 0 | present, CNF = 1, TRE = 0 |
| Track Update                                           | not present or RP = 0 | present, CNF = 0, TRE = 0 |
| Track Termination                                      | not present or RP = 0 | present, TRE = 1          |

CFN (Data Item I048/170): Confirmed vs. Tentative Track. cf part 4.6.8.13.

TRE (Data Item I048/170): Signal for End\_of\_Track. cf part 4.6.8.13.

|                                                |                     |                                             |                |                  |
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#### 4.6.7 I048 Special Purpose Field

To provide additional information for targets, namely X-Pulse information (required in some applications and areas), the ASTERIX data message can be extended by a system specific Special Purpose Item. This is in accordance with the ASTERIX standards and does not violate the interface compatibility.

The special purpose field always represents an explicit length field with one octet containing the overall length of the item. This allows standard applications not recognising the special purpose item to simply ignore its contents and to carry on processing data with the next record or block.

The special purpose field is added to a message if and only if at least one of the information bits is set. In the other cases the absence of the special purpose field implicitly identifies that none of the bits (e.g. X-pulses) was recognised in the message.

The following structure shows the octet sequence for the I048 Special Purpose Item of the MSSR 2000 ID.

##### 4.6.7.1 Data Item I048/SPF, Special Purpose Length Indicator

**Definition:** MSSR 2000 ID Special Purpose Item:

**Format:** Special Purpose field (explicit length)

Structure of Length Indicator:

| Octet No. 1 |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| LEN         |   |   |   |   |   |   |   |

Bit 8/1LEN Length of SPF in octets, including the Length Indicator itself.

= 3, if E40 = 0 and E42 = 0 and BPP = 0

= 8, if E40 = 1 and E42 = 0 and BPP = 0

= 14, if E40 = 1 and E42 = 1 and BPP = 0

Note:

Bit E42 is not set, if E40 is not set

Note: If MkX is set, the length is increased by 6.

Note: If BPP is set, the length is increased by 3.

##### 4.6.7.2 Data Item I048/SPF, Special Purpose Data Field

**Definition:** CAT 048 Special Purpose Data Field.

**Format:** Compound Data Item, comprising a primary subfield, followed by defined subfields.

Note:

For compatibility reasons, the primary subfield contains some data bits and an additional octet (see below) is sent.

**Structure of Primary Subfield:**

| Octet No. 1 |    |    |    |     |     |     |    |
|-------------|----|----|----|-----|-----|-----|----|
| 8           | 7  | 6  | 5  | 4   | 3   | 2   | 1  |
| XA          | X2 | RP | SL | E40 | E42 | MkX | FX |
| Octet No. 2 |    |    |    |     |     |     |    |
| 8           | 7  | 6  | 5  | 4   | 3   | 2   | 1  |
| BPP         | 0  | 0  | 0  | 0   | 0   | 0   | FX |

Bit 8, octet 1 XAX pulse detected in Mode 3/A reply (1 = Mode 3/A X pulse)

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If the E42 Bit is set, the following 6 Byte Item is included as calculated position of an aircraft in Cartesian co-ordinates:

**Structure of Subfield # 2 (extended I042):**

This subfield can only be part of Track messages.

| Octet No. 1          |    |    |    |    |    |    |    | Octet No. 2 |    |    |    |    |    |    |   | Octet No. 3 |   |   |   |   |   |   |   |
|----------------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| 24                   | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| X component extended |    |    |    |    |    |    |    |             |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |

Bit 1 LSB =  $1/2^{14}$  NM Max Range =  $\pm 2^{23} / 16384 = 512$  NM

| Octet No. 4          |    |    |    |    |    |    |    | Octet No. 5 |    |    |    |    |    |    |   | Octet No. 6 |   |   |   |   |   |   |   |
|----------------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| 24                   | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Y component extended |    |    |    |    |    |    |    |             |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |

Bit 1 LSB =  $1/2^{14}$  NM Max Range =  $\pm 2^{23} / 16384 = 512$  NM

**Structure of Subfield # 3 Part 1 (MkX info M3/A code):**

| Octet No. 1 |    |    |    |    |    |    |    | Octet No. 2 |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8           | 7  | 6  | 5  | 4  | 3  | 2  | 1  |
| NE          | V  | G  | L  | A4 | A2 | A1 | B4 | B2          | B1 | C4 | C2 | C1 | D4 | D2 | D1 |

Bit 16 (NE) Not Empty = 0 The content of MkX M3/A is not filled and has to be ignored.

= 1 The content of MkX M3/A code is filled in. The code and the V,G,L bits are derived from a MkX reply.

On Mixed mode interrogation (Mode S and MkX), the content of CAT 048/070 item (M3/A Code) is derived from Mode S.

Bit 15 (V) = 0 Code validated

= 1 Code not validated

Bit 14 (G) = 0 Default (Code not garbled)

= 1 Code garbled

Bit 13 (L) = 0 Mode 3/A code derived from the MkX reply of the transponder

= 1 Mode 3/A code not extracted during the last scan

Bit 12/1 Mode 3/A code in octal representation

**Structure of Subfield # 3 Part 2 (MkX info Flight Level):**

| Octet No. 1 |    |              |    |    |    |    |   | Octet No. 2 |   |   |   |   |   |   |   |
|-------------|----|--------------|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| 16          | 15 | 14           | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| NE          | G  | Flight Level |    |    |    |    |   |             |   |   |   |   |   |   |   |

Bit 16 (NE) Not Empty = 0 The content of MkX Flight Level not filled and has to be ignored.

= 1 The content of MkX Flight Level is filled in. The Flight Level and the G bits are derived from a MkX reply.

On Mixed mode interrogation (Mode S and MkX), the content of CAT 048/090 item (Flight level) is derived from Mode S.

Bit 15 (G) = 0 Default (Code not garbled)

= 1 Code garbled

Bit 14/1 Flight Level LSB =  $1/4$  FL (25ft)

Note: Part 2 of the subfield # 3 is only filled (NE = 1), if the MkX M3/MC code is validated and decodable. Otherwise Part 3 of Subfield # 3 is filled. It is always present when Subfield #3 is present.

**Structure of Subfield # 3 Part 3 (MkX info M3/MC code):**

| Octet No. 1 |    |    |    |    |    |    |   | Octet No. 2 |   |   |   |   |   |   |   |
|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

|                                                |                     |                                             |                |                  |
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|    |   |   |     |    |    |    |    |    |    |    |    |    |    |    |    |
|----|---|---|-----|----|----|----|----|----|----|----|----|----|----|----|----|
| NE | V | G | res | C1 | A1 | C2 | A2 | C4 | A4 | B1 | D1 | B2 | D2 | B4 | D4 |
|----|---|---|-----|----|----|----|----|----|----|----|----|----|----|----|----|

Bit 16 (NE) Not Empty = 0 The content of MkX M3/MC is not filled and has to be ignored.  
 = 1 The content of MkX M3/MC code is filled in. The code and the V,G bits are derived from a MkX reply.

Bit 15 (V) = 0 Code validated

= 1 Code not validated

Bit 14 (G) = 0 Default (Code not garbled)

= 1 Code garbled

Bit 13 = reserved (set to 0)

Bit 12/1 Mode 3/Mode C code in octal representation

Note: Part 3 of the subfield # 3 is only filled (NE = 1), if the MkX M3/MC code is not validated or undecodable. Otherwise Part 2 of Subfield # 3 is filled. It is always present when Subfield #3 is present.

Note:

The MkX Info can be deactivated by a config parameter. If activated, the subfield MkX Info is only filled with M3/A code and/or M3/MC code, if the data was received by MkX interrogation.

The special purpose field can be part of Raw Plots, Processed Plots and Track messages (with exceptions for some subfields as defined above).

#### Structure of Subfield # 4 (BPP):

| Octet No. 1 |   |   |   |   |   |   |   | Octet No. 2 |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|-------------|---|---|---|---|---|---|---|
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| START BPP   |   |   |   |   |   |   |   | STOP BPP    |   |   |   |   |   |   |   |

Bit 8/1, Octet 1 START\_BPP Unsigned Integer, Number of the start BPP of a range of BPPs a target was detected in, in a clockwise orientation

Bit 8/1, Octet 2 STOP\_BPP Unsigned Integer, Number of the stop BPP of a range of BPPs a target was detected in, in a clockwise orientation

The Subfield # 4 for Beam Pointing Positions is only sent when in a NESIS 4000 or Steer Function 1 system configurations where phased array antennas are used.

In NESIS 4000 BPP values can range from 0 to 127. The start BPP can be higher than the stop BPP when the north azimuth is in the range of the BPPs for one target. The target was detected in the start BPP through the stop BPP.

Note: Subfield # 4 is only sent with raw plots as of now.

#### 4.6.8 Specific Content of the Data Items

The contents of the items of the CAT 048 Messages is in accordance to the relevant ASTERIX documentation.

For a better understanding, the following items will be described in more detail.

##### 4.6.8.1 Handling of the Bit V, L, G in the appropriate Data Items

The description in this chapter is valid for all Data Items, with the exception of V-Bit in I048/REF, containing the V-, L- and G- Bits.

The V-Bit (code validated) is set to 1 (code not validated) when at least one of the bits of the corresponding reply is declared as "Low Confident". For the Extended Mode 1 Code in I048/REF the V-Bit is set to 0 (code not validated).

The L-Bit is = 0 when the value is derived from the reply of the transponder. The L-Bit is = 1 if the value is calculated by the MSSR 2000 ID internal tracker.

|                                                |                     |                                             |                |                  |
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The G-Bit indicates whether the data is generated out of a garbling situation or not. The G-Bit = 1 if the value is calculated from replies, possibly in a garbling situation. The G-Bit = 0 if the value is calculated from replies which are not influenced by a garbling situation. The G-Bit = 1 indicates the garbling situation only, it does not mean that the value is automatically degraded.

#### 4.6.8.2 Content of Data Item I048/020, Target Report Descriptor

This item is part of the following CAT 048 Messages of MSSR 2000 ID:

- Raw Plots,
- Processed Plots,
- Track Creation/Update Message.

When Mode 4 information has been received, the IFF bits are set to 0. The output of the Mode 4 Evaluator is reported in I048/REF.

When Mode 4 has been interrogated but no M4 reply has been received, the IFF bits are coded as follows:

11 ... no reply

##### 4.6.8.2.1 The First Part of Data Item I048/020

The subfield TYP is handled as follows:

If the message which contains this item is combined from both, MK XA Replies and Mode S Replies, a "quality" ranking will be used.

Subfield Type in Data Item I048/020, Target Report Descriptor:

*Table 4-17. Subfield Type in Data Item I048/020, Target Report Descriptor*

| Mode S Roll Call Reply | Mode S All Call Reply | MK XA Reply | TYP                                                        |
|------------------------|-----------------------|-------------|------------------------------------------------------------|
| present                | don't care            | don't care  | 101 = Single Mode S Roll-Call detection                    |
| not present            | present               | don't care  | 100 = Single Mode S All-Call detection                     |
| not present            | not present           | present     | 010 = Single SSR detection                                 |
| not present            | not present           | not present | 000 = no detection, report is generated from coasted track |

The SPI bit is handled as follows:

This bit indicates the presence or absence of the SPI pulse in a MK XA reply, or in case of a Mode S Downlink, it is derived from the content of the Flights Status Field. The Flights Status information will be in Item I048/230, so the SPI information is hold in both Items. The following table 4-20 represents the mapping for the SPI bit. If the SPI Pulse position is used for an identity or emergency response, the SPI Bit is not set.

Subfield SPI in Data Item I048/020, Target Report Descriptor:

*Table 4-18. Subfield SPI in Data Item I048/020, Target Report Descriptor*

| Mode S flight status in DF4, DF5, DF20 or DF21 | MK XA Reply | SPI                             |
|------------------------------------------------|-------------|---------------------------------|
| not present                                    | no SPI      | 0 = Absence of SPI              |
| not present                                    | SPI present | 1 = Special Position indication |
| flight status <= 3                             | don't care  | 0 = Absence of SPI              |
| flight status > 3                              | don't care  | 1 = Special Position indication |

|                                                |                     |                                             |                |                  |
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Interpretation of the other subfields of Data Item I048/020:

**Table 4-19. Subfields of the First Part of Data Item I048/020**

| TYP | subfield TYP                                                              | See detailed description above.                                                                                                              |
|-----|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| SIM | indicates a simulated target                                              | This bit is used to indicate simulated targets.<br>1: the report is generated from a simulated target                                        |
| RDP | indicates the source of the report                                        | in single chain configuration always 0<br>in dual redundant configurations, the content depends on the chain from which the data are coming. |
| SPI | Indicates the presence or absence of the SPI (Special Position Indicator) | See detailed description above.                                                                                                              |
| RAB | Indicates if the report is generated from replies of a Far Field Monitor  | 1: the report is generated from the replies of a Far Field Monitor (FFM). The position of the FFM has been declared by using the FFM command |

#### 4.6.8.2.2 The First Extend of Data Item I048/020

The First Extend of the Data Item I048/020 will only be present, if information is available (at least one bit is set = 1).

Interpretation of the content of the First Extend of Data Item I048/020:

**Table 4-20. Subfields in First Extend of Data Item I048/020**

|           |                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TST       | indicates if the report is generated from replies of an internal test target.                                            | 1: report is generated from replies of an internal test target.                                                                                                                                                                                                                                                                                                                                                                                                          |
| XPP       | Indicates the presence or absence of X-Pulse                                                                             | 1: X-Pulse is present                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ME        | Indicates the presence or absence of military emergency information in the reply (as defined in STANAG 4193 Part 1)      | 1: military emergency information is present                                                                                                                                                                                                                                                                                                                                                                                                                             |
| MI        | Indicates the presence or absence of military identification information in the reply (as defined in STANAG 4193 Part 1) | 1: military identification information is present                                                                                                                                                                                                                                                                                                                                                                                                                        |
| FOE / FRI | Indicated the results of the Mode 4 processing and evaluation                                                            | Interpretation for Processed Plot, Track and Raw Plot Messages!<br>00: This item I048/020 contains no Mode 4 information. Note: Mode 4 information may still be reported in I048/REF.<br>01: the target is declared friendly. Note: Value not reported in item I048/020 as it is reported in I048/REF.<br>10: the target is declared unknown target. Note: Value not reported in item I048/020 as it is reported in I048/REF.<br>11: the target has not answered on Mode |

|                                                |                     |                                             |                |                  |
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|  |  |                                                                                                                                                                                                             |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | 4 interrogations (not possible for coasted Tracks)<br><br>Note:<br>A target is marked as "unknown target" when the system had received Mode 4 replies but evaluator criteria for a friend is not fulfilled. |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|                                                |                     |                                      |                |                  |
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#### 4.6.8.3 Content of Data Item I048/REF, Reserved Expansion Field

This item is part of the following CAT 048 Messages of MSSR 2000 ID:

- Raw Plots,
- Processed Plots,
- Track Creation/Update Message.

This field uses the M5N coding format. The MD5 coding format is not used.

Note:

M4E field contains the output of the Mode 4 Evaluator output. "No Reply" is not an output of the Mode 4 Evaluator. The information "No Reply" is coded in I048/020.

Specifically, the M4E field is only coded when a Mode 4 reply has been received.

The M5 bit of the Mode 5 Summary field (Subfield #1) is configurable depending on the parameter AST\_TX\_CONTROL and has the following meanings:

If (AST\_TX\_CONTROL BIT 2 = 0)

0 when M5 is not interrogated or no statement

1 when M5 is interrogated with valid response, Friend.

Note:

*The M5 bit of the Mode 5 Summary field (Subfield #1) is set to 0 when the Mode 5 data does not fulfil the authentication conditions as defined in Addendum 2 /AIMS 17-1000 A2/, and the stale bit is set in the Time Synchronisation Status of the CAT 253 Extended System Status message - even when M5 is being interrogated at this azimuth.*

OR if

(AST\_TX\_CONTROL BIT 2 = 1)

0 when M5 is not interrogated at the target azimuth

1 when M5 is interrogated at the target azimuth

Note, this is only implemented for config 1x.05.02 or higher. For config 1x.05.01, the setting for AST\_TX\_CONTROL BIT 2 = 0 is fixed.

The ID bit of the Mode 5 Summary field (Subfield #1) is the output of the evaluator and set to

0 when the evaluator output is no friend

1 when the evaluator output is friend

The DA bit of the Mode 5 Summary field (Subfield #1) is the output of the evaluator and is set to

0 when the evaluator output is no friend

1 when the evaluator output is friend

Notes:

- Only authenticated data is used in the evaluator.
- In surveillance operation only authenticated data is output in the target report according to AIMS 17-1000.

#### 4.6.8.4 Content of Data Item I048/070, Mode 3/A Code in Octal Representation

This item is part of the following CAT 048 Messages of MSSR 2000 ID:

- Raw Plots,
- Processed Plots,
- Track Creation/Update Message.

In Raw Plot Messages the Mode 3/A code is derived from the reply and not calculated from older information. The L-Bit is always set to 0. If no Mode 3/A reply is available, this item is not present.

If Code-Improvement is activated (default) in Processed Plot Messages the Mode 3/A code is derived from the reply if available or calculated from the track. The L-Bit is set to 0 if the value is derived from the reply or is set to 1 if the value is calculated from the tracking information. In case of Mode S, Mode 3/A code will not be received every scan but must be reported with the Mode S report.

|                                                |            |                  |          |          |
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If Code-Improvement is disabled, the Mode 3/A code in Processed Plot Messages is handled as described for Raw Plot Messages.

In Track Messages the Mode 3/A code is derived from the reply if available or calculated from the track. The L-Bit is set to 0 if the value is derived from the reply or is set to 1 if the value is calculated from the tracking information. In case of Mode S, Mode 3/A code will not be received every scan but must be reported with the Mode S report.

It is possible to have different values for this item from the same target in Raw Plot Messages and in Processed Plot Messages and Track Messages (e.g. after a code change at the target).

#### 4.6.8.5 Content of Data Item I048/090 and I048/100

These items are both handled as described in /EC CAT 048/.

Data Item I048/090 will be present when the Mode C data can be decoded and the flight level can be calculated. In this case, Data Item I048/100 will not be present.

Data Item I048/100 will be present when the Mode C data cannot be decoded and the flight level cannot be calculated. In this case, Data Item I048/090 will not be present.

Data Item I048/090 and Data Item I048/100 will be both present if:

- the Mode C code can be decoded and
- at least one Bit of the Mode C reply is marked as "Low Confidence".

In this case the V-Bit in Data Item I048/090 will then be marked as "non validated".

For data item I048/090 no Mode C code smoothing is performed and negative values are expressed in two's complement form.

Note:

In Processed Plot and Track messages the V-Bit is always marked as "non validated" in Data Item I048/100.

In Raw Plot Messages, the Data Item I048/100 will be always present, regardless if decode able or not.

#### 4.6.8.6 Content of Data Item I048/220, Aircraft Address.

If the Aircraft Address is not part of the actual reply, this item will not be sent with the Plot Messages.

In the Track Messages, this information will be derived from the received information part of the Mode S reply, and from the internal track if no Mode S reply has been received.

This Data Item is handled as described in /EC CAT 048/

#### 4.6.8.7 Content of Data Item I048/240, Aircraft Identification.

The aircraft identification is derived from BDS 2,0.

This Data Item is handled as described in /EC CAT 048/.

The information of this item will be kept as part of the Track Messages as well as for processed plots for which code improvement has been applied.

For Raw Plots and Processed Plots which are not associated with an existing track, this item will only be filled if data has been extracted from the replies in the actual scan.

#### 4.6.8.8 Content of Data Item I048/250, Mode S MB Data.

This Data Item is handled as described in /EC CAT 048/.

|                                                |                     |                                             |                |                  |
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This Data Item will be transmitted only if Mode S MB Data has been extracted in the actual scan. The information will be transmitted only once.

**Note:**

The content of BDS Register 1,0 will be transmitted when the register has been interrogated in the current scan. Information extracted from this register will be also transmitted via the Data Item I048/230 where applicable.

#### 4.6.8.9 Content of Data Item I048/260, ACAS Resolution Advisory Report.

The content of BSD Register 3,0 will be transmitted if the register has been extracted in the actual scan.

The content will be transmitted one by one, no processing will be applied on this data.

It is within the responsibility of the receiver of these data to interpret the content.

#### 4.6.8.10 Content of Data Item I048/230, Communications/ACAS Capability and Flight Status

Properties which have not been extracted yet shall be set to zero.

The information of this item will be kept as part of the Track Messages as well as for Processed Plots for which code improvement has been applied.

For Raw Plots and Processed Plots which are not associated with an existing track, this item will only be filled if data has been extracted from the replies in the actual scan.

Re-extraction of BDS 1.0, 1.7, 2.0 is autonomously initiated after

- first acquisition of the target
- 18 seconds misses of the target
- after the flight through the Cone Of Silence

The repetitive re-extraction of BDS 17 can be configured to be every N seconds.

(  $0 \leq N \leq 3600$  )

N = 0 means no repetitive re-extraction (disabled)

The COM Field of these data item will be set as:

*Table 4-21. Coding of the COM Field in Data Item I048/230*

| I048/230<br>COM<br>(Communication<br>capability of the<br>transponder):<br>Bits 16-14 | DF 11 Reply<br>CA Field | BDS 1.0 in DF20 or DF21 reply |                       |                         | Transponder<br>Level |
|---------------------------------------------------------------------------------------|-------------------------|-------------------------------|-----------------------|-------------------------|----------------------|
|                                                                                       |                         | Bit 24                        | Bits<br>26,27 &<br>28 | Bits 29, 30,<br>31 & 32 |                      |
| 0                                                                                     | 0                       | n.a.                          |                       |                         | 1                    |
| 1                                                                                     | =1, 2 or 3              | n.a.                          |                       |                         | 2                    |
| 1                                                                                     | $\geq 4$                | n.a.                          | 0                     | 0                       | 2                    |
| 2                                                                                     | $\geq 4$                | n.a.                          | > 0                   | 0                       | 3                    |
| 3                                                                                     | $\geq 4$                | 0                             | > 0                   | > 0                     | 4                    |
| 4                                                                                     | $\geq 4$                | 1                             | > 0                   | > 0                     | 5                    |

The SI-Field of these data item will be set as defined in the following table "Operation with II Code and SI Code".

|                                                |                     |                                             |                |                  |
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The transponder which replies "with matching II Code" is indicated as non SI equipped transponder (SI = 1), independent of the content of BDS 1,0.

The transponder which replies "not SI Code capable" is indicated as non SI equipped transponder (SI = 1), independent of II- or SI-Code operating.

The transponder from which "BDS 1,0 is not extracted" is indicated as non SI equipped transponder (SI = 1).

Within the raw plot, the SI-bit is set to 1 as soon as one of the received all-call replies used to build the raw Plot contains the matching II Code, otherwise the SI-bit is not set to 1. The extraction or no extraction of the BDS1,0 has no impact on the setting of the SI-bit of the raw plot."

Within the processed plot, the SI-bit is set to 1 depending upon different criteria:

- one of the received all-call replies used to build the processed plot contain the „matching II Code“
- the non extraction of the BDS1,0 (the delay of the internal processing within the tracker has to be considered)
- the content of the extracted BDS1,0 indicate "SI Code not capable" (the processing result of the tracker will apply)
- the configuration of some parameters (e.g. plot completion)

Within the track, the SI-bit will be set to 1 differently depending upon the different phases of the track life cycle:

- For the first track based on all-calls, the SI bit will be a "wired or" of the SI bit set within the corresponding target reports used to initiate the track. If the SI-bit is set to 1, it will be further set to 1 in the next scans.
- For maintained tracks (based on roll-calls; all-calls might still be received) the SI-bit will be set to 1 if the BDS1,0 is not extracted, or if the extracted BDS1,0 contain the information "SI Code not capable" or if at least one all-call reply has been received, which contain the matching II Code.

The SI-bit is set to 0 again in case the BDS1,0 has been extracted with information „SI Code capable“ in the current scan and no all-call replies containing the „matching II Code“ has been received

**Table 4-22. Operation with II Code and SI Code**

| Operation with II Code resp. SI Code | BDS 1,0 not extracted | BDS 1,0 extracted |                     | All- Call with SI Code resp. with matching II Code |                       | SI Bit Data Item I048/230 |
|--------------------------------------|-----------------------|-------------------|---------------------|----------------------------------------------------|-----------------------|---------------------------|
|                                      |                       | SI Code Capable   | Not SI Code Capable | With II Code                                       | With matching II Code |                           |
| II Code                              | x                     | n/a               | n/a                 | n/a                                                | n/a                   | 1                         |
| II Code                              | n/a                   | x                 | n/a                 | n/a                                                | n/a                   | 0                         |
| II Code                              | n/a                   | n/a               | x                   | n/a                                                | n/a                   | 1                         |
| SI Code                              | x                     | n/a               | n/a                 | x                                                  | n/a                   | 1                         |
| SI Code                              | x                     | n/a               | n/a                 | n/a                                                | x                     | 1                         |
| SI Code                              | n/a                   | x                 | n/a                 | x                                                  | n/a                   | 0                         |
| SI Code                              | n/a                   | x                 | n/a                 | n/a                                                | x                     | 1                         |
| SI Code                              | n/a                   | n/a               | x                   | x                                                  | n/a                   | 1                         |
| SI Code                              | n/a                   | n/a               | x                   | n/a                                                | x                     | 1                         |

|                                                |                     |                                             |                |                  |
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#### 4.6.8.11 Content of Mode 1 Data Items I048/055 and I048/065

##### 4.6.8.11.1 I048/055, Mode 1 Code in Octal Representation

Output of Item 55 must be activated in system parameter AST\_TX\_CONTROL, see /Parameter/. If output of Item 55 is activated and an old, short Mode 1 Code is received (only pulses A1 to B2, max. 73 (oct)), this short Mode 1 Code is output in Item 55 and Item 55 is handled in the same way as described for I048/070. Otherwise (Item 55 not activated or extended Mode 1 code received), Item 55 is not used.

Note 1/2: The Mode 1 code is always transmitted in the REF, irrespective of system parameter AST\_TX\_CONTROL.

Note 2/2: Mode 1 code smoothing is supported. If Mode 1 code smoothing is performed and Item 55 is output, the L-Bit of Item 55 will be set accordingly.

##### 4.6.8.11.2 I048/065, Mode-1 Code Confidence Indicator

If Item 55 is output and one or more Mode 1 bits are of low confidence, this confidence information is output in Item 65 as described in the Asterix standard. Otherwise, Item 65 is not used.

#### 4.6.8.12 Content of Data Item I048/050, Mode 2 Code in Octal Representation

The handling is the same as described for I048/070.

Mode 2 code smoothing is supported and if performed the L-Bit will be correspondingly set.

#### 4.6.8.13 Content of Data Item I048/140, Time of Day

In a Target Message the Item I048/140 represents the time where the target has sent the reply. The value is calculated from the single replies building the plot corrected by the runtime.

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#### 4.6.8.14 Content of Data Item I048/170, Track Status

The fields of this data item are handled as described below:

*Table 4-23. Subfields in Data Item I048/170*

|     |                                                                                             |                                                                                                                                                     |
|-----|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| CNF | Confirmed vs. Tentative Tracks                                                              | 0 = Confirmed Track<br>1 = Tentative Track                                                                                                          |
| RAD | Type of Sensor(s) maintaining Track                                                         | Always = 10 (SSR/Mode S Track)                                                                                                                      |
| DOU | Signal level of confidence in plot to track association                                     | Always = 0                                                                                                                                          |
| MAH | Manoeuvre detection in Horizontal Sense                                                     | 0 = No horizontal man.sensed<br>1 = Horizontal man. sensed                                                                                          |
| CDM | Climbing / Descending Mode                                                                  | 00 = Maintaining<br>01 = Climbing<br>10 = Descending<br>11 = Unknown                                                                                |
| TRE | Signal for End_of_Track                                                                     | 0 = Track Creation/Update<br>1 = Track Termination                                                                                                  |
| GHO | Ghost vs. True track                                                                        | 0 = True target track.<br>1 = Ghost target track.                                                                                                   |
| SUP | Track maintained with track information from neighbouring Node B on the cluster, or network | 0 = no<br>1 = yes                                                                                                                                   |
| TCC | Type of plot coordinate transformation mechanism:                                           | Always = 1 (Slant range correction and a suitable projection in a 2D reference plane, tangential to the earth model at the Radar Site co-ordinates) |

|                                                |                     |                                             |                |                  |
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#### 4.6.8.15 Content of Data Item I048/030, Warning/Error Conditions

The Warning/Error Conditions in Data Item I048/030 give additional information about the target, handled by this report.

Usage of the Warning/Error Condition in CAT 048 Messages:

**Table 4-24. Usage of the Warning/Error Condition in CAT 048 Messages**

| Warning/Error Code | Description                                                                  | Usage with MSSR 2000 ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                  | Not defined; never used.                                                     | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1                  | Multipath Reply (Reflection)                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2                  | Reply due to sidelobe interrogation/reception                                | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3                  | Split plot                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4                  | Second time around reply                                                     | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5                  | Angle                                                                        | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6                  | Slow moving target correlated with road infrastructure (terrestrial vehicle) | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7                  | Fixed PSR plot                                                               | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8                  | Slow PSR target                                                              | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 9                  | Low quality PSR plot                                                         | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 10                 | Phantom SSR plot                                                             | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11                 | Non-Matching Mode-3/A Code                                                   | A target has different Mode 3/A codes resulting from Mode S and MkX interrogations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12                 | Mode C code / Mode S altitude code abnormal value compared to the track      | <p>Either in SSR or in Mode S, when the variation between the previous and current reported altitude exceeds a parameterised threshold. It is just a matter of history and has nothing to do with geometrical calculations.</p> <p>For processed plots, error code 12 is set when one of the following conditions holds true:</p> <ul style="list-style-type: none"> <li>- flight level &lt; -10</li> <li>- flight level &gt; 1267</li> <li>- reported altitude &gt; measured range (this does usually not happen)</li> <li>- difference between reported altitude and altitude saved in track is too large</li> </ul> |
| 13                 | Target in Clutter Area                                                       | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 14                 | Maximum Doppler Response in Zero Filter                                      | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 15                 | Transponder anomaly detected                                                 | Transponder anomaly for MS transponder replies with Mode A/C to Mode A/C-only all-call.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 16                 | Duplicated or Illegal Mode S Aircraft Address                                | This WE Condition regards only Mode S targets. The error condition shall be raised when an illegal Mode S Aircraft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                |                     |                                             |                |                  |
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|           |                                                |                                                                                                                                                                                                                                                                                                                                                  |
|-----------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                                | Address has been detected (either 24 ZEROES or 24 ONES) when within the coverage of the interrogator at least 2 Aircraft with the same Mode S addresses have been detected                                                                                                                                                                       |
| 17        | Mode S error correction applied                | This WE Condition regards only Mode S targets. The error condition shall be raised when an error correction has been applied by the interrogator on at least one of the replies received from the Mode S aircraft during the beam dwell.                                                                                                         |
| 18        | Undecodable Mode C code / Mode S altitude code | This WE Condition regards SSR targets ("undecodable Mode C Code") or Mode S targets ("undecodable Mode S altitude"). The error condition shall be raised for SSR targets or for Mode S targets when the received Mode C code does not correspond to an ICAO pressure altitude. It has to be noted that empty brackets are thrown away from RTCC. |
| 19 ... 30 | various description                            | Not used                                                                                                                                                                                                                                                                                                                                         |
| 31        | Duplicated Mode 5 Pair NO/PIN detected         | Indicates duplicate ID condition when multiple targets possess identical NO-PIN codes.                                                                                                                                                                                                                                                           |
| 32 ... 36 | various description                            | Not used                                                                                                                                                                                                                                                                                                                                         |
|           |                                                |                                                                                                                                                                                                                                                                                                                                                  |
|           |                                                |                                                                                                                                                                                                                                                                                                                                                  |

#### 4.6.8.16 Content of Data Item I048/130, Radar Plot Characteristic

Subfields #4 (Primary Plot Runlength), #5 (Amplitude of Primary Plot), #6 (Difference in Range between PSR and SSR plot) and #7 (Difference in Azimuth between PSR and SSR plot) will never be present in MSSR 2000 ID Target Messages because the MSSR 2000 ID does not have information from the primary radar.

##### 4.6.8.16.1 Subfield #1: SSR Plot Runlength (SRL)

This subfield will not be used with MSSR 2000 ID.

##### 4.6.8.16.2 Subfield #2: Number of Received Replies for (M)SSR (SRR)

This Subfield holds the number of Replies which are used for the calculation of this target.

##### 4.6.8.16.3 Subfield #3: Amplitude of (M)SSR Reply (SAM)

This Subfield holds the amplitude of the reply with the highest amplitude, used to calculate this report.

The value is calculated as the average of all pulses within the reply.

The LSB value = 1 dBm.

#### 4.6.8.17 Content of Data Item I048/042, Calculated Position in Cartesian Coordinates

This data item contains the calculated position of the CAT48 extended plot or track.

The reference point used for this calculation is set by the TRACK\_POS\_REFERENCE parameter, and can be either the Radar Head, or the SIRP. Refer /SysArc M23/ for the details on transformation from ITEM 40 (Range,Azimuth) to ITEM 42 (X,Y).

Note: If the target altitude is unknown, a target elevation of 0° above the reference plane will be assumed to calculate cartesian coordinates.

|                                                |            |                  |          |          |
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#### 4.6.8.18 Content of Data Item I048/040, Polar Position

This data item contains the measured polar position of the CAT48 raw plot, processed plot, extended plot or track.

In the case of Mode 5 Level 2 reports a polar position is not measured. A WGS84 position is delivered by the target. Because the Mode 5 Level 2 Report can be triggered by interrogations, the MSSR 2000 ID uses the Category 048 to report the information. In this case this item contains the position delivered by the target, but converted (by the MSSR 2000 ID) to a polar position relative to the position of the radar-head.

|                                                |                     |                                             |                |                  |
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#### 4.6.9 Interface for Directed Interrogation via SPF

**Note:**

The first version of the Directed Interrogation interface uses the SPF of CAT 048.

The second version of the Directed Interrogation interface uses the CAT 048M as described in /MSSR AST Ext. for DI/.

The first and the second version of the Directed Interrogation interface are outdated. For backwards compatibility, these interface versions are still available, but it is strongly recommended not to use the first or second Directed Interrogation interface version for newly designed systems.

The first interface version, Directed Interrogation via CAT048 SPF, is no longer described in this document. Please refer to previous document versions.

The current (third) version of the Directed Interrogation interface uses CAT 007 as described in /EC CAT 007/, and in the CAT 007 chapter of this document.

Only the CAT 007 Directed Interrogation interface is compliant to the ASTERIX standard.

It is strongly recommended to use only one version of the Directed Interrogation interface.

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## 4.7 Directed Interrogation Messages (CAT 007)

### 4.7.1 General

The usage of CAT 007 Messages is based on /EC CAT 007/.

Therefore, this chapter is readable only in collaboration with the referenced Document.

The standard UAP for CAT 007 is described in Table 4-4 and Table 4-5.

This chapter will describe:

- how the MSSR 2000 ID uses the CAT 007 Messages
- how the MSSR 2000 ID interprets some specifications of the CAT 007 messages.

Dual redundant system:

All transactions shall only be performed with the MSSR 2000 ID Master chain.

The Master chain will reply to a DI request with an appropriate DI (Acknowledge or Reject eventually).

Note:

The reason for a reject is included within the data item Warning Error conditions I007/030.

The Slave chain will reply to a DI request with an Interrogation Reject and ignore the DI request. It is in the responsibility of the client to ensure that the Slave chain does not receive any DI request.

All target information resulting from a directed interrogation shall be transmitted through CAT 048 by the Master chain and by the Slave chain.

The Slave chain will not transmit any CAT 007 message containing target information.

Target Output:

The defined target reports (Raw Plots, Processed Plots, Tracks) for the CAT 048 and the output enable filters (AST\_TX\_RAW\_PLOT, AST\_TX\_PLOT, AST\_TX\_TRACK) are also valid for the CAT 007 output at the dedicated port. In Comparison with CAT 048, the CAT 007 target reports contains some additional data items (I007/025, I007/400, I007/410).

### 4.7.2 Differences in the Usage of the Message Types

Summary of message types

| Type     | A<br>(pos.)                           | B<br>(wind.) | C<br>(track) | BDS | DI<br>Ack.                 | DI<br>reject | DI<br>finish | Target<br>Report | DI completed |
|----------|---------------------------------------|--------------|--------------|-----|----------------------------|--------------|--------------|------------------|--------------|
| I007/410 | 5                                     | 6            | 7            | 8   | 0                          | 1            | 2            | 4                | 3            |
|          | Uplink: Client to Sensor (DI request) |              |              |     | Downlink: Sensor to Client |              |              |                  |              |
| I007/010 | M                                     | M            | M            | M   | M                          | M            | M            | M                | M            |
| I007/025 | M                                     | M            | M            | M   | M                          | M            | M            | M                | M            |
| I007/410 | M                                     | M            | M            | M   | M                          | M            | M            | M                | M            |
| I007/140 | M                                     | M            | M            | M   | M                          | M            | M            | M                | M            |
| I007/400 | M                                     | M            | M            | M   | M                          | M            | M            | M                | M            |
| I007/020 | -                                     | -            | -            | -   | -                          | -            | -            | M                | -            |
| I007/040 | M                                     | -            | -            | -   | -                          | -            | -            | O                | -            |
| I007/070 | -                                     | -            | -            | -   | -                          | -            | -            | O                | -            |
| I007/090 | -                                     | -            | -            | -   | -                          | -            | -            | O                | -            |
| I007/130 | -                                     | -            | -            | -   | -                          | -            | -            | O                | -            |

|                                                |                     |                                             |                |                  |
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|          |       |       |       |   |   |      |   |       |       |
|----------|-------|-------|-------|---|---|------|---|-------|-------|
| I007/220 | O     | O     | O     | M | - | -    | - | O     | -     |
| I007/240 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/250 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/161 | -     | -     | M     | - | - | -    | - | O     | -     |
| I007/042 | M     | -     | -     | - | - | -    | - | O     | -     |
| I007/200 | O     | -     | -     | - | - | -    | - | O     | -     |
| I007/170 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/210 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/030 | -     | -     | -     | - | O | M(1) | O | O     | O     |
| I007/080 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/100 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/110 | -     | -     | -     | - | - | -    | - | - (2) | -     |
| I007/120 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/230 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/260 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/055 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/050 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/065 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/060 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/085 | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/415 | M(1)  | M(1)  | M(1)  | O | - | -    | - | -     | -     |
| I007/420 | -     | M     | -     | - | - | -    | - | -     | -     |
| I007/440 | - (2) | - (2) | - (2) | M | - | -    | - | -     | -     |
| I007/450 | -     | -     | -     | - | - | -    | M | -     | - (2) |
| I007/SPF | -     | -     | -     | - | - | -    | - | O     | -     |
| I007/REF | -     | -     | -     | - | - | -    | - | O     | -     |

M = Mandatory O = Optional - = not allowed

Difference to Standard:

- (1) Mandatory for MSSR 2000 ID (Optional in Standard)
- (2) not used for MSSR 2000 ID (Optional in Standard)

#### 4.7.3 Specific to Uplink Messages

##### Reaction of the MSSR 2000 ID to a DI request containing the same SIC/SAC code as the MSSR itself in the Data Source Identifier

The MSSR 2000 ID shall accept a DI request containing the same SIC/SAC code as the MSSR itself in the Data Source Identifier, because the SIC/SAC code applies to the overall sensor. Within this sensor, there can be several client entities, that need to send DI requests to the MSSR. It is of course the responsibility of these client entities to avoid ambiguity. Besides these client entities must be part of the sensor.

##### Directed Position Interrogation (A)

The directed interrogation window is configurable (DIRECTED\_AZIMUTH\_GATE, DIRECTED\_RANGE\_GATE). The default values are 5° and 5 NM.

##### Directed Interrogation Window (B)

|                                                |                     |                                             |                |                  |
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In case of moving platform, the window borders will be adjusted due to the movement if necessary, an exact transformation cannot be performed (due to the lack of the height) by the MSSR 2000 ID.

### Directed Track-Number Interrogation (C)

The track number of the MSSR 2000 ID is used for a directed interrogation. The external track interrogation via the external interface is similar to the processing of an external plot. However, MSSR 2000 ID uses the additional speed and heading information to determine a smaller window for the interrogation and the expected replies.

### Selective BDS-Request

The MSSR 2000 ID does not verify the contents of the BDS 1,7 for CAT 007. It is in the responsibility of the requesting client.

The system shall have the capability to send a DI Selective BDS-request containing a request for extraction of up to 2 BDS registers.

#### Note:

When 2 BDS registers are requested through a single DI (see I007/440), it is highly recommended to analyse the impact of such a request before sending it. It is therefore recommended to send such a request when particular conditions are fulfilled in order to avoid unnecessary Mode S transactions (such as verification of the availability of the requested BDS register, sufficient beam dwell time for the Mode S transaction, early sending of this request, automatic extraction of the BDS disabled within the configuration set of the interrogator).

### Limit for number of Directed Interrogation Requests

It is only possible to send one DI per aircraft per scan.

The total number of DI requests per scan shall not exceed 100. If this limit is exceeded, the MSSR 2000 ID shall send an indication in data item I007/030 (Warning/Error; Code 68).

### Timing Requirement of Directed Interrogation

The Time of Day to be used within this data item shall lie at maximum 2 scans in the past.

A directed **all-call** interrogation, received from the system at least 100 ms prior to the time where the beam of the antenna (boresight -  $\frac{1}{2}$  antenna beamwidth) reaches the given azimuth shall generate interrogations in the current scan.

Directed Interrogation Messages which are too late to be performed within the actual scan will be executed with the next scan at the given position.

A directed Mode S **BDS** Register request commanded to the system more than 180° (configurable SU\_TFC\_ANGLE) before the beam of the antenna (boresight -  $\frac{1}{2}$  antenna beamwidth) reaches the given azimuth (start in case of a window) shall generate interrogations in the current scan.

A directed Mode S interrogation commanded to the system less than the above mentioned angle may be discarded.

### Waiting for a DI complete before sending a new request for the same aircraft:

It is extremely important that the equipment having sent a DI request to the MSSR 2000 ID for a particular aircraft, waits for reception of the DI Complete message, before sending again in the next scan another DI Request message for the same aircraft.

Otherwise the MSSR 2000 ID will reply with an Ambiguous Acknowledge message and it will not consider this new DI Request!

|                                                |            |                  |          |          |
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#### 4.7.4 Specific to Downlink Messages

##### Transmission of information via CAT 007 and CAT 048

Communication messages (Type: DI Acknowledge, DI Finished , DI Rejected, DI Completed) are sent only to the requesting client through CAT 007.

After a successful directed interrogation the SW shall generate a CAT 007 Target Report and send it to those ports as configured by the target-port routing logic.

The target information resulting from a directed interrogation will be transmitted through CAT 048 to all connected clients.

##### Interrogation Acknowledge/Reject Message:

Upon receipt of an interrogation request the MSSR 2000 ID response immediately with a corresponding Acknowledge or Reject Message. The Item I007/030 shall be present in case of Reject.

##### Directed Position Interrogation (A)

Outside the configured window centred around the position within the corresponding sector (see figure 4-6), no target reports will be transmitted either in ASTERIX CAT 007 or in ASTERIX CAT 048 for the directed interrogation.

##### Directed Interrogation Window (B)

Outside the directed interrogation window within the corresponding sector (see figure 4-6), no target reports will be transmitted either in ASTERIX CAT 007 or in ASTERIX CAT 048 for the directed interrogation.

With this function, it is not possible to perform 360° directed interrogation. Indeed, a 360° directed interrogation is not a directed interrogation anymore. Hence, the maximum sector that can be used is defined by: 360°-Beamwidth (Mode S) of the antenna.

##### Directed Track-Number Interrogation (C)

Outside the interrogation window as forecasted by the internal tracker within the corresponding sector (see figure 4-6), no target reports will be transmitted either in ASTERIX CAT 007 or in ASTERIX CAT 048 for the directed interrogation.

##### Selective BDS-Request

Only the target report related to this Mode S aircraft as a result of this BDS-request is transmitted in ASTERIX CAT 007. In ASTERIX CAT 048, all target reports for all aircraft within the directed sector are transmitted including all supplementary information for this Mode S aircraft resulting from the BDS request.

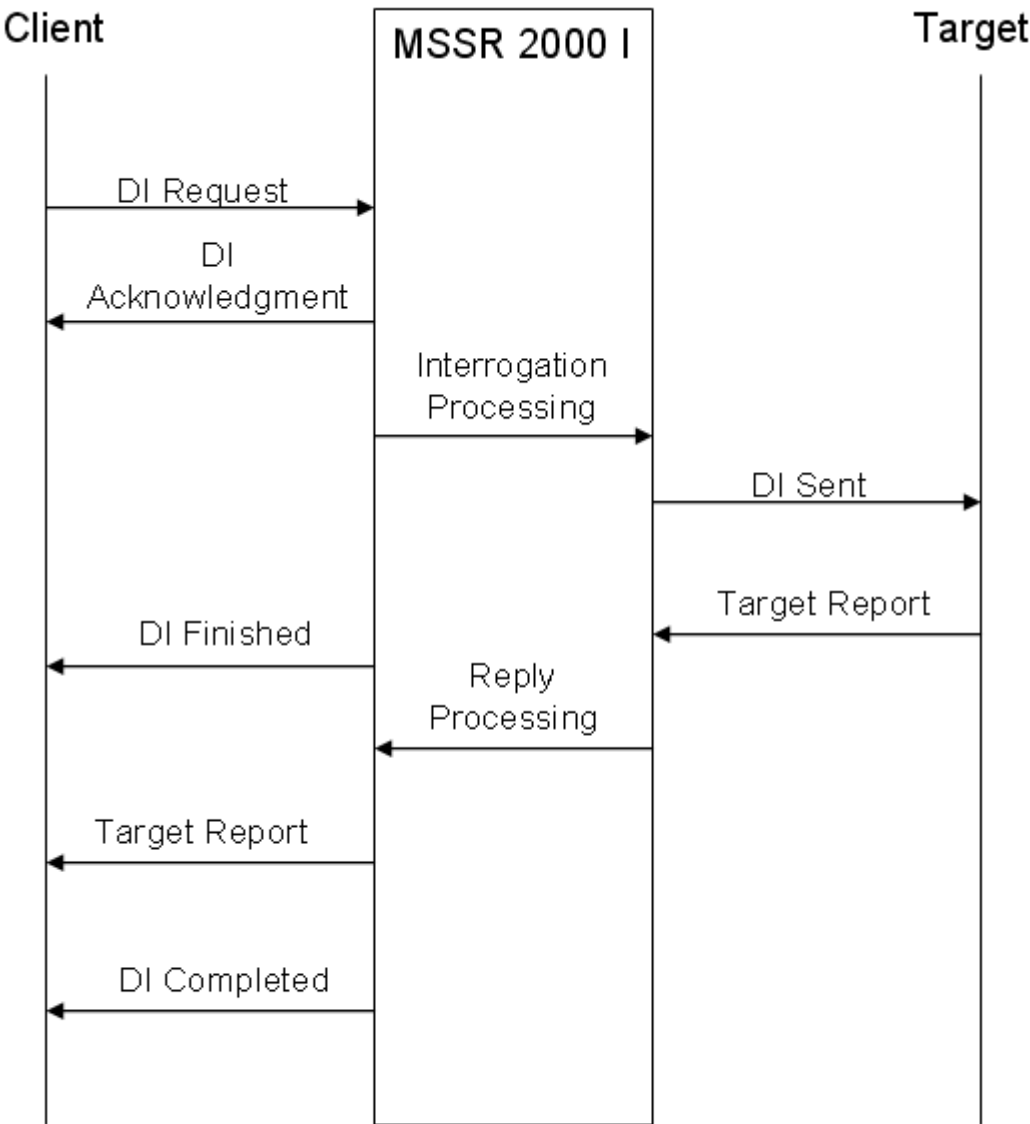
|                                                |            |                  |          |          |
|------------------------------------------------|------------|------------------|----------|----------|
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4.7.5 Communication process

4.7.5.1 Successful DI

Figure "Successful DI" describes the communication messages exchanged between the Client, the MSSR 2000 ID and the selected Target during a successful Directed Interrogation. The Interrogation Finished message will be generated by the MSSR 2000 ID as soon as the target interrogations have been completely transmitted into the air. It shall be noted, that depending on the geometry of a directed interrogation, i.e. the azimuthal extent of an expectation window, the Interrogation Finished message may arrive after the first target reports related to this interrogation. Furthermore, target reports may appear after the Interrogation Finished.

Figure 4-3. Successful DI



|                                                |                     |                                      |                |                  |
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4.7.5.2 Rejected DI

A "Reject-Message" can arise in the following cases:

- Wrong information requested in the DI (for example: inexistent MIP): Then a "Reject-Message" arises before the DI request is processed by the MSSR 2000 ID, as illustrated in Figure "DI failure: wrong information requested".
- DI requested for a target in a silent sector: A "Reject-Message" arises instead of a "Finished-Message" after the DI is processed by the MSSR 2000 ID, when the interrogation cannot be sent in the Silent Sector, as illustrated in Figure "DI failure: Target in Silent Sector".

Figure 4-4. DI failure: wrong information requested

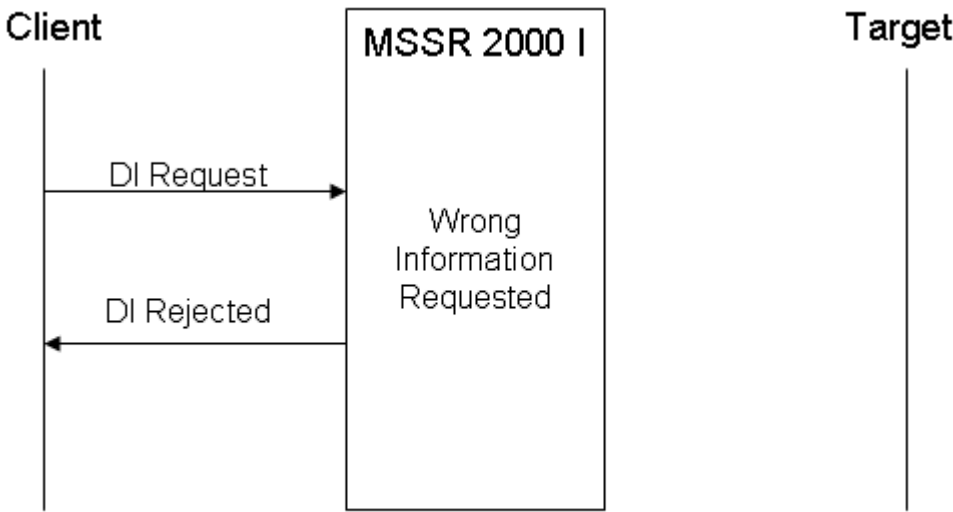
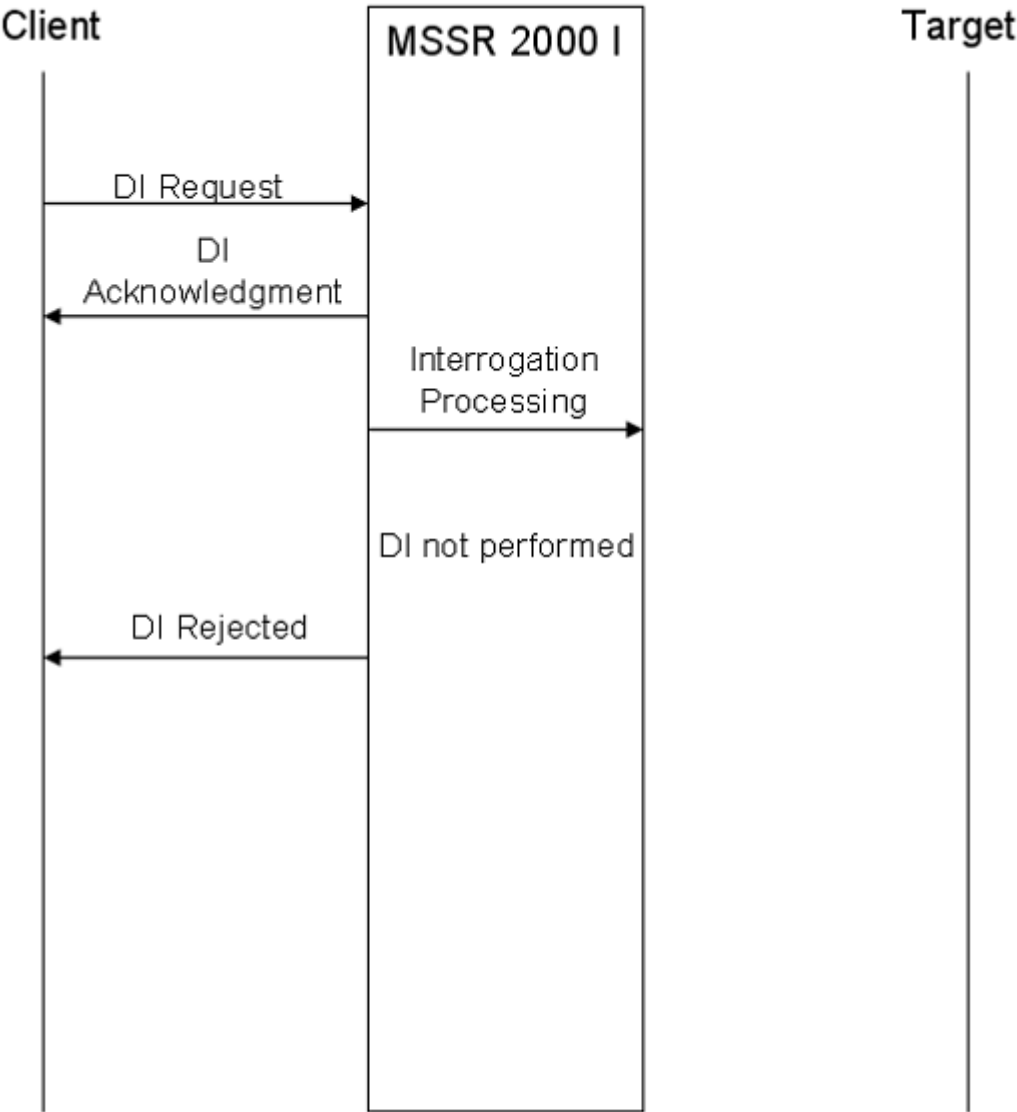


Figure 4-5. DI failure: Target in Silent Sector

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|                                                |                     |                                      |                |                  |
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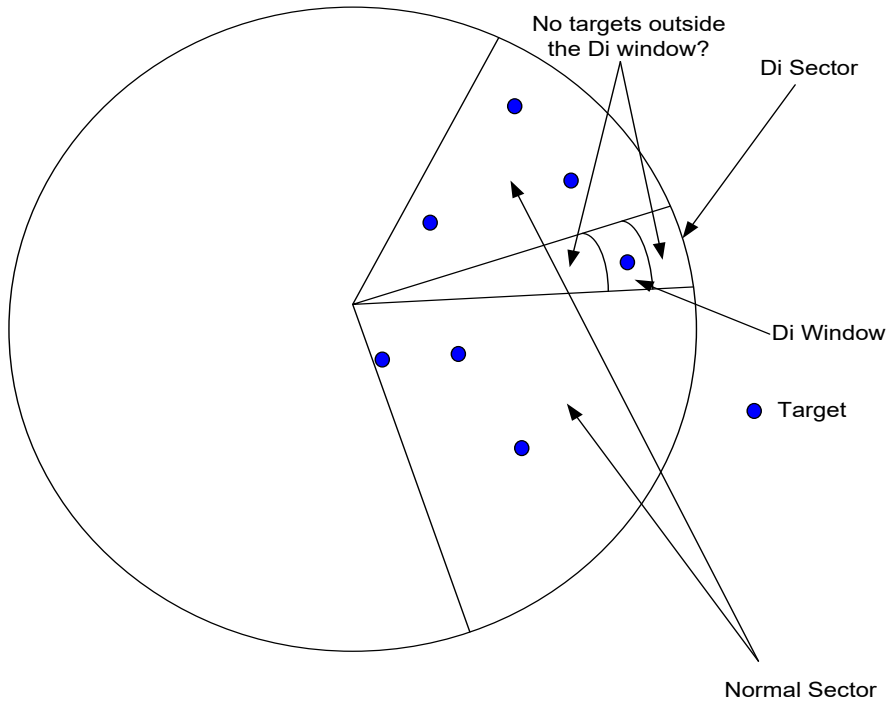


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|                                                |            |                  |          |          |
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#### 4.7.6 Definition

Figure 4-6. Directed Interrogation window and sector



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|                                                |                     |                                             |                |                  |
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#### 4.7.7 Specific Content of the Data Items

##### 4.7.7.1 Content of Data Item I007/020, Target Report Descriptor

This item is part of the following CAT 007 Messages of MSSR 2000 ID:

- Raw Plots,
- Processed Plots,
- Track Creation/Update Message.

When Mode 4 information has been received, the IFF bits are set to 0. The output of the Mode 4 Evaluator is reported in I007/REF.

When Mode 4 has been interrogated but no M4 reply has been received, the IFF bits are coded as follows:

11 ... no reply

##### 4.7.7.2 Content of Data Item I007/REF, Reserved Expansion Field

This item is part of the following CAT 007 Messages of MSSR 2000 ID:

- Raw Plots,
- Processed Plots,
- Track Creation/Update Message.

This field uses the M5N coding format. The MD5 coding format is not used.

Note:

M4E field contains the output of the Mode 4 Evaluator output. "No Reply" is not an output of the Mode 4 Evaluator. The information "No Reply" is coded in I048/020.

Specifically, the M4E field is only coded when a Mode 4 reply has been received.

The M5 bit of the Mode 5 Summary field (Subfield #1) is configurable depending on the parameter AST\_TX\_CONTROL and has the following meanings:

If (AST\_TX\_CONTROL BIT 2 = 0)

1 when M5 is interrogated at the target azimuth

0 when the Mode 5 data does not fulfil the authentication conditions as defined in Addendum

2 /AIMS 17-1000 A2/, and the stale bit is set in the Time Synchronisation Status of the CAT

253 Extended System Status message - even when M5 is being interrogated at this azimuth.

OR if

(AST\_TX\_CONTROL BIT 2 = 1)

0 when M5 is not interrogated at the target azimuth

1 when M5 is interrogated at the target azimuth

Note, this is only implemented for config 1x.05.02 or higher. For config 1x.05.01, the setting for AST\_TX\_CONTROL BIT 2 = 0 is fixed.

##### 4.7.7.3 Data Item I007/400, Directed Interrogation Request Number

Bit 16 (PRI) Priority of Command:

Not used for MSSR 2000 ID

Directed interrogation has priority over sector interrogation.

Bit 15/1 Request Number: used as defined in /EC CAT 007/.

##### 4.7.7.4 Data Item I007/415, Required Interrogation Modes

Bit 3 (RIM) Subfield #1: Required Interrogation Modes:

Not used for MSSR 2000 ID

Subfield #1 is ignored if received

|                                                |            |                  |          |          |
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#### 4.7.7.5 Data Item I007/440, Directed Interrogation BDS Register Request

Bit 8/1 (REP) Repetition Factor:

The REP field can take the values 1 or 2 for MSSR 2000 ID.

Therefore it is possible to request the extraction of up to 2 BDS registers within one DI, but precautions shall be taken.

#### 4.7.7.6 Data Item I007/450, Directed Interrogation Result

Bit 3 (SMS) Subfield #6:

Selective Mode S Interrogations:

Not used for MSSR 2000 I

#### 4.7.7.7 Content of Mode 1 Data Items I007/055 and I007/065

Category 007 Data Items 055 and 065 are output under the same conditions as Category 048 Data Items 055 and 065. Please see the description of I048/055 and I048/065 in this document.

#### 4.7.8 I007 Special Purpose Field

##### 4.7.8.1 I007/SPF Downlink Message

The I007/SPF in the downlink message (target report) is used according to the I048/SPF.

##### 4.7.8.2 I007/SPF Uplink Message

When the system is configured to operate with Steer Function 1 the system expects a Directed Position Interrogation (A) that includes this item I007/SPF in the uplink message.

Notes:

- The DI request Type A plus this item I007/SPF is called DI Type A+.
- A DI request Type A+ gets rejected when Steer Function 1 is not configured.
- All DI requests except Type A+ get rejected when Steer Function 1 is configured.

The special purpose field always represents an explicit length field with one octet containing the overall length of the item. This allows standard applications not recognising the special purpose item to simply ignore its contents and to carry on processing data with the next record or block.

The following structure shows the octet sequence for the I048 Special Purpose Item of the MSSR 2000 ID.

##### 4.7.8.2.1 Data Item I007/SPF, Special Purpose Length Indicator

**Definition:** MSSR 2000 ID Special Purpose Item:

**Format:** Special Purpose field (explicit length)

Structure of Length Indicator:

|             |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|
| Octet No. 1 |   |   |   |   |   |   |   |
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| LEN         |   |   |   |   |   |   |   |

Bit 8/1LEN Length of SPF in octets, including the Length Indicator itself.

= 4, BPP = 1

= 6, BPP and BSP = 1

##### 4.7.8.2.2 Data Item I007/SPF, Special Purpose Data Field

**Definition:** CAT 007 Special Purpose Data Field (Uplink Message).

**Format:** Compound Data Item, comprising a primary subfield, followed by defined subfields.

##### Structure of Primary Subfield

|             |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|
| Octet No. 1 |   |   |   |   |   |   |   |
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

|                                                |                     |                                             |                |                  |
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|     |     |   |   |   |   |   |    |
|-----|-----|---|---|---|---|---|----|
| BPP | BSP | 0 | 0 | 0 | 0 | 0 | FX |
|-----|-----|---|---|---|---|---|----|

Bit 8 octet 1, BPP

= 0 Absence of Subfield #1

= 1 Subfield #1 is used to provide the BPP to the system

Bit 7 octet 1, BSP

= 0 Absence of Subfield #2

= 1 Subfield #2 is used to provide the BSP to the system

Bit 6 to Bit 2 octet 1 is not used

Bit 1 octet 1, FX = 0 End of Primary Subfield

= 1 Extension of Primary Subfield (future use)

#### Structure of Subfield # 1 (BPP):

| Octet No. 1 |   |   |   |   |   |   |   | Octet No. 2 |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|-------------|---|---|---|---|---|---|---|
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| START BPP   |   |   |   |   |   |   |   | STOP BPP    |   |   |   |   |   |   |   |

Bit 8/1, Octet 1 START\_BPP Unsigned Integer (0-255), Number of the start BPP of a range of BPP's the antenna steers

Bit 8/1, Octet 2 STOP\_BPP Unsigned Integer (0-255), Number of the stop BPP of a range of BPP's the antenna steers

The Subfield #1 for Beam Pointing Positions is only required when the system is configured for Steer Function 1

#### Notes:

- Subfield #1 has the capability to define start and stop BPP. But in case of Steer Function 1 both values must have the same value.
- BPP = 0 is the first BPP. In normal case this is the most negative steer position that is possible.
- Subfield #1 must be send when the value is zero.

#### Structure of Subfield # 2 (BSP):

| Octet No. 1 |    |    |    |    |    |    |   | Octet No. 2 |   |   |   |   |   |   |   |
|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| BSP         |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |

Bit 16/1 BSP LSB =  $360^\circ/2^{16} = 0.0055^\circ$

#### Note:

- Subfield #2 is the Bore Site Position (BSP) or mechanical orientation of the antenna system to north.
- Subfield #2 must be send when the value is zero.

|                                                |                     |                                             |                |                  |
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## 4.8 Surveillance Co-ordination Messages (CAT 017)

### 4.8.1 General

Surveillance Co-ordination Messages are generated from a radar station being part of a SCN (Surveillance Co-ordination Network) and sent to one or more radar stations inside this SCN.

Surveillance Co-ordination Messages are transferred via Data Category 017. In case of SCF twelve types of Surveillance Co-ordination Messages are supported and sent by the SCF. For detailed information about specific CAT 017 items refer to /EC CAT 017/.

Dual redundant system:

All transactions of a node shall only be performed with the MSSR 2000 ID Master chain.

In case of a chain switch, the connection to the Slave is closed and the connection to the new Master is established automatically.

### 4.8.2 Network Information Message

The Network Information Message indicates all reachable radar stations inside the SCN.

Processing delay:

This message is processed repetitive every 0.1 s to 2.0 s.

Output in state:

TX

The Message is composed of the following data items:

*Table 4-25. Network Information Message*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 000 fixed                  |
| I017/350  | Cluster Station / Node List | n.a.         | ---                        |

|                                                |                     |                                             |                |                  |
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### 4.8.3 Track Data Message

The Track Data Message is used to announce a Mode S track entering the area of responsibility of a neighbouring radar station or to support it with measured track data if available.

Processing delay:

This message is processed immediately after receiving a Track Update Message (indicating to the next radar station that is responsible for a maintained Mode S track) or after receiving a Track Request Message.

Output in state:

TX

The Message is composed of the following data items:

*Table 4-26. Track Data Message*

| Data Item | Description                             | System Units                   | Information                                                                                                              |
|-----------|-----------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| I017/010  | Data Source Identifier                  | n.a.                           | From system configuration.                                                                                               |
| I017/012  | Data Destination Identifier             | n.a.                           | From system configuration.                                                                                               |
| I017/000  | Message Type                            | n.a.                           | 010 fixed.                                                                                                               |
| I017/220  | Aircraft Address                        | n.a.                           | 24 bit Mode S address                                                                                                    |
| I017/221  | Duplicate address reference number      | n.a.                           | if needed                                                                                                                |
| I017/140  | Time of day                             | Second                         | LSB = 1/128s                                                                                                             |
| I017/045  | Aircraft position in WGS84 co-ordinates | Latitude & Longitude in Degree | Latitude Range: -90° .. +90°<br>Longitude Range: -180° .. +180°<br>LSB = $(180 / 2^{23})^\circ = 2.145767 \cdot 10^{-5}$ |
| I017/070  | Mode 3/A code in octal representation   | n.a.                           | ---                                                                                                                      |
| BI017/050 | Flight level in binary representation   | Feet                           | LSB = 1/4 FL = 25ft                                                                                                      |
| I017/200  | Track velocity in polar co-ordinates    | Speed: NM/s<br>Heading: deg    | Speed: $1 / (2^{14})$ NM/s = 0.22 kt<br>Heading: $360^\circ / (2^{16}) = 0.0055^\circ$                                   |
| I017/230  | Transponder capability                  | n.a.                           | ---                                                                                                                      |
| I017/240  | Track status                            | n.a.                           | CST/FLT                                                                                                                  |

Note:

Data Item I017/230 Transponder capability

The CA mapping between I048/230 and I017/230 is performed as follows:

CAT048\_230.CA = 0 -> ITEM\_017\_230.CA = 0  
 CAT048\_230.CA = 1 -> ITEM\_017\_230.CA = 6  
 CAT048\_230.CA = 2 -> ITEM\_017\_230.CA = 6  
 CAT048\_230.CA = 3 -> ITEM\_017\_230.CA = 6  
 CAT048\_230.CA = 4 -> ITEM\_017\_230.CA = 6  
 CAT048\_230.CA = 5 -> ITEM\_017\_230.CA = 6  
 CAT048\_230.CA = 6 -> ITEM\_017\_230.CA = 6  
 CAT048\_230.CA = 7 -> ITEM\_017\_230.CA = 6

Besides I017/230 bit 5 (spare) will be used to indicate the SI capability:

Bit 5=0 indicates that the aircraft is SI capable

Bit 5=1 indicates that the aircraft is not SI capable

Notes:

|                                                |                     |                                             |                |                  |
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1) Data Item I017/240 Track Status Contain the information CST (Coasted Track) and FLT (Flight Level Tracking)

2) If the CST bit and the FLT bit are "0", the item not be send.

#### 4.8.4 Track Data Request Message

The Track Data Request Message indicates a Mode S track received as "No detection".

Processing delay:

This message is processed immediately after receiving a Track Update Message marked as "No Detection".

Output in state:

TX

The Message is composed of the following data items:

*Table 4-27. Track Data Request Message*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 020 fixed                  |
| I017/220  | Aircraft Address            | n.a.         | 24 bit Mode S address      |

#### 4.8.5 Cancellation of Track Data Message

The Cancellation of Track Data Message indicates that a radar station has (re)acquired a track received from another radar station.

Processing delay:

This message is processed immediately after receiving a track update not marked as "No Detection".

Output in state:

TX

The Message is composed of the following data items:

*Table 4-28. Cancellation of Track Data Message*

| Data Item | Description                        | System Units | Information                |
|-----------|------------------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier             | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier        | n.a.         | From system configuration. |
| I017/000  | Message Type                       | n.a.         | 022 fixed                  |
| I017/220  | Aircraft Address                   | n.a.         | 24 bit Mode S address      |
| I017/221  | Duplicate address reference number | n.a.         | if needed                  |

|                                                |                     |                                             |                |                  |
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#### 4.8.6 Track Data Stop Message

The Track Data Stop Message indicates a Mode S track dropped by the sending radar station.

Processing delay:

This message is processed immediately after dropped a Mode S track.

Output in state:

TX

The Message is composed of the following data items:

*Table 4-29. Track Data Stop Message*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 021 fixed                  |
| I017/220  | Aircraft Address            | n.a.         | 24 bit Mode S address      |

#### 4.8.7 Track Data Stop Acknowledgement Message

The Track Data Stop Acknowledgement Message is the reply on the Track Data Stop Message.

Processing delay:

This message is processed immediately after receiving a Track Data Stop Message.

Output in state:

TX

The Message is composed of the following data items:

*Table 4-30. Track Data Stop Acknowledgement*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 023 fixed                  |
| I017/220  | Aircraft Address            | n.a.         | 24 bit Mode S address      |

|                                                |                     |                                             |                |                  |
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#### 4.8.8 NNCO (initial or intermediate message segment)

The "New Node / Change Over initial or intermediate message segment" indicates all Mode S tracks in an area of common responsibility with neighbouring radar stations.

Processing delay:

This message is processed immediately after SCF changed to "Distributed State".

Output in state:

TX

The Message is composed of the following data items:

*Table 4-31. New Node / Change Over initial or intermediate Message segment*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 030 fixed                  |
| I017/210  | Mode S Address List         | n.a.         | ---                        |

#### 4.8.9 NNCO (final or only message segment)

The "New Node / Change Over final or only message segment" indicates all Mode S tracks in an area of common responsibility with neighbouring radar stations.

Processing delay:

This message is processed immediately after SCF changed to "Distributed State"..

Output in state:

TX

The Message is composed of the following data items:

*Table 4-32. New Node / Change Over final or only Message segment*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 031 fixed                  |
| I017/210  | Mode S Address List         | n.a.         | ---                        |

|                                                |                     |                                             |                |                  |
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#### 4.8.10 NNCO Reply (initial or intermediate message segment)

The "New Node / Change Over initial or intermediate message segment reply" indicates all Mode S tracks in an area of common responsibility with neighbouring radar stations.

Processing delay:

This message is processed immediately after receiving a New Node / Change Over initial or intermediate Message segment reply.

Output in state:

TX

The Message is composed of the following data items:

*Table 4-33. New Node / Change Over initial or intermediate Message segment reply*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 032 fixed.                 |
| I017/210  | Mode S Address List         | n.a.         | ---                        |

#### 4.8.11 NNCO Reply (final or only message segment)

The "New Node / Change Over final or only message segment reply" indicates all Mode S tracks in an area of common responsibility with neighbouring radar stations.

Processing delay:

This message is processed immediately after receiving a New Node / Change Over final or only Message segment reply.

Output in state:

TX

The Message is composed of the following data items:

*Table 4-34. New Node / Change Over final or only Message segment reply*

| Data Item | Description                 | System Units | Information                |
|-----------|-----------------------------|--------------|----------------------------|
| I017/010  | Data Source Identifier      | n.a.         | From system configuration. |
| I017/012  | Data Destination Identifier | n.a.         | From system configuration. |
| I017/000  | Message Type                | n.a.         | 033 fixed.                 |
| I017/210  | Mode S Address List         | n.a.         | ---                        |

|                                                |                     |                                             |                |                  |
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## 4.9 Mode S Datalink Function Messages (CAT 018)

### 4.9.1 General

The usage of CAT 018 Messages is based on the documents /EC CAT 018/ and /EC GDLP ICD/ (e.g. description for connection timer). Therefore, this chapter is readable only in collaboration with the referenced documents!

This chapter will describe:

- how the MSSR 2000 ID uses the CAT 018 Messages
- how the MSSR 2000 ID interprets some specifications of the CAT 018 messages.

The standard UAP for CAT 018 is described in Table "Standard UAP for Category 018".

The DLF function is part of the MSSR 2000 ID Interrogator Unit.

Extended Length Messages are not implemented with MSSR 2000 ID. Therefore no message which requires an extended length message will be accepted by the MSSR 2000 ID.

The MSSR 2000 ID is completely passive, regarding the building-up of this connection (Establishment or Disabling).

Dual redundant system:

All transactions shall only be performed with the MSSR 2000 ID Master chain. If a CMS is trying to establish a connection with the slave chain (Associate\_req message), the Slave will send a message Associate\_Resp with data item I018/001 with CAUSE=1h (Rejected) and DIAG=0h (No diagnostic available).

In case of a switch over from Master to Slave, pending transactions are discharged. It is in the responsibility of the CMS to repeat the transactions for the new Master.

#### Constraints to be respected:

Attempt to establish a connection which is already running.

If a CMS is trying to establish a connection (using a SIC/SAC code), which is already running (with the same SIC/SAC code), the Master removes existing aircraft data associated with that link, resets internal timers/counters and replies with an Associate\_resp message. The D\_result field of the Associate\_resp is set to (CAUSE=3h(finished),DIAG=0h(No diagnostic available)).

Time to send a GICB extraction.

To allow proper GICB extraction, the GICB Extraction Message shall be received from the Master at least half a scan prior to the time where the beam of the antenna (boresight - ½ antenna beamwidth) reaches the given azimuth. GICB Extraction messages which are too late to be performed within the actual scan will be executed with the next scan at the given position.

Maximum number of GICB extraction per aircraft per scan.

To allow proper GICB extraction, a maximum number of 2 GICB extractions shall be performed per aircraft per scan.

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#### 4.9.2 Category 018 Message Types

Supported transactions:

- GICB extraction
- Broadcast (uplink and Downlink)
- Uplink Comm-A and linked uplink Comm-A

For CAT 018 a number of message types is defined. For each message type, a direction of the dataflow between the MSSR 2000 ID and the GDLP or the Local User is defined.

For some message types, the direction as described in /EC GDLP ICD/ is different to the implementation at the MSSR 2000 ID.

CAT 018 Message Types:

*Table 4-35. CAT 018 Message Types*

| Message                 | Message Type | Message Purpose           | Original direction<br>GDLP <-> DLF | Implemented<br>direction<br>GDLP <-> MSSR |
|-------------------------|--------------|---------------------------|------------------------------------|-------------------------------------------|
| Associate_req           | 00h          | Connection establishment  | <-->                               | -->                                       |
| Associate_resp          | 01h          | Connection establishment  | <->                                | <--                                       |
| Release_req             | 02h          | Connection disabling      | <->                                | -->                                       |
| Release_resp            | 03h          | Connection disabling      | <->                                | <--                                       |
| Abort_req               | 04h          | Connection disabling      | <->                                | -->                                       |
| Keep_alive              | 05h          | Keep alive procedure      | <->                                | <->                                       |
| Aircraft_report         | 10h          | Routing information       | <--                                | <--                                       |
| Aircraft_command        | 11h          | Routing information       | -->                                | -->                                       |
| Il_code_change          | 12h          | Routing information       | <--                                | <--                                       |
| Uplink_packet           | 20h          | Manage uplink data flow   | -->                                | -->                                       |
| Cancel_uplink_packet    | 21h          | Manage uplink data flow   | -->                                | -->                                       |
| Uplink_packet_ack       | 22h          | Manage uplink data flow   | <--                                | <--                                       |
| Downlink_packet         | 23h          | Manage downlink data flow | <--                                | <--                                       |
| Data_XON                | 26h          | Flow control              | <--                                | Not used for<br>MSSR 2000 I               |
| Data_XOFF               | 27h          | Flow control              | <--                                | Not used for<br>MSSR 2000 I               |
| Uplink_broadcast        | 30h          | Manage broadcast service  | -->                                | -->                                       |
| Cancel_uplink_broadcast | 31h          | Manage broadcast service  | -->                                | -->                                       |
| Uplink_broadcast_ack    | 32h          | Manage broadcast service  | <--                                | <--                                       |
| Downlink_broadcast      | 34h          | Manage broadcast service  | <--                                | <--                                       |
| GICB_extraction         | 40h          | Manage GICB service       | -->                                | -->                                       |

|                                                |                     |                                             |                |                  |
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|                        |     |                     |     |     |
|------------------------|-----|---------------------|-----|-----|
| Cancel_GICB_extraction | 41h | Manage GICB service | --> | --> |
| GICB_extraction_ack    | 42h | Manage GICB service | <-- | <-- |
| GICB_response          | 43h | Manage GICB service | <-- | <-- |

#### 4.9.3 Differences in the Usage of the Messages Types

See /EC GDLP ICD/, Appendix B - Messages

##### 4.9.3.1 Associate\_req

The MSSR 2000 ID will never send an Associate\_req Message.

The Associate\_req Message will be sent by the GDLP / Local User.

##### 4.9.3.2 Associate\_resp

The MSSR 2000 ID will never accept an Associate\_resp Message.

##### 4.9.3.3 Abort\_req

The MSSR 2000 ID will never send an Abort\_req Message.

The Abort\_req Message will be sent by the GDLP / Local User.

##### 4.9.3.4 Release\_req

The MSSR 2000 ID will never send an Release\_req Message.

The Release\_req Message will be sent by the GDLP / Local User.

##### 4.9.3.5 Release\_resp

The MSSR 2000 ID will never accept an Release\_resp Message.

##### 4.9.3.6 Downlink\_packet

Used only for pilot initiated Comm-B Messages.

##### 4.9.3.7 XON / XOFF Protocol

The XON / XOFF protocol is not supported by MSSR 2000 I

|                                                |                     |                                             |                |                  |
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#### 4.9.4 Specific Content of the Data Items

##### 4.9.4.1 Data Item I018/019

Extended Length Messages are not supported by the MSSR 2000 ID. Therefore the parameter LENGTH in this message is limited to 28 octets (4 segments of a linked message).

##### 4.9.4.2 Data item I018/025 GICB Number

The GDLP shall uniquely assign a GICB number to each GICB extraction with this data item. Within the Cancel\_GICB\_extraction and within the GICB\_extraction Ack, the data item I018/025 GICB Number might be or not present. If the data item I018/025 is present within the Cancel\_GICB\_extraction message, it means that the cancellation refers to a specific GICB extraction request. If it is absent all the GICB extractions for the aircraft identified within I018/005 are concerned by the cancellation.

##### 4.9.4.3 Data Item I018/028 GICB Periodicity

In order to extract once a BDS register, the data item I018/028 GICB Periodicity shall be absent from the GICB\_extraction message.

In order to extract each scan a BDS register, the data item I018/028 GICB Periodicity shall be present and contain a number of seconds smaller than the antenna turn time. In this case the lifetime of the GICB extraction will be until:

- the connection between GDLP and MSSR is terminated
- the aircraft is leaving the datalink coverage
- the GICB extraction is cancelled by the GDLP using the Cancel\_GICB\_extraction message

There are three ways of cancelling GICB extractions as indicated within the table "Different types of GICB extraction cancellation" below:

*Table 4-36. Different types of GICB extraction cancellation*

| Type of GICB extraction cancellation                                         | GDLP -> MSSR 2000 ID Master Chain                                                                                                                                             | MSSR 2000 ID Master Chain -> GDLP                                                                                                                                                                                   |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cancellation of one GICB extraction for a particular Mode S aircraft</b>  | GDLP sends to MSSR a message<br>Cancel_GICB_extraction containing the data item I018/025.<br>Data item I018/005 contains the Mode S address of the concerned aircraft         | MSSR replies normally to the GDLP with a GICB_extraction_ack with CAUSE=Cancelled.<br>The Cause=Rejected could be sent by the MSSR if for example the GICB number is not valid or there is no more pending request. |
| <b>Cancellation of all GICB extractions for a particular Mode S aircraft</b> | GDLP sends to MSSR a message<br>Cancel_GICB_extraction but the data item I018/025 is not present.<br>Data item I018/005 contains the Mode S address of the concerned aircraft | MSSR replies normally to the GDLP with a GICB_extraction_ack with CAUSE=Cancelled<br>The Cause=Rejected could be sent by the MSSR if for example the GICB number is not valid or there is no more pending request.  |
| <b>Cancellation of all GICB extractions for all Mode S aircraft</b>          | GDLP sends to MSSR a message<br>Cancel_GICB_extraction but the data item I018/025 is not present.                                                                             | MSSR replies normally to the GDLP with a GICB_extraction_ack with CAUSE=Cancelled.<br>The Cause=Rejected could be                                                                                                   |

|                                                |                     |                                             |                |                  |
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|  |                                                                                           |                                                                                                   |
|--|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  | Data item I018/005 contains FFFFFFFF(Hex), meaning that all Mode S aircraft are concerned | sent by the MSSR if for example the GICB number is not valid or there is no more pending request. |
|--|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|

#### 4.9.4.4 Data item I018/030 GICB Properties

The following values shall be set by the GDLP within the GICB\_extraction message:

- Priority = 4 (This value ensures that other more critical information like ACAS is extracted with a higher priority) )
- PC (Periodicity Constraint) = 0
- AU (Asynchronous Update) = 0
- NE (Non Extraction) = 0
- RD (Reply destination) = 2

|                                                |                     |                                             |                |                  |
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## 4.10 ADS-B Messages (CAT 021)

### 4.10.1 General

The usage of CAT 021 Messages is based on /EC CAT 021/.

Therefore, this chapter is readable only in collaboration with the referenced Document. MSSR 2000 ID specifics and differences stated in this chapter.

Dual redundant system:

Only the Master chain delivers ADS-B reports.

### 4.10.2 List of the Data Items used by MSSR 2000 ID

| Data Item | Description                                             | Resolution    | Used by MSSR 2000 ID |
|-----------|---------------------------------------------------------|---------------|----------------------|
| I021/008  | Aircraft Operational Status                             | n.a.          | -                    |
| I021/010  | Data Source Identification                              | n.a.          | X                    |
| I021/015  | Service Identification                                  | n.a.          | -                    |
| I021/016  | Service Management                                      | n.a.          | -                    |
| I021/020  | Emitter Category                                        | n.a.          | X                    |
| I021/040  | Target Report Descriptor                                | n.a.          | X                    |
| I021/070  | Mode 3/A Code                                           | n.a.          | X                    |
| I021/071  | Time of Applicability for Position                      | 1/128 s       | -                    |
| I021/072  | Time of Applicability for Velocity                      | 1/128 s       | -                    |
| I021/073  | Time of Message Reception for Position                  | 1/128 s       | X                    |
| I021/074  | Time of Message Reception for Position - High Precision | 2-30 s        | -                    |
| I021/075  | Time of Message Reception for Velocity                  | 1/128 s       | X                    |
| I021/076  | Time of Message Reception for Velocity - High Precision | 2-30 s        | -                    |
| I021/077  | Time of Message Transmission                            | 1/128 s       | -                    |
| I021/080  | Target Address                                          | n.a.          | X                    |
| I021/090  | Quality Indicators                                      | n.a.          | X                    |
| I021/110  | Trajectory Intent                                       | n.a.          | -                    |
| I021/130  | Position in WGS-84 co-ordinates                         | 180/223 °     | X                    |
| I021/131  | Position in WGS-84 co-ordinates, high resolution        | 180/230 °     | -                    |
| I021/132  | Message Amplitude                                       | 1 dBm         | -                    |
| I021/140  | Geometric Height                                        | 6.25 ft       | X                    |
| I021/145  | Flight Level                                            | ¼ FL          | X                    |
| I021/146  | Intermediate State Selected Altitude                    | 25 ft         | -                    |
| I021/148  | Final State Selected Altitude                           | 25 ft         | -                    |
| I021/150  | Air Speed                                               | n.a.          | X                    |
| I021/151  | True Air Speed                                          | n.a.          | X                    |
| I021/152  | Magnetic Heading                                        | 360/216 °     | X                    |
| I021/155  | Barometric Vertical Rate                                | 6.25 ft / min | X                    |
| I021/157  | Geometric Vertical Rate                                 | 6.25 ft / min | X                    |

|                                                |                     |                                             |                |                  |
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|          |                                          |          |   |
|----------|------------------------------------------|----------|---|
| I021/160 | Airborne Ground Vector                   | n.a.     | X |
| I021/161 | Track Number                             | n.a.     | - |
| I021/165 | Track Angle Rate                         | 1/32 °/s | - |
| I021/170 | Target Identification                    | n.a.     | X |
| I021/200 | Target Status                            | n.a.     | X |
| I021/210 | MOPS Version                             | n.a.     | X |
| I021/220 | Met Report                               | n.a.     | - |
| I021/230 | Roll Angle                               | 0.01 deg | - |
| I021/250 | Mode S MB Data                           | n.a.     | - |
| I021/260 | ACAS Resolution Advisory Report          | n.a.     | X |
| I021/271 | Surface Capabilities and Characteristics | n.a.     | - |
| I021/295 | Data Ages                                | n.a.     | - |
| I021/400 | Receiver ID                              | n.a.     | - |

The RE (Reserved Expansion Field) and SP (Special Purpose Field) are not supported by the MSSR 2000 ID.

#### 4.10.3 Specific Content of the Data Items

##### 4.10.3.1 Data item I021/040 Target Report Descriptor

The Bits in the Subfield are fixed to the listed values or set variable:

ATP = variable 0 or 1  
 ARC = variable 0 or 1  
 RC = 0  
 RAB = 0  
 DCR = 0  
 GBS = variable  
 SIM = variable  
 TST = variable  
 SAA = 0  
 CL = 2  
 EC7 = 0  
 EC6 = 0  
 EC5 = 0  
 NOGO = 0  
 CPR = 0  
 LDJP = 0  
 RCF = 0

##### 4.10.3.2 Data item I021/090 Quality Indicators

The following Bits in the Subfield are set variable, all others not used:

NUCr = variable  
 NUCp = variable  
 NACp = variable  
 all others not set

##### 4.10.3.3 Data item I021/150 Airspeed

This item is only present if the "Velocity Over Ground" information is not available from the aircraft and the aircraft reports the "indicated Airspeed" in the Airborne Velocity Message.

##### 4.10.3.4 Data item I021/151 True Airspeed

This item is only present if the "Velocity Over Ground" information is not available from the aircraft and the aircraft reports the "TRUE Airspeed" in the Airborne Velocity Message..

|                                                |                     |                                             |                |                  |
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#### 4.10.3.5 Data item I021/152 Magnetic Heading

This item is only present if the "Velocity Over Ground" information is not available from the aircraft and the aircraft reports the "indicated Airspeed" or the "TRUE Airspeed" in the Airborne Velocity Message.

#### 4.10.3.6 Data item I021/210 MOPS Version

The Bits in the Subfield are fixed to the listed values or set variable:

VNS = 0 The MOPS Version is supported by the GS  
VN = variable  
LTT = 2 1090 ES

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## 4.11 System Control and Monitoring (CAT 253)

### 4.11.1 General

MSSR 2000 ID provides a CAT 253 compliant interface for remote system monitoring and control (SMC). Via this interface full operational control of the MSSR 2000 ID interrogator can be achieved. This allows the local and remote monitoring and control of the system either via direct connections or via a network.

Dual redundant system:

The Master and the Slave should be connected to a CMS. In Principle, commands sent to the Master and an internal synchronisation mechanism forwards the command to the Slave if necessary. Exceptions listed in the detailed Parameter description below.  
Status and monitoring information delivered from the Master and the Slave.

#### 4.11.1.1 Limitations to ASTERIX Category 253

While ASTERIX CAT 253 represents a dedicated sub-suite of the ASTERIX standard designed for the monitoring and control of surveillance related equipment, unfortunately the ASTERIX CAT 253 standardisation was implementation driven and not co-ordinated. Consequently some shortcomings of the current released standard need to be overcome.

In case of the MSSR 2000 ID the reference document /EC CAT 253/ is used as basis for the SMC interface. However compatible extensions to the standard were made, which make use of common ASTERIX practice.

#### 4.11.1.2 Rules for the design of CMS SW

##### 4.11.1.2.1 No parameter polling

The CMS (Control and Monitoring System) shall not constantly query parameters via Data-Request (see section 'Get Parameter from MSSR') but rather rely on the delta status transfer message. Requesting parameters shall only be done in the initialisation phase after CAT-253 connection, but not during a CAT-253 session with a permanent connection.

Exceptions apply to:

- SYSTEM TIME (Ast. ID: 0x0320),
  - TEMPERATURES (Ast. ID 0x32E, 0x32F),
  - ONLINE\_STATUS (Ast. ID 0x0387) and
  - POWER\_MONITORING (Ast. ID 0x0413)
- which can be polled once per second.

##### 4.11.1.2.2 No cyclic commanding of configuration parameters

The CMS (Control and Monitoring System) shall not perform cyclic commanding of parameters just to be sure that the MSSR is using the correct value. The correct setting of the parameter can be evaluated by reading the Delta Status Message automatically following the Command Message when the parameter has been changed.

Each setting of a configuration parameter (see Section "Description of Configuration Parameters") can cause an action inside the MSSR (i.e. initialization) which can affect operation.

##### 4.11.1.2.3 Prevent flooding of CAT-253 messages

To prevent flooding of the MSSR CAT-253 interface, the number of messages per client connection (e.g. CMS (Control and Monitoring System)) shall not exceed 5 messages averaged over one second. Exception is once at startup/initialisation phase, for query of all needed parameters. When establishing a connection, in total, i.e. summed up over all clients, the number of data requests must not exceed 50 data requests per second.

|                                                |            |                  |          |          |
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#### 4.11.2 Category 253 Layout and Data Items

ASTERIX CAT 253 will have the following layout:

| CAT = 253 | LEN | FSPEC | Items of the first record | FSPEC | Items of the second record |
|-----------|-----|-------|---------------------------|-------|----------------------------|
|-----------|-----|-------|---------------------------|-------|----------------------------|

Where:

- Data Category (CAT) = 253 is a one-octet field indicating that the Data Block contains remote control data.
- Length Indicator (LEN) is a two octet field indicating the total length in octets of the Data Block, including the CAT and LEN fields.
- FSPEC is the Field Specification.

The following Data Items shall be used for the exchange of command and status messages to monitor and control radar stations.

Data Items of Category 253:

*Table 4-37. Data Items of Category 253*

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | X                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | X                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         |                     |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         |                     |
| I253/060  | Blocking Information                  | n.a.         |                     |
| I253/070  | Time of Day                           | 1/128 s      | X                   |
| I253/080  | Application Data Structure            | n.a.         | X                   |
| I253/090  | Application Data                      | n.a.         |                     |
| I253/100  | Transparent Application Data 1        | n.a.         | X                   |
| I253/110  | Transparent Application Data 2        | n.a.         | X                   |

|                                                |                     |                                             |                |                  |
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### 4.11.3 Specific Content of the Data Items

#### 4.11.3.1 Data Item I253/010, Data Source Identifier

**Definition:** Identification of the system from which the data are received.

**Format:** Two octets fixed length Data Item.

**Structure:**

| Octet No. 1 |    |    |    |    |    |    |   | Octet No. 2 |   |   |   |   |   |   |   |
|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| SAC         |    |    |    |    |    |    |   | SIC         |   |   |   |   |   |   |   |

Bit 16/9 (SAC) = System Area Code

Bit 8/1 (SIC) = System Identification Code

Notes:

- 1) MSSR 2000 ID allows the configuration of the SAC / SIC via HMI.
- 2) The up-to-date list of SACs is published on the Eurocontrol Web Site (<<http://www-eurocontrol.int>>).
- 3) The responsibility for the issuing of the SICs rest with the national administration.
- 4) The selected SIC / SAC number is not necessarily the SIC / SAC code as defined in the ASTERIX documentation. For use within a system self defined SIC / SAC codes can be used. These messages must not be transferred to any official site.

#### 4.11.3.2 Data Item I253/015, Local Data Source Identifier

**Definition:** Identification of the local system from which the data are received.

**Format:** One octet fixed length Data Item.

**Structure:**

| Octet no. 1 |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|---|
| 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Local ID    |   |   |   |   |   |   |   |

bits-8/1 (Local ID) = Local Identifier per SAC/SIC

Notes:

- 1) This item is optional and only used in the "Token Request" message. It is used, to determine a client, if serveral clients are connected with the same SAC/SIC. The value of the Local ID is replied from the MSSR in the messages "Token Assign" and "Alive" in item I253/025.
- 2) If the item is not send, the MSSR set the local identifier internally to "0".

#### 4.11.3.3 Data Item I253/020, Data Destination Identifier

**Definition:** Identification of the system to which the data must be sent.

**Format:** Repetitive Data Item, starting with a one-octet Field Repetition Indicator (REP) followed by at least one destination code of two-octet length.

**Structure:**

|             |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|
| Octet No. 1 |    |    |    |    |    |    |    |
| 24          | 23 | 22 | 21 | 20 | 19 | 18 | 17 |
| REP         |    |    |    |    |    |    |    |

|             |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |
|-------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| Octet No. 2 |    |    |    |    |    |    |   | Octet No. 3 |   |   |   |   |   |   |   |
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| SAC         |    |    |    |    |    |    |   | SIC         |   |   |   |   |   |   |   |

Bit 24/17 (REP) = Number of SAC/SICs

Bit 16/9 (SAC) = System Area Code

Bit 8/1 (SIC) = System Identification Code

Notes:

- 1) See Notes: for Data Item I253/010 for SAC SIC definition.

|                                                |            |                  |          |          |
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2) The SAC/SIC is used to uniquely identify the destinations of a CAT 253 message. The typical destination of a message from a server (MSSR 2000 ID) is a general one, a subset or all concerned clients (CMS). The destination of a message originating at a client is generally the server.

3) The underlying network, e.g. RADNET, will in all cases provide the routing function to distribute a message to the designated SAC/SICs.

#### 4.11.3.4 Data Item I253/025, Data Destination and Local Identifier

**Definition:** Identification of the client holding the token.

**Format:** Repetitive Data Item, starting with a one-octet Field Repetition Indicator (REP) followed by at least one destination code.

**Structure:**

|             |    |    |    |    |    |    |    |             |    |    |    |    |    |    |   |
|-------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|---|
| Octet No. 1 |    |    |    |    |    |    |    |             |    |    |    |    |    |    |   |
| 32          | 31 | 30 | 29 | 28 | 27 | 26 | 25 |             |    |    |    |    |    |    |   |
| REP         |    |    |    |    |    |    |    |             |    |    |    |    |    |    |   |
| Octet No. 2 |    |    |    |    |    |    |    | Octet No. 3 |    |    |    |    |    |    |   |
| 24          | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9 |
| SAC         |    |    |    |    |    |    |    | SIC         |    |    |    |    |    |    |   |
| Octet No. 4 |    |    |    |    |    |    |    |             |    |    |    |    |    |    |   |
| 8           | 7  | 6  | 5  | 4  | 3  | 2  | 1  |             |    |    |    |    |    |    |   |
| Local ID    |    |    |    |    |    |    |    |             |    |    |    |    |    |    |   |

Bit 32/25 (REP) = Number of destination identifiers

Bit 24/17 (SAC) = System Area Code

Bit 16/9 (SIC) = System Identification Code

Bit 8/1 (Local ID) = Local identifier per SAC/SIC

Notes:

1) The Data Destination and Local Identifier are only used by the MSSR 2000 ID in the Alive Message and the Token Assign Message to determine which SMC is currently holding the Command Token. Because there is only one Token, the REP field is set to 1.

2) If the item is received from a management station the item is ignored.

#### 4.11.3.5 Data Item I253/030, Source Application Identifier

**Definition:** Identification of the Source Application Identifier the message originates from.

**Format:** Two octet fixed length Data Item.

**Structure:**

|                               |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |
|-------------------------------|----|----|----|----|----|----|---|-------------|---|---|---|---|---|---|---|
| Octet No. 1                   |    |    |    |    |    |    |   | Octet No. 2 |   |   |   |   |   |   |   |
| 16                            | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Source Application Identifier |    |    |    |    |    |    |   |             |   |   |   |   |   |   |   |

Bit 8/1(SAI) = Source Application Identifier:

1 = DFS/REMP

2 = UK

3 = MSSR 2000 ID

Notes:

1) The SAI is used to distinguish between different applications of the CAT 253 service. Whereas the basic structure of ASTERIX CAT 253 is the same for all applications, different applications may use only a subset of the items provided by the CAT 253 standard. Additionally, a customisation of the used items is feasible.

2) MSSR 2000 ID will use SAI 3 only. If a message with no SAI or a different SAI is received the message is discarded.

|                                                |                     |                                             |                |                  |
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#### 4.11.3.6 Data Item I253/040, Message Type Identifier

**Definition:** The Message Type Identifier defines what kind of CAT 253 message was sent and identifies some basic properties of the message.

**Format:** One octet fixed length Data Item.

**Structure:**

|             |   |                         |   |   |   |   |   |
|-------------|---|-------------------------|---|---|---|---|---|
| Octet No. 1 |   |                         |   |   |   |   |   |
| 8           | 7 | 6                       | 5 | 4 | 3 | 2 | 1 |
| PI          | D | Message Type Identifier |   |   |   |   |   |

Bit 8 (PI) = Priority Identifier

0 = low priority

1 = high priority

Bit 7 (D)= Delivery Bit

0 = no explicit acknowledgement compulsory

1 = explicit acknowledgement compulsory

Bit 6/1(MTI) = Message Type Identifier

1 = Time and Day

2 = Connect request

3 = Connect Response

4 = Connect Release

5 = Command Token Request

6 = Command Token Release

7 = Command Token Assign

8 = Command Message

9 = Complete Status Transfer Message

10 = Delta Status Transfer Message

11 = Complete Status Transfer Request Message

12 = Centre Exchange Message

13 = Radar Monitoring Start Message

14 = Radar Monitoring Stop Message

15 = Transparent Data Message

16 = Acknowledgement

17 = Alarm

18 = Alive Message

19 = Data Request

Notes:

1) The priority of the message can be marked as “high” or “normal” (PI Bit). The D -Bit contains the information whether the current message shall be acknowledged by the recipient.

2) The MSSR 2000 ID does not use the PI-Bit. The MSSR 2000 ID will ignore the PI-Bit in CAT 253 messages.

3) The MSSR 2000 ID does not use the D-Bit. The MSSR 2000 ID will ignore the D-Bit in CAT 253 messages.

4) The MSSR 2000 ID uses the following message types:

2 = Connect request

3 = Connect Response

4 = Connect Release

5 = Command Token Request

6 = Command Token Release

7 = Command Token Assign

8 = Command Message

10 = Delta Status Transfer Message

17 = Alarm

18 = Alive Message

19 = Data Request

5) The MSSR 2000 ID does not use the following message types:

1 = Time and Day

|                                                |            |                  |          |          |
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- 9 = Complete Status Transfer Message
- 11 = Complete Status Transfer Request Message
- 12 = Centre Exchange Message
- 13 = Radar Monitoring Start Message
- 14 = Radar Monitoring Stop Message
- 15 = Transparent Data Message
- 16 = Acknowledgement

#### 4.11.3.7 Data Item I253/050, Message Sequence Identifier

**Definition:** The Item Message Sequence Identifier contains consecutively incremented message counters for each receiving application. The ID counters shall be incremented by the originator application.

**Format:** Repetitive Data Item, starting with a one-octet Field Repetition Indicator (REP) followed by at least one Message Sequence Identifier of two octet length.

**Structure:**

|             |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |
|-------------|----|----|----|----|----|----|----|-------------|---|---|---|---|---|---|---|
| Octet No. 1 |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |
| 24          | 23 | 22 | 21 | 20 | 19 | 18 | 17 |             |   |   |   |   |   |   |   |
| REP         |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |
| Octet No. 2 |    |    |    |    |    |    |    | Octet No. 3 |   |   |   |   |   |   |   |
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| MSID        |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |

Bit 24/17 (REP) = Number of MSIDs

Bit 16/1 (MSID) = Message Sequence Identifiers

Notes:

1) The Message Sequence Identifier can be used by an application to associate acknowledgements or responses to the corresponding request. This may be important in case the issuing of requests is not synchronous, i.e. if an application has multiple outstanding requests at a time.

2) The Message Sequence Identifier is additionally used for applications that use an underlying network that potentially re-orders messages on the way to the receiver. For messages with multiple destinations (multicast transmissions) the sending application can specify a sequence number for each of the receiving instances. The order of the sequence numbers must correspond to the order and number of SAC/SICs in I253/020 or I253/025.

3) This Data Item is not used for MSSR 2000 ID. If this item is part of a received message the content of this Data Item is discarded.

#### 4.11.3.8 Data Item I253/070, Time of Day

**Definition:** Absolute time stamping expressed as UTC time.

**Format:** Three octets fixed length Data Item.

**Structure:**

|             |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |
|-------------|----|----|----|----|----|----|----|-------------|---|---|---|---|---|---|---|
| Octet No. 1 |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |
| 24          | 23 | 22 | 21 | 20 | 19 | 18 | 17 |             |   |   |   |   |   |   |   |
| Time of Day |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |
| Octet No. 2 |    |    |    |    |    |    |    | Octet No. 3 |   |   |   |   |   |   |   |
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Time of Day |    |    |    |    |    |    |    | Time of Day |   |   |   |   |   |   |   |
|             |    |    |    |    |    |    |    |             |   |   |   |   |   |   |   |

Bit 24/1 = Time Of Day

Bit 1 (LSB) =  $2^{-7} \text{ s} = 1/128 \text{ s}$

Note:

The time of day value is reset to zero each day at midnight. The Time of Day item shall contain the originator's time stamp.

|                                                |            |                  |          |          |
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#### 4.11.3.9 Data Item I253/080, Application Data Structure

**Definition:** Data item I253/080 conveys the structure and identity of the user data that is contained in the data item I253/100 or I253/110. The format of the transferred information is described below.

**Format:** Variable length Data Item comprising a first part of four octets, followed by one-octet extensions as necessary.

**Structure:**

|             |    |    |    |    |    |    |    |             |    |    |    |    |    |     |    |
|-------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|-----|----|
| Octet No. 1 |    |    |    |    |    |    |    | Octet No. 2 |    |    |    |    |    |     |    |
| 32          | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24          | 23 | 22 | 21 | 20 | 19 | 18  | 17 |
| START INDEX |    |    |    |    |    |    |    |             |    |    |    |    |    | LSB |    |
| Octet No. 3 |    |    |    |    |    |    |    |             |    |    |    |    |    |     |    |
| 16          | 15 | 14 | 13 | 12 | 11 | 10 | 9  |             |    |    |    |    |    |     |    |
| COUNT       |    |    |    |    |    |    |    |             |    |    |    |    |    |     |    |
| Octet No. 4 |    |    |    |    |    |    |    |             |    |    |    |    |    |     |    |
| 8           | 7  | 6  | 5  | 4  | 3  | 2  | 1  |             |    |    |    |    |    |     |    |
| STATUS      |    |    |    |    |    |    |    | FX          |    |    |    |    |    |     |    |

Bit 32/17 (START INDEX) Provides identity of first object value to be transferred as part of the application data item. Index refers to an ordered list of objects previously agreed between client and server. (see Notes)

Bit 16/9 (COUNT) Indicates the number of object values with the same status to be transferred as part of the data field, starting with "Start Index" and using the consecutive numbers following this index. Thus, objects with the same status are bundled.

(see Notes)

Bit 8/1 (STATUS) Provides common status information for all objects in the data field:

- Bit 8 =
- 0 non-stale data
  - 1 stale data
- Bit 7 =
- 0 real data
  - 1 simulated data
- Bit 6 =
- 0 monitored object under remote control
  - 1 monitored object under local control
- Bit 5 =
- 0 no data included
  - 1 data included
- Bit 4/2 =
- 0 reserved for application specific data
  - 1 stale data
- Bit 1 =
- 0 End of item
  - 1 Extension into another status segment

Notes:

- 1) Stale data are data that may not be up-to-date, e.g. because the acquisition processor has failed.
- 2) Simulated data may be data that an operator "manually" sets for various purposes.
- 3) The "Data included" bit determines whether there exists a corresponding data field in I253/090, I253/100 or I253/110 of this record.
- 4) Monitored objects under local control indicates that the object is not in an operational mode but in a maintenance mode.
- 5) For MSSR 2000 ID the START INDEX represents the number of the parameter which is to be transferred.
- 6) For MSSR 2000 ID the COUNT field is always set to 1 - one parameter per record only can be transferred.

|                                                |            |                  |          |           |
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7) The field "STATUS" is not used for MSSR 2000 ID. Received values will be neglected.

#### 4.11.3.10 Data Item I253/100, Transparent Application Data 1

**Definition:** The Transparent Application Data item contains application data not further standardised by ASTERIX CAT 253.

**Format:** Explicit length Data Item.

**Structure:**

|                         |     |     |     |     |     |     |     |
|-------------------------|-----|-----|-----|-----|-----|-----|-----|
| Octet No. 1             |     |     |     |     |     |     |     |
| ...                     | ... | ... | ... | ... | ... | ... | ... |
| LEN                     |     |     |     |     |     |     |     |
| Octet No. 2             |     |     |     |     |     |     |     |
| ...                     | ... | ... | ... | ... | ... | ... | ... |
| TAD octet 1             |     |     |     |     |     |     |     |
| Octet No. LEN           |     |     |     |     |     |     |     |
| 8                       | 7   | 6   | 5   | 4   | 3   | 2   | 1   |
| TAD Data octet (LEN -1) |     |     |     |     |     |     |     |

Bit 8/1(LEN) = Total Length of the Data Item including itself.

(TAD) = Sequence of bytes containing transparent application data for various purposes.

Notes:

- 1) MSSR 2000 ID uses this item to transfer data contents.
- 2) It tags the messages in order to identify the message Data Type.

The following overview shows the used Octets for different supported Data Types.

signed / unsigned integer (8 bits)

|     |      |       |
|-----|------|-------|
| 1   | 2    | 3     |
| LEN | type | value |

signed / unsigned integer (16 bits)

|     |      |       |   |
|-----|------|-------|---|
| 1   | 2    | 3     | 4 |
| LEN | type | value |   |

signed / unsigned integer (32 bits)

|     |      |       |   |   |   |
|-----|------|-------|---|---|---|
| 1   | 2    | 3     | 4 | 5 | 6 |
| LEN | type | value |   |   |   |

floating point

|     |      |          |   |   |   |          |   |
|-----|------|----------|---|---|---|----------|---|
| 1   | 2    | 3        | 4 | 5 | 6 | 7        | 8 |
| LEN | type | mantissa |   |   |   | exponent |   |

The value is calculated as

$$value = \frac{mantissa}{2^{31}} \cdot 2^{exponent}$$

The following DataTypes are defined:

- 0 Special
- 1 Signed integer 8 bit
- 2 Unsigned integer 8 bit
- 3 Signed integer 16 bit

|                                                |                     |                                             |                |                   |
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- 4 Unsigned integer 16 bit
- 5 Signed integer 32 bit
- 6 Unsigned integer 32 bit
- 7 Floating Point
- 8 Text string (8 bit ASCII)

All signed Data Types are in 2ers-complement representation as well as the mantissa and exponent of the floating point type.

The length of the item determines if an array (dimension = 1) is conveyed. In this case the tag is not repeated. In the following example, an array of 4 values (type = signed integer 16 bit) is presented.

|    |   |       |   |       |   |       |   |       |    |
|----|---|-------|---|-------|---|-------|---|-------|----|
| 1  | 2 | 3     | 4 | 5     | 6 | 7     | 8 | 9     | 10 |
| 10 | 3 | value |   | value |   | value |   | value |    |

#### 4.11.3.11 Data Item I253/110, Transparent Application Data 2

**Definition:** The Transparent Application Data item contains application data not further standardised by ASTERIX CAT 253.

**Format:** Explicit length Data Item.

**Structure:**

|                         |
|-------------------------|
| Octet No. 1             |
| ... ..                  |
| LEN                     |
| Octet No. 2             |
| ... ..                  |
| TAD octet 1             |
| Octet No. LEN           |
| 8 7 6 5 4 3 2 1         |
| TAD Data octet (LEN -1) |

Bit 8/1(LEN) = Total Length of the Data Item including itself.

(TAD) = Sequence of bytes containing transparent application data for various purposes.

|                                                |                     |                                             |                |                   |
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#### 4.11.4 Encoding Rules for Message Types

Even though ASTERIX in general enables the user to combine an arbitrary set of items in a message, the MSSR 2000 ID application makes use of a comprehensive set of "messages". The following section will provide detailed information of the construction of messages and how the messages are intended to be used.

##### 4.11.4.1 Type 2 = Connect Request

Connect Requests are expected from the SMC console before any further interaction between the SMC position and MSSR 2000 ID is possible. MSSR 2000 ID will respond to the connect request with a Connect Response Message.

If a connection is already established MSSR 2000 ID still will respond with a Connect Response Message but clears all pending operations connected with the previous connection.

Connect Requests make use of the following items:

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | optional            |
| I253/080  | Application Data Structure            | n.a.         | -                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | -                   |
| I253/110  | Transparent Application Data 2        | n.a.         | -                   |

**Direction:** MSSR 2000 ID ← SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the requesting SMC
- I253/020 holds the SAC/SIC from the MSSR 2000 ID.
- I253/030 SAI shall be 3
- I253/040 Message Type is 2 = Connect Request
- I253/070 Optional, ignored from the MSSR 2000 ID if received.

**Note:**

If the MSSR 2000 ID has not sent a "Connect Response" message within 1s since this message has been sent, this message shall be repeated.

|                                                |                     |                                             |                |                   |
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#### 4.11.4.2 Type 3 = Connect Response

Connect Responses are sent by MSSR 2000 ID in response to a Connect Request Message. After this message MSSR 2000 ID will send Alive Message, Status Update Messages and Alarm Messages to the connection partner without further request.

Further, the SMC position will be in the position to request further information on demand as well as it may request the command token. If no connection is established, any request will be discarded.

**Note:**

The connect response is sent to the newly connected system only. The other connected system can derive the fact of a new connection with the next message containing the Item 253/020, e.g. the Alive Message.

Connect Response Messages make use of the following items:

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | X                   |
| I253/080  | Application Data Structure            | n.a.         | -                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | -                   |
| I253/110  | Transparent Application Data 2        | n.a.         | -                   |

**Direction:** MSSR 2000 ID → SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the MSSR 2000 I
- I253/020 holds the destination SAC/SIC of SMC positions
- I253/030 SAI shall be 3
- I253/040 Message Type is 3 = Connect Response
- I253/070 holds the time of the message.

|                                                |                     |                                             |                |                   |
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#### 4.11.4.3 Type 4 = Connect Release

Connect Release Message may be sent by either MSSR 2000 ID or by the SMC console, in order to indicate the disconnection. After this Message MSSR 2000 ID will no longer send spontaneous messages to the disconnected system nor will it accept any commands.

Connect Release Messages make use of the following items.

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | X                   |
| I253/080  | Application Data Structure            | n.a.         | -                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | -                   |
| I253/110  | Transparent Application Data 2        | n.a.         | -                   |

**Direction:** MSSR 2000 ID ↔ SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the MSSR 2000 ID or SMC dependent on the direction
- I253/020 holds the destination SAC/SIC.
- I253/030 SAI shall be 3
- I253/040 Message Type is 4 = Connect Release
- I253/070 Time of the message. Optional, ignored from the MSSR 2000 ID if received.

|                                                |                     |                                             |                |                   |
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#### 4.11.4.4 Type 5 = Command Token Request

With the Command Token Request the SMC position requests the MSSR 2000 ID to grant access to write parameter.

Command Token Request Messages make use of the following items:

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | optional            |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | optional            |
| I253/080  | Application Data Structure            | n.a.         | -                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | -                   |
| I253/110  | Transparent Application Data 2        | n.a.         | X                   |

**Direction:** MSSR 2000 ID ← SMC Console

**Encoding Rules:**

- I253/010 SAC/SIC of the requesting SMC Console
- I253/015 Local Identifier of the requesting SMC Console (optional see note)
- I253/020 holds the SAC/SIC from the MSSR 2000 ID.
- I253/030 SAI shall be 3
- I253/040 Message Type is 5 = Command Token Request
- I253/070 Optional, ignored from the MSSR 2000 ID if received.
- I253/110 holds the access level and the password.

The password is left aligned. Not used bits are set to 0x00. For detailed information about the contents and structure, see chapter PASSWORD of the Description of Configuration Parameters section.

**Note:**

Item I253/015 is used for the identification of a client, if several clients with the same SAC/SIC are connected. The information is only used, to tag the token holder via item I253/025 in the messages "Command Token Assign" and "Alive Message".

If item I253/015 is missing, the MSSR will set the local identifier internally to "0". The local identifier in item I253/025 of the two messages "Token Assign" and "Alive" will be "0" as well.

**Note:**

When, during system power up, the token is requested before the PBIT has finished, the system will not assign the token because it is not yet ready to execute remote commands. Only token requests received after the PBIT can result in a successful token assignment.

|                                                |                     |                                             |                |                   |
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#### 4.11.4.5 Type 6 = Command Token Release

With the Command Token Release the SMC position returns the command token to the MSSR 2000 ID.

This message makes use of the following items:

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | optional            |
| I253/080  | Application Data Structure            | n.a.         | -                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | -                   |
| I253/110  | Transparent Application Data 2        | n.a.         | -                   |

**Direction:** MSSR 2000 ID ← SMC Console

**Encoding Rules:**

- I253/010 SAC/SIC of the requesting SMC Console
- I253/020 holds the SAC/SIC from the MSSR 2000 ID.
- I253/030 SAI shall be 3
- I253/040 Message Type is 6 = Command Token Release
- I253/070 Optional, ignored from the MSSR 2000 ID if received.

|                                                |                     |                                             |                |                   |
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#### 4.11.4.6 Type 7 = Command Token Assign

This message informs the connected SMCs about the new system holding the command token.

The message is sent to all connected SMC positions.

In case a token request is not granted for various reasons, still a command token assign message is generated. It identifies the SMC position which holds the token. If no SMC position is addressed (I253/025 is empty or not sent) the token is held locally at the HMI.

Command Token Assign Messages make use of the following items:

| Data Item | Description                                  | System Units | Used by MSSR 2000 I |
|-----------|----------------------------------------------|--------------|---------------------|
| I253/010  | <b>Data Source Identifier</b>                | n.a.         | X                   |
| I253/015  | <b>Local Data Source Identifier</b>          | n.a.         | -                   |
| I253/020  | <b>Data Destination Identifier</b>           | n.a.         | X                   |
| I253/025  | <b>Data Destination and Local Identifier</b> | n.a.         | X                   |
| I253/030  | <b>Source Application Identifier</b>         | n.a.         | X                   |
| I253/035  | <b>Data Origin Identifier</b>                | n.a.         | -                   |
| I253/040  | <b>Message Type</b>                          | n.a.         | X                   |
| I253/050  | <b>Message Sequence ID</b>                   | n.a.         | -                   |
| I253/060  | <b>Blocking Information</b>                  | n.a.         | -                   |
| I253/070  | <b>Time of Day</b>                           | 1/128 s      | X                   |
| I253/080  | <b>Application Data Structure</b>            | n.a.         | -                   |
| I253/090  | <b>Application Data</b>                      | n.a.         | -                   |
| I253/100  | <b>Transparent Application Data 1</b>        | n.a.         | -                   |
| I253/110  | <b>Transparent Application Data 2</b>        | n.a.         | -                   |

**Direction:** MSSR 2000 ID → SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the MSSR 2000 I
- I253/020 holds the destination SAC(s)/SIC(s) of SMC positions
- I253/025 The destination list contains only one entry which holds the SAC/ SIC currently holding the token. The item is missing if MSSR 2000 ID is under local control, or if the password is incorrect.
- I253/030 SAI shall be 3
- I253/040 Message Type is 7 = Command Token Assign
- I253/070 holds the time of the message.

**Note:**

The local identifier in item I253/025 holds the local identifier of the token holder, if the client (which holds the token), requested it with a message "Token Request" including item I253/015.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.4.7 Type 8 = Command Message

The SMC position controlling the token transmit a parameter (or command) to the MSSR 2000 ID.

These messages make use of the following items:

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | optional            |
| I253/080  | Application Data Structure            | n.a.         | X                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | X                   |
| I253/110  | Transparent Application Data 2        | n.a.         | X                   |

**Direction:** MSSR 2000 ID ← SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the SMC
- I253/020 holds the SAC/SIC from the MSSR 2000 ID.
- I253/030 SAI shall be 3
- I253/040 Message Type is 8 = Command Message
- I253/070 Optional, ignored from the MSSR 2000 ID if received.
- I253/080 Holds the Parameter number. Only one parameter will be conveyed by one message.
- I253/100 The value of the parameter, must match the criteria defined in the Parameter Description.
- I253/110 The value of the parameter, must match the criteria defined in the Parameter Description.

The message contains either Item I253/100 and / or Item I253/110 depending on the format of the parameter.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.4.8 Type 10 = Delta Status Transfer Message

The MSSR 2000 ID transmit this message to all connected systems.

The message may be sent spontaneous or on request via the Data Request Message.

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | X                   |
| I253/080  | Application Data Structure            | n.a.         | X                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | X                   |
| I253/110  | Transparent Application Data 2        | n.a.         | X                   |

**Direction:** MSSR 2000 ID → SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the MSSR 2000 I
- I253/020 holds the destination SAC(s)/SIC(s) of SMC positions
- I253/030 SAI shall be 3
- I253/040 Message Type is 10 = Delta Status Transfer Message
- I253/070 holds the time of the message.
- I253/080 Holds the Parameter number. Only one parameter will be conveyed by one message.
- I253/100 holds the data (see chapters for Parameter Description)
- I253/110 holds the data (see chapters for Parameter Description)

The message contains either Item I253/100 or Item I253/110 depending on the format of the parameter.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.4.9 Type 17 = Alarm Message

Alarm messages are sent spontaneously by the MSSR 2000 ID to all connected systems.

The Alarm Messages make use of the following items:

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | X                   |
| I253/080  | Application Data Structure            | n.a.         | -                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | X                   |
| I253/110  | Transparent Application Data 2        | n.a.         | X                   |

**Direction:** MSSR 2000 ID → SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the MSSR 2000 I
- I253/020 holds the destination SAC(s)/SIC(s) of SMC positions.
- I253/030 SAI shall be 3
- I253/040 Message Type is 17 = Alarm
- I253/070 holds the time of the message.
- I253/100 holds the alarm as free text. This includes the alarm classification (message, warning, error, fatal), the alarm code and the alarm text
- I253/110 optional item.

If available, this item holds additional information of the alarm message as free text.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.4.10 Type 18 = Alive Message

Alive messages are sent periodically by MSSR 2000 ID and are expected from the connected SMC consoles in order to supervise the connection. Alive Messages rate is configurable.

If MSSR 2000 ID does not receive an Alive Message from a SMC position in the connected state it will shut down the connection. This leads to an explicit Connect Release Message by the MSSR 2000 ID. The failure of more than two consecutive Alive Messages will lead to this reaction.

Alive Messages make use of the following items.

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | X                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | X                   |
| I253/080  | Application Data Structure            | n.a.         | -                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | -                   |
| I253/110  | Transparent Application Data 2        | n.a.         | -                   |

X(MSSR) = available for the Messages output by the MSSR 2000 ID.

**Direction:** MSSR 2000 ID ↔ SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the MSSR 2000 ID or SMC dependent on the direction
- I253/020 holds the destination SAC(s)/SIC(s) of SMC positions
- I253/025 the destination list contains only one entry which holds the SAC / SIC currently holding the token. The item is missing if MSSR 2000 ID is under local control. This item is available for Alive Messages output by the MSSR 2000 ID.
- I253/030 SAI shall be 3
- I253/040 Message Type is 18 = Alive
- I253/070 holds the time of the message.

|                                                |                     |                                             |                |                   |
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#### 4.11.4.11 Type 19 = Data Request Message

This message enables the SMC to request a specific parameter. Prerequisite for this operation is an established connection.

The message contains the following items.

| Data Item | Description                           | System Units | Used by MSSR 2000 I |
|-----------|---------------------------------------|--------------|---------------------|
| I253/010  | Data Source Identifier                | n.a.         | X                   |
| I253/015  | Local Data Source Identifier          | n.a.         | -                   |
| I253/020  | Data Destination Identifier           | n.a.         | X                   |
| I253/025  | Data Destination and Local Identifier | n.a.         | -                   |
| I253/030  | Source Application Identifier         | n.a.         | X                   |
| I253/035  | Data Origin Identifier                | n.a.         | -                   |
| I253/040  | Message Type                          | n.a.         | X                   |
| I253/050  | Message Sequence ID                   | n.a.         | -                   |
| I253/060  | Blocking Information                  | n.a.         | -                   |
| I253/070  | Time of Day                           | 1/128 s      | optional            |
| I253/080  | Application Data Structure            | n.a.         | X                   |
| I253/090  | Application Data                      | n.a.         | -                   |
| I253/100  | Transparent Application Data 1        | n.a.         | -                   |
| I253/110  | Transparent Application Data 2        | n.a.         | -                   |

**Direction:** MSSR 2000 ID ← SMC Console

**Encoding Rules:**

- I253/010 holds the configured SAC/SIC of the SMC console
- I253/020 holds the SAC/SIC from the MSSR 2000 ID.
- I253/030 SAI shall be 3
- I253/040 Message Type is 19 = Data Request
- I253/070 Optional, ignored from the MSSR 2000 ID if received.
- I253/080 Holds the Parameter number. Only one parameter will be requested by one message.

All parameters can be read, by means of a Data Request message

|                                                |                     |                                             |                |                   |
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#### 4.11.5 Connection for CAT 253

##### 4.11.5.1 Connection Establishment

MSSR 2000 ID requires the SMC positions to establish a connection on Session Layer. Note that this requires a connection on LAN to be already established. The Session Layer connection establishment identifies the SMC position to the MSSR 2000 ID SMC position administration. From this time on until the connection is removed, MSSR 2000 ID provides the following services without further interactions:

- Periodic Alive Message (the periodicity is configurable, default value is 3 seconds)
- Spontaneous Error Message
- Special Status / Diagnostics Values

Once a connection is established, the SMC position can request additional parameters, either from the configuration parameter list or from the diagnostics values. These request leads to a single read access to the respectively managed object.

In order to get write access to the configuration parameters, the SMC position must obtain a token (see below). This prevents multiple SMC positions to initiate conflicting transactions. As soon as the SMC position has obtained the token, it can use Command Messages to change configuration parameters. This then will influence the system operation.

##### 4.11.5.2 Alive Messages

Alive Messages are used to supervise an established connection. Either side of a connection (MSSR 2000 ID, SMC Position) is required to send an Alive Message at least once every configured time. If one side fails to send the Alive Message, the other will tear down the connection.

The period of the Alive Message is configurable (CMS\_ALIVE\_CYCLE), the default value is 3 seconds.

The MSSR tolerates one miss in the reception of Alive Messages from a SMC position. In case of 2 misses the MSSR will close the connection, and take again the token if the concerned SMC owned it previously. The concerned SMC will have to send a Connect Request message, if it wants to build up the connection again.

##### 4.11.5.3 Alarm Messages

Alarm Messages are sent spontaneously by MSSR 2000 ID to all connected systems. Actually they are the translation of a message, warning, error or a fatal error

##### 4.11.5.4 Token Management

MSSR 2000 ID supports multiple SMC positions. These can access the MSSR 2000 ID concurrently to monitor status and configuration of the interrogator. The access is even concurrent to the local HMI.

However, at any time only one SMC or the local HMI is allowed to control the MSSR 2000 ID, i.e. change the configuration or the operational state. This is granted by the token control of the MSSR 2000 ID.

In LOCAL mode, the token always is on the local HMI. In REMOTE mode, a SMC can request the token if no other SMC position controls the token. All SMC controls can be made ineffective by switching to LOCAL mode on the local HMI. This returns the token to the HMI immediately.

A SMC position or the local HMI can only give control the MSSR 2000 ID if it successfully obtained the token. Unless a SMC position releases the token, no other SMC position can obtain the token with exception of the local HMI.

##### 4.11.5.5 Authorisation

Due to the civil nature of ASTERIX CAT 253 authorisation of a SMC position is not necessary. Hence SMC can directly connect to MSSR 2000 ID without further authorisation procedure. Nevertheless MSSR 2000 ID checks the physical interface and the SAC/SIC in order to limit potential misuse of the interface.

|                                                |            |                  |          |           |
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Using TCP, the CAT253 connection between client and the MSSR 2000 ID is assigned to an unique internal line. Only on this line command and control is possible even if another or the same client uses the same IP address or SAC/SIC.

Using UDP unicast for a CAT253 connection, the IP address and the SAC/SIC of the client is checked by the MSSR 2000 ID prior to allow command and control of it. It is recommended to use TCP connection for CAT253 since it has a more secure connection.

As a basic and simple protection a password needs to be provided when a SMC position requests a control token. Only if the password is correct, the token is granted. The password is pre-configured but changeable by the CAT 253 interface.

In a future version of the MSSR 2000 ID, an enhanced authorisation mechanism of the MSSR 2000 ID needs to be implemented. However, this extension shall be limited to a software change.

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#### 4.11.6 Transactions for CAT 253

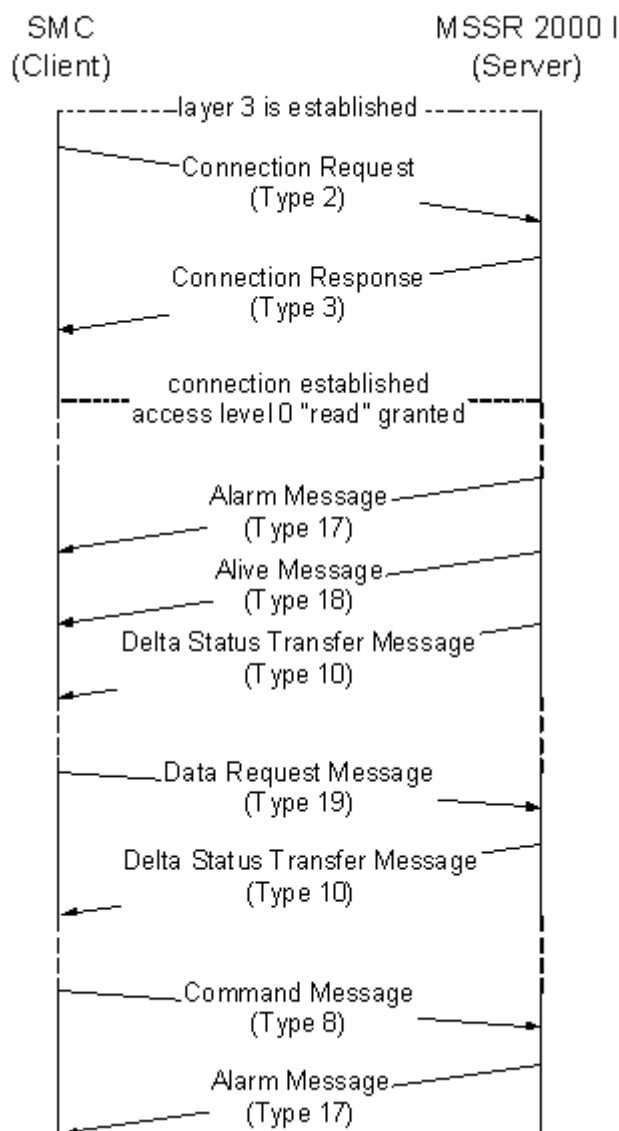
##### 4.11.6.1 Connection Establishment

In all cases the MSSR 2000 ID is considered as server and the SMC positions are considered as clients.

As a prerequisite to all the following transactions, the connection on Transport (TCP/UDP) layer(4) shall be already established.

The Session Establishment is performed following the transactions given in Figure "Transaction - Session Establishment".

Figure 4-7. Transaction - Session Establishment



If there is no Connection Response Message, the connection is not established. The SMC has to repeat the Connection Request Message.

After the connection to a CMS has been established, the MSSR 2000 ID transmits spontaneously a MSR\_STATE message to this CMS

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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After the connection has been established, the spontaneous messages (Alarm Message, Alive Message, Delta Status Transfer Message) from the MSSR 2000 ID are sent to the SMC position. The given order is an example only. The order depends on the current system status.

After the connection has been established, the SMC can request the current status of the MSSR 2000 ID by sending the Data Request Message. The MSSR 2000 ID will answer this message with the appropriate Delta Status Transfer Message.

During this state of the connection (no Command Token is assigned to the SMC) Command Messages will be rejected by the MSSR 2000 ID and answered with an Alarm Message "No Command Token Assigned".

The connection for the CAT 253 session is closed by sending the Connect Release (Type 4). If the MSSR 2000 ID receives this message, no further messages are sent to this SMC. Messages from this SMC are discarded.

If the MSSR 2000 ID wants to close a connection it will send the Connect Release (Type 4).

**Note:**

The connected SMC positions shall have unique SIC/SAC - codes. The MSSR 2000 ID manages the SIC/SAC - codes of connected SMC positions dynamically. There is no need to predefine the SIC/SAC codes of the SMC positions.

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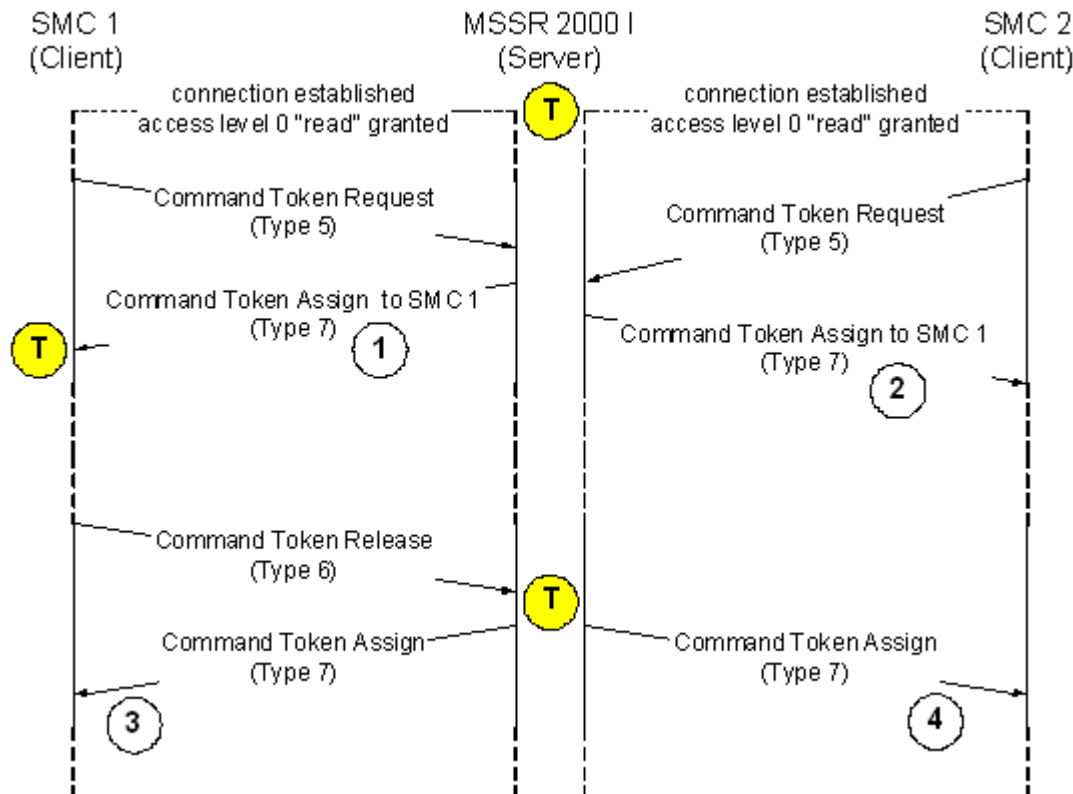


#### 4.11.6.2 Command Token Handling

The MSSR 2000 ID, in addition to the embedded HMI, can handle multiple SMC positions to be connected to it. But only one (the HMI or one of the SMC) can have permission to command the MSSR 2000 ID.

The holder of the Command Token has this permission. The handling of the Command Token is described below.

Figure 4-8. Transactions - Command Token Assignment



In Figure "Transactions - Command Token Assignment" the transactions necessary to assign the Command Token to a SMC are described.

##### Situation:

Connection on CAT 253 level is established.  
The Command Token is not requested by the local HMI (LOCAL LED is off).

##### Description:

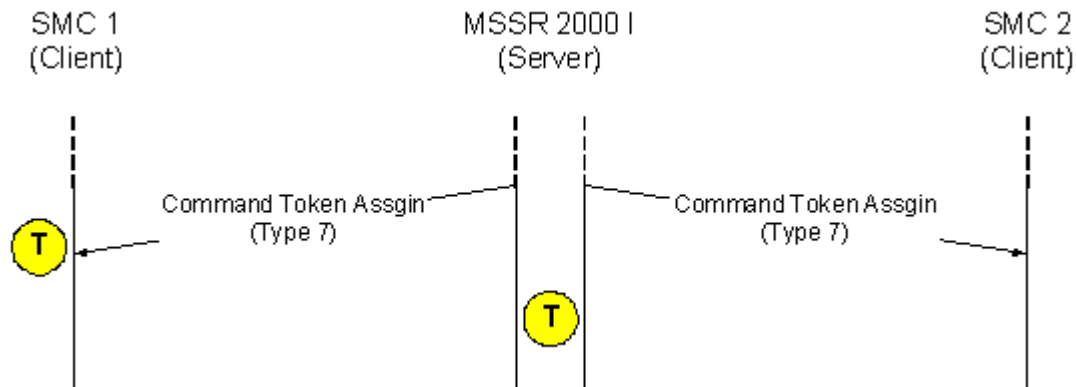
SMC 1 sends the Command Token Request to the MSSR 2000 ID. As the Token is held by the MSSR 2000 ID free to be assigned to one SMC, the MSSR 2000 ID will assign the Command Token to SMC 1 by sending the Command Token Assign.  
The Item I253/025 (Data Destination and Local Identifier) holds the SIC/SAC code of the SMC, the Command token has been assigned to.  
In case of Message (1), the item I253/025 holds the value of the Item I253/010 (Data Source Identifier) of the Command Token Request from SMC 1.  
As SMC 2 requests the Command Token after the Command Token Request from SMC 1 has been received (or the Command Token has been already assigned to SMC 1), the command Token Request will be rejected by sending a Command Token Assign (2) with the I253/025 (Data Destination and Local Identifier) containing the SIC / SAC code of SMC 1

|                                                |                     |                                             |                |                   |
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SMC 1 gives back the Command Token to the MSSR 2000 ID by sending the Command Token Release.  
 The MSSR 2000 ID will accept the command token by sending a Command Token Assign to SMC 1 and all other connected SMC.  
 The Item I253/025 (Data Destination and Local Identifier) is missing in this message, informing the SMCs that the MSSR 2000 ID HMI is holding the Command Token now.

The only way to revoke the Command Token from a SMC is to press the LOCAL / REMOTE Key at the embedded keyboard.

Figure 4-9. Transaction - Revocation of the Command Token



After pressing the LOCAL / REMOTE Key at the embedded keyboard the MSSR 2000 ID will send a Command Token Assign to SMC 1 and all other connected SMC.

The Item I253/025 (Data Destination and Local Identifier) is missing in this message, informing the SMCs that the MSSR 2000 ID HMI is holding the Command Token.

As long as the LOCAL LED is on at the embedded keyboard, the Command Token is assigned to the local HMI and cannot be assigned to a SMC.

|                                                |            |                  |          |           |
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#### 4.11.6.2.1 Data Item I253/110 Transparent Application Data 2 in the Command Token Request Message

Data Item I253/110 contains the requested access level and the appropriate password.

The structure of the Data Item I253/110 is the following:

|       |     |
|-------|-----|
| Octet |     |
| 1     | 2   |
| len   | ACL |
| 0x12  | y   |

|                    |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |
|--------------------|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|
| Octet              |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |
| 3                  | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| x                  | x | x | x | x | x | x | x  | x  | x  | x  | x  | x  | x  | x  | x  |
| Password for ACL y |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |

len = implicit length of the data item, is always 0x12 DataType = UINT\_8

ACL = Access Level Data Type = UINT\_8

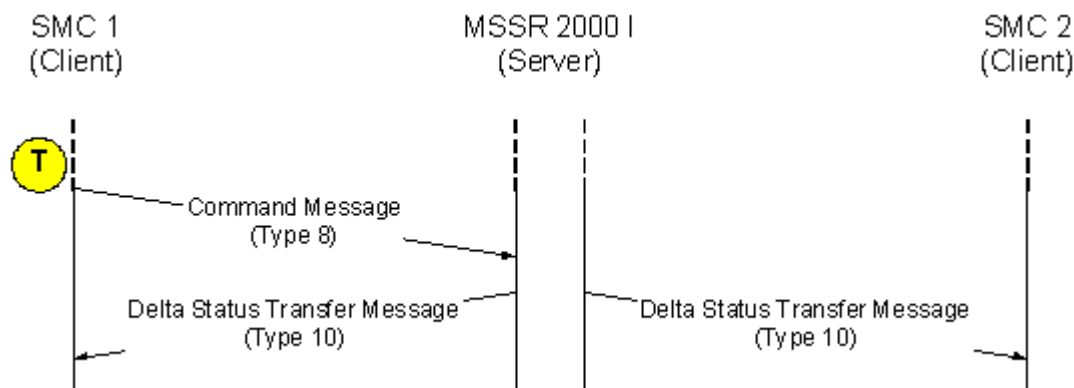
- 0: Access Level 0
- 1: Access Level 1
- 2: Access Level 2
- 3: Access Level 3

Password for ACL y: see Section about Configuration Parameter PASSWORD.

#### 4.11.6.3 Set Parameters at the MSSR 2000 I

If a SMC holds the Command Token, it can set parameters within the MSSR 2000 ID. Figure "Transactions - Set Parameter" shows the transactions necessary to set a parameter.

Figure 4-10. Transactions - Set Parameter



SMC 1 is holding the Command Token.

The Command Message contains the parameter to be changed and the value the parameter shall be set to.

If the SMC has the access rights to this parameter and if the parameter value is within the defined range, the MSSR 2000 ID will set the parameter to the given value.

As an acknowledgement to the command, a Delta Status Transfer Message, containing the parameter and its new value is sent to all connected SMC.

|                                                |                     |                                             |                |                   |
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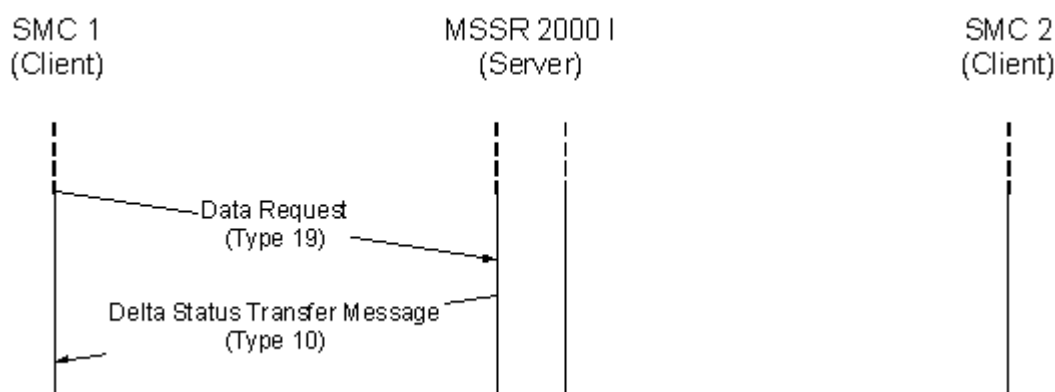
If the SMC 1 does not have the access right to this parameter, the MSSR 2000 ID will reject the Command Message and will send an Alarm Message to SMC 1. The value of the parameter within the MSSR 2000 ID is not changed.

If the parameter value is not within the defined range, the MSSR 2000 ID will reject the Command Message and a Delta Status Transfer Message, containing the parameter and its actual value is sent to the requesting SMC.

#### 4.11.6.4 Get Parameter from the MSSR 2000 I

Every connected SMC can get parameter values from the MSSR 2000 ID. The Command Token is not necessary for this action. Figure "Transactions - Get Parameter" shows the transactions necessary to get a parameter.

Figure 4-11. Transactions - Get Parameter



The Data Request contains the number of the parameter to be read.

As an acknowledgement to the Data Request, a Delta Status Transfer Message, containing the parameter and its actual value is sent to the requesting SMC (addressed by the SIC / SAC code).

If the requested parameter does not exist on the MSSR 2000 ID the MSSR 2000 ID will send an Alarm Message "Parameter not available".

#### 4.11.6.5 Get crypto status, Key status and TOD status

For a detailed description see /MSSR ICD AI - Annex/.

|                                                |                     |                                             |                |                   |
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#### 4.11.7 Parameter Management

##### 4.11.7.1 General

To provide a monitoring and control system with a comprehensive set of information and capabilities, the SMC interface of MSSR 2000 ID supports access to all configuration and status information of Level 0, 1, 2 and 3 (Level 0 does not need password).

All parameters are identified by a 16 bit encoded information (parameter number). This parameter number is transferred via the data item I253/080 (Application Data Structure) as the field "START INDEX".

Internally this identifier links to the internal parameter identification.

The following parameters are used with the following message types:

- 8 = Command Message
- 10 = Delta Status Transfer Message
- 19 = Data Request

Command Messages are used to set the selected parameter (see figure **Transactions - Set Parameter**).

Data Request Messages are used to read the contents of the selected parameter. The MSSR 2000 ID will answer to a Data Request Message with a Delta Status Transfer Message (see figure **Transactions - Get Parameter**).

To avoid system overload situations during the request of the selected parameter, the total amount of requests per seconds is limited to 10 per client.

The Delta Status Transfer Message is used by the MSSR 2000 ID to report status changes within the MSSR 2000 ID to the connected stations.

After a change in the status, Delta Status Transfer Messages will be automatically transmitted. The Complete Status Transfer Message (9) and the Complete Status Transfer Request Message (11) are not defined yet.

Some parameters do not appear anymore within this document because new more efficient parameters have replaced them.

##### 4.11.7.2 General Parameter Properties

Each parameter has a set of properties which are common to all types of Parameter:

###### Use of Parameter

Configuration:

All these parameters are stored internal and persistent in terms of changes.

All Configuration Parameter from this type.

This type is although used for some Operational and Diagnostic Parameter.

Commands:

Used for temporary actions.

This type is used for Operational Parameter.

Status:

Used for current system status and performance information.

This type is used for Diagnostic Parameter.

###### Parameter ID

A symbolic identifier which is used to identify the parameter. The Parameter ID is used as heading or/and assigned to the unique parameter number.

###### Number

A unique number (16 bit) which is used internally to identify the parameter / value (corresponding to START INDEX of I253/080 (see chapter: Data Item I253/080)

###### Synchronise

|                                                |            |                  |          |           |
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"Synchronise" stands for synchronisation, meaning that the parameter value is transferred from the Master to the Slave during the synchronisation process. This indicates the default value. It is possible to override the default value by an internal configuration file (class\_param.txt)

#### Access Level

Defines the access of the parameters:

r/w read and write access  
 read only read access  
 write only write access (it is not possible to read the command back)

Defines the access level which is necessary to have write access to this parameter:

Level 0 no password is required

Note:

Functional roles with additional password protection can be defined for the CMS (e.g. ROSA), where the access rights to the different operational parameters can be managed.

Level 1 password level 1 is required

Level 2 password level 2 is required

Level 3 password level 3 is required

Note: For read access is no password required, except the PASSWORD parameter.

#### Time of Activation (Update)

Defines the point in time when a change of the parameter becomes effective.

Restart:

The change becomes effective after a Power Up or Restart of the MSSR 2000 ID.

Status Change:

The update becomes effective with the transition from STANDBY to TRANSMIT

Always:

(Default) If not otherwise stated the change of the parameter becomes effective with the reception of the new value (about 100 ms).

#### 4.11.7.3 Parameter Properties for Application Data

##### Data Type

The values transferred via the item I253/100 (Transparent Application Data 1) can have the following Data Types (see chapter: Data Item I253/100):

- UINT\_8 unsigned integer 8 bit wide
- INT\_8 signed integer 8 bit wide
- UINT\_16 unsigned integer 16 bit wide
- INT\_16 signed integer 16 bit wide
- UINT\_32 unsigned integer 32 bit wide
- INT\_32 signed integer 32 bit wide
- REAL floating point value
- TEXT Text string (8 bit ASCII)

The values transferred via the item I253/110 (Transparent Application Data 2) may be a structure which is defined in detail at the according chapter.

##### Value(s)

Defines allowed values of the Data (Enumeration, Min/Max, Default, Unit, ...)

|                                                |            |                  |          |           |
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#### 4.11.7.4 General Parameter Classification

##### Operational Parameters

Operational Parameters are system parameters which are necessary to operate the MSSR 2000 ID.

##### Configuration Parameters

Configuration Parameters are necessary to adapt the MSSR 2000 ID site and customer specific. It is possible to set and read the value of all these parameters.

All these parameters (except ACTUAL\_TIME) are persistent in terms of changes even if the system is powered down.

##### Diagnostic Parameters

Diagnostic Parameters are momentary values automatically determined by the system which characterise the current system conditions.

Examples: load conditions etc...

|                                                |            |                  |          |           |
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#### 4.11.8 Description of Operational Parameters

##### 4.11.8.1 TX\_ON

This message commands the MSSR 2000 ID the operational states TX or STANDBY.

Note: This message is acknowledged by the system with a delta status transfer message of type MSR\_STATE, EXTENDED\_SYS\_STATE

|                                                           |                                                         |
|-----------------------------------------------------------|---------------------------------------------------------|
| I253/080                                                  |                                                         |
| Number                                                    | 0x0001                                                  |
| Synchronise                                               | yes                                                     |
| Access                                                    | write Level 0                                           |
| Content of Item I253/100 - Transparent Application Data 1 |                                                         |
| DataType                                                  | UINT_8                                                  |
| Value                                                     | 0x00: Command system state STANDBY to stop transmission |
| Value                                                     | 0x01: Command system state TX to transmit               |

Content of Item I253/110 - Transparent Application Data 2  
This item is not used for this message.

##### 4.11.8.2 CALIBRATION

This command starts several calibration actions. During calibration process, the system state is switched to CALIBRATION.

Note: This message is acknowledged by the system with a delta status transfer message of type MSR\_STATE, EXTENDED\_SYS\_STATE

|                                                           |                                                                                          |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------|
| I253/080                                                  |                                                                                          |
| Number                                                    | 0x0003                                                                                   |
| Synchronise                                               | no                                                                                       |
| Access                                                    | write Level 2                                                                            |
| Content of Item I253/100 - Transparent Application Data 1 |                                                                                          |
| DataType                                                  | UINT_8                                                                                   |
| Value                                                     | 0x00: Calibration OFF                                                                    |
| Value                                                     | 0x01: Start monopulse Calibration with FFM                                               |
| Value                                                     | 0x02: Start monopulse Calibration with real targets (live traffic)                       |
| Value                                                     | 0x03: Start phase adjustment with real targets (only configurations higher than 9.01.03) |
| Value                                                     | 0x04: Start transmitter calibration                                                      |
| Value                                                     | 0x05: Start receiver calibration                                                         |
| Value                                                     | 0x06: Start BIT power threshold calibration (for transmitter)                            |
| Values                                                    | 0x07 - 0xFF reserved for future use                                                      |

Content of Item I253/110 - Transparent Application Data 2  
This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.8.3 BITE

This command starts the BITE of MSSR 2000 ID. When the BITE test is started, the Interrogator is set in a state which does not allow any operational interrogations.

Note: This message is acknowledged by the system with a delta status transfer message of type MSR\_STATE, EXTENDED\_SYS\_STATE

|                                                           |                                                                                              |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------|
| I253/080                                                  |                                                                                              |
| Number                                                    | 0x0006                                                                                       |
| Synchronise                                               | no                                                                                           |
| Access                                                    | write Level 0                                                                                |
| Content of Item I253/100 - Transparent Application Data 1 |                                                                                              |
| DataType                                                  | UINT_8                                                                                       |
| Value                                                     | 0x00: no action (note the execution of the BITE cannot be interrupted by setting this value) |
| Value                                                     | 0x01: Start the BITE of the MSSR 2000 I                                                      |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

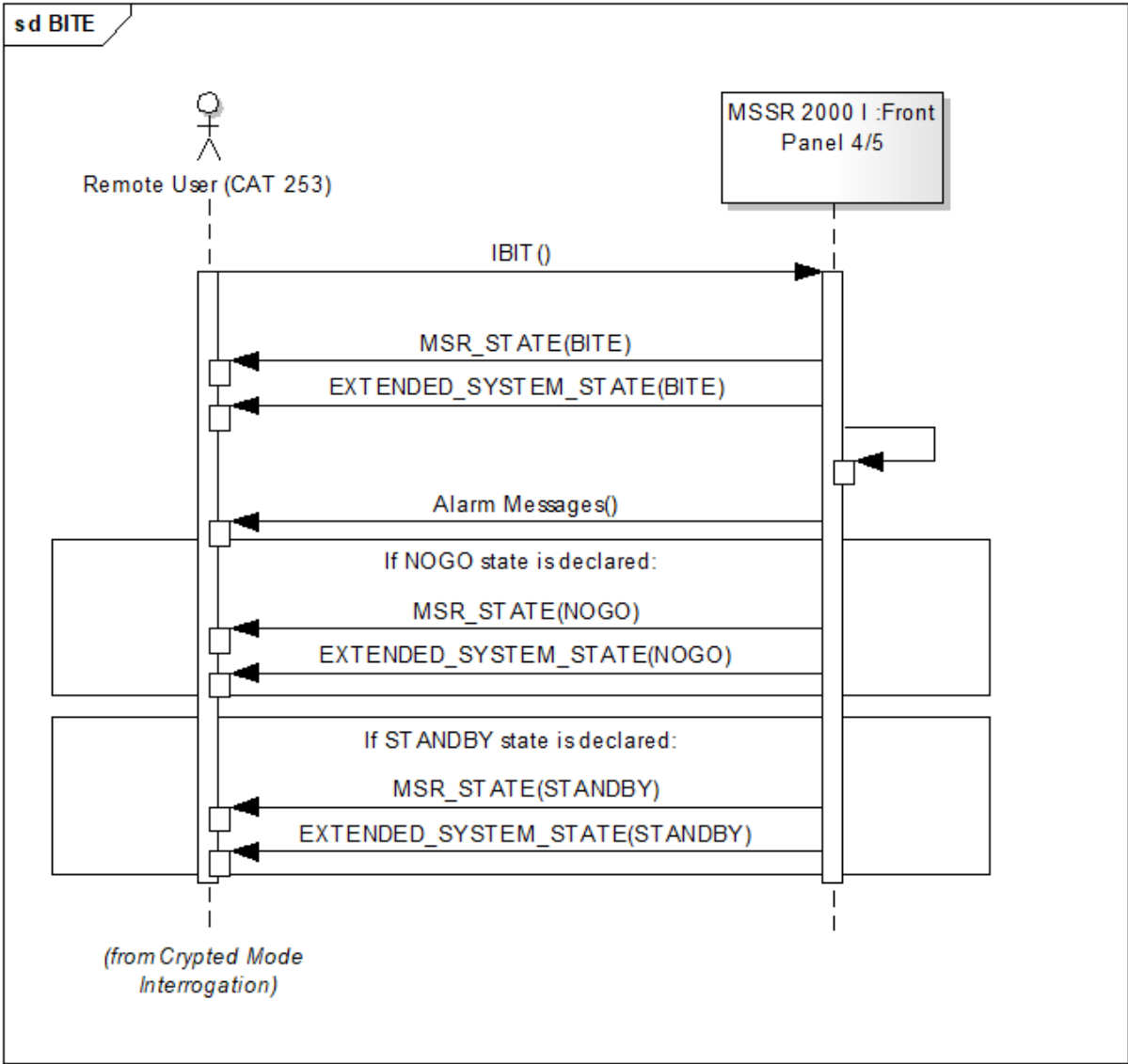
Note:

The BITE Bit is temporary only. After the action has been performed by the MSSR 2000 ID, this Bit will be reset automatically.

The purpose of the BITE test is to enable a failure recognition and failure localisation (IBIT-Initiated BIT). The following diagram shows an example of the sequence of resulting messages:

*Figure 4-12. IBIT and Localisation*

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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|                                                |            |                  |          |           |
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#### 4.11.8.4 RESTART

This command restart the MSSR 2000 ID.

The MSSR 2000 ID kills all application SW and performs a restart of the SW. All connections to connected devices are closed. After about 30s the restart procedure is completed, external connections e.g. for CAT 253, can be established again.

|             |               |
|-------------|---------------|
| I253/080    |               |
| Number      | 0x0007        |
| Synchronise | no            |
| Access      | write Level 0 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------|
| DataType | UINT_8                                                                                          |
| Value    | 0x00: no action (note the execution of the RESTART cannot be interrupted by setting this value) |
| Value    | 0x01: RESTART the MSSR 2000 I                                                                   |

Notes:

- 1) The RESTART is temporary only. After the action has been performed by the MSSR 2000 ID, this Bit will be reset automatically.
- 2) To avoid unintentional use of this command the remote control station shall install security measures such as additional confirmation of the command. Only after confirmation by the user the command shall be sent to the MSSR 2000 ID.
- 3) The acceptance of a RESTART command will be indicated by an alarm message (warning).

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                         |          |           |
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#### 4.11.8.5 SHUT DOWN

This message commands the MSSR 2000 ID to turn off the application SW. This ensures that the application SW is terminated in an appropriate manner (i.e open config files are closed, etc...). The operating system is not shut down through this command.

Besides this message causes the setting of a particular register on the adaptation board which makes possible afterwards the immediate "turn-off" of power.

|                                                           |                                                                                                   |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| I253/080                                                  |                                                                                                   |
| Number                                                    | 0x0008                                                                                            |
| Synchronise                                               | no                                                                                                |
| Access                                                    | write Level 0                                                                                     |
| Content of Item I253/100 - Transparent Application Data 1 |                                                                                                   |
| DataType                                                  | UINT_8                                                                                            |
| Value                                                     | 0x00: no action (note the execution of the SHUT DOWN cannot be interrupted by setting this value) |
| Value                                                     | 0x01: SHUT DOWN the MSSR 2000 I                                                                   |

Note:

- 1) The SHUT DOWN is temporary only. After the action has been performed by the MSSR 2000 ID, this Bit will be reset automatically.
- 2) To avoid unintentional use of this command the remote control station shall install security measures such as additional confirmation of the command. Only after confirmation by the user the command shall be sent to the MSSR 2000 ID.

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.8.6 Silence Sectors, North Referenced

This parameter sets or resets silent sectors.

It defines number of sectors and start/stop values in degrees with an increment of 1 degree. Within the defined silence sectors no interrogations are sent.

The reference for silence sectors is true north.

There is one set of sectors available at the MSSR 2000 ID. If the user should have the choice of several predefined sets of silent sectors, this shall be implemented at the remote control station.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x02D6      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|         |                                         |           |
|---------|-----------------------------------------|-----------|
| 1. Byte | Overall Length of Item                  | = 3 + N*4 |
| 2. Byte | Number of silent sector (high byte) (N) |           |
| 3. Byte | Number of silent sector (low byte) (N)  |           |

|                                                |            |                  |          |           |
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|                    |                                    |  |
|--------------------|------------------------------------|--|
| (4 + (N-1)*4) Byte | Start of sector in deg (high byte) |  |
| (5 + (N-1)*4) Byte | Start of sector in deg (low byte)  |  |
| (6 + (N-1)*4) Byte | End of sector in deg (high byte)   |  |
| (7 + (N-1)*4) Byte | End of sector in deg (low byte)    |  |

$0 \leq N \leq 63$

$0 \leq \text{"Start of Sector"} \leq 360$

$0 \leq \text{"End of Sector"} \leq 360$

If "Start Of Sector" > "End Of Sector" and sent to MSSR then the MSSR responds to client (on request) with splitted sector information as below:

Sector 1: "Start Of Sector" to 360°

Sector 2: "0°" to "End Of Sector"

LSB of Start of Sector and of End of Sector = 1°.

Examples

|                                    |  |                                                                |
|------------------------------------|--|----------------------------------------------------------------|
| Set one silent sector              |  | Contents of Item I253/110                                      |
| Start 1 = 10 deg, End 1 = 170 deg  |  | 07 00 01 00 0a 00 aa                                           |
| Set two silent sectors             |  | Contents of Item I253/110                                      |
| Start 1 = 10 deg, End 1 = 20 deg   |  | 0b 00 02 00 0a 00 14 00 1e 00 28                               |
| Start 2 = 30 deg, End 2 = 40 deg   |  |                                                                |
| Set four silent sectors            |  | Contents of Item I253/110                                      |
| Start 1 = 10 deg, End 1 = 20 deg   |  | 13 00 04 00 0a 00 14 00 1e 00 28 00 c8<br>01 5e<br>01 63 00 05 |
| Start 2 = 30 deg, End 2 = 40 deg   |  |                                                                |
| Start 3 = 200 deg, End 3 = 350 deg |  |                                                                |
| Start 4 = 355 deg, End 4 = 5 deg   |  |                                                                |
| Set zero / reset silent sectors    |  | Contents of Item I253/110                                      |
| Number of silent sector = 0        |  | 03 00 00                                                       |

#### Handling of Overlapping Sectors

Overlapping sectors are handled as one sector. The start of the sector will be the start of the first sector. The end of the sector will be the end of the last overlapping sector.

#### Risks in Using this Parameter

This parameter can inhibit the transmission of the MSSR 2000 ID if a resulting sector is defined which covers 360°.

#### 4.11.8.7 Silent Sectors, Bow Referenced

This parameter sets or resets silent sectors.

It defines number of sectors and start/stop values in degrees.

Within the defined silence sectors no interrogations are sent.

Silence sectors referenced to the bow in case of a mobile platform.

There is one set of sectors available at the MSSR 2000 ID. If the user should have the choice of several predefined sets of silent sectors, this shall be implemented at the remote control station.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x06CE      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2  
See Silence Sectors, North Referenced

#### 4.11.8.8 Silent Sectors, Dual Rotator Mode

This parameter set or reset silent sectors that are active in Dual Rotator Mode additional to SILENCE\_SECTOR and SILENCE\_SECTOR\_BOW.  
Within the defined silence sectors no interrogations are sent.

Silence sectors referenced to the bow in case of a mobile platform.

Note: To configure a surveillance sector for dual rotator operation, the inverse of this sector must be configured as dual-rotator silence sector.

There is one set of sectors available at the MSSR 2000 ID. If the user should have the choice of several predefined sets of silent sectors, this shall be implemented at the remote control station.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x06CF      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2  
See Silence Sectors, North Referenced

#### 4.11.8.9 Sector Handling for Interrogations

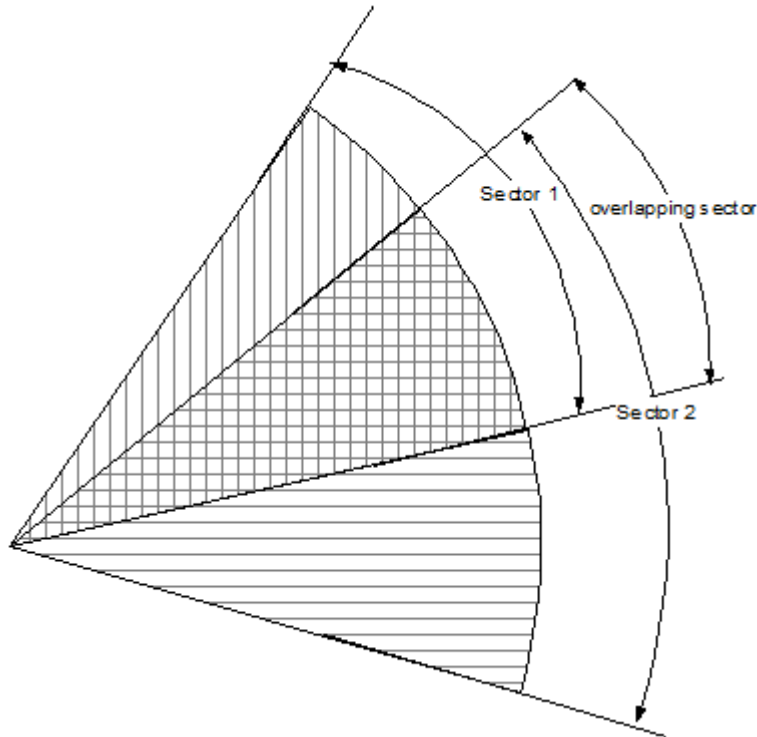
##### Handling of Overlapping Sectors

Overlapping Sectors are handled as follows:

The start of a new sector overwrites the old sector. The size of the old sector is reduced by the size of the overlapping area (see Figure 4-13). Overlapping rules for interrogation sectors also apply to overlapping directed interrogations.

*Figure 4-13. Handling of Overlapping Sectors*

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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The prioritisation of interrogations is from highest to lowest:

1. Silent sectors are not interrupted by any other type of interrogation.
2. Directed interrogations are performed in any defined or non defined sector (except silent) without respect to underlying normal sectors.

**Note:**

Within a silent sector the DI interrogation are not sent. This means that when a DI is completely within a silent sector, the REJECT comes (no FINISHED) with the error condition "UNABLE to PROCESS". If the interrogation could be partly sent, then the FINISHED message comes.

Whether target messages are seen or not, depends if enough hits received. In a silent Sector no power is transmitted and in the overlapping the replays are missing and no target may be generated.

**3. Sector Interrogation**

**Note:**

In the case of overlapping of normal sectors the one with the larger azimuth start (turnwise later, clockwise turning is expected) has priority over the previous sector. In the case the overlapping ends before the underlying sector, the underlying is continued.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.8.10 SCAN\_SEQ\_PROGRAM

The purpose of this parameter is to assign predefined interrogation programs (including the interrogation pattern) to a sector and scan based interrogation scheme.

This parameter substitutes the SECTOR\_MODE parameter.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0345      |
| Synchronise | yes         |
| Access      | r/w Level 0 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|                                            |                                             |                                                |
|--------------------------------------------|---------------------------------------------|------------------------------------------------|
| 1.Byte                                     | Overall length of item                      | OctN                                           |
| 2.Byte                                     | Total number of defined scans               | ScanN                                          |
| (3 + prevScanSize).Byte                    | Current scan number                         | curScan(x): 1-8                                |
| (4 + prevScanSize).Byte                    | Number of defined sectors in current scan   | SecN(x)                                        |
| (5 + prevScanSize + (curSec - 1) * 4) Byte | Start azimuth and End Azimuth of the sector | Start_End<br>Start: Bit 23/12<br>End: Bit 11/0 |
| (6 + prevScanSize + (curSec - 1) * 4) Byte |                                             |                                                |
| (7 + prevScanSize + (curSec - 1) * 4) Byte |                                             |                                                |
| (8 + prevScanSize + (curSec - 1) * 5).Byte | Program ID                                  | Prog_ID                                        |

ScanN = 0:

OctN = 2 (Byte 1)

ScanN = 0 (Byte 2)

no other Byte transferred

1 <= ScanN <= 8

OctN = 2 + fullScanSize

1 <= CurScan(x) <= 8

is the number of the current scan definition in the ASTERIX message. This value must start with 1 and must be incremented by one in every following scan definition. So

CurScan(x) = x.

0 <= SecN (x) <= 16

Every scan can have a different number of defined sectors.

So SecN(x) means number of defined sectors in scan x (1 <= x <= 8). A Sector is well-defined as the combination of curScan/curSec, where curScan is the position of the scan it is in (starting with 1) and curSec is the position of the sector in this scan (starting with 1). If SecN(x) of a scan is 0, no interrogations will be performed in this scan.

Start\_End

This field contains the start and end azimuth of the interrogation sector.

0 <= Sector\_Start <= 360, LSB = 1°

0 <= Sector\_End <= 360, LSB = 1°

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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If "Start Of Sector" > "End Of Sector" and sent to MSSR then the MSSR responds to client (on request) with splitted sector information as below:

Sector 1: "Start Of Sector" to 360°

Sector 2: "0°" to "End Of Sector"

For NESIS 4000, sectors can basically be set in the same way as described above with a step size of 1 degree. However, in the NESIS 4000 system, there are only 128 Beam Pointing Positions (BPPs) defined (corresponding to an ACP resolution of 7 bit), giving an azimuth resolution of 2.8125 degrees. The adjusted sectors are not automatically adjusted to the boundaries of BPPs, since the MSSR SW internally still works with 14bit ACP resolution. Since the TX beamwidth (about 6 deg) is larger than the width of one BPP anyway, a distinct change of settings at the boundary of adjacent BPPs is not possible anyway. So it may happen that the sector change (and with that the change of settings) takes place while pointing to one BPP.

The coding of the Start Azimuth and the End Azimuth of the sector are as follows.

| 1. Byte                 |     | 2. Byte |                       | 3. Byte |   |
|-------------------------|-----|---------|-----------------------|---------|---|
| Start Azimuth of Sector |     |         | End Azimuth of Sector |         |   |
| 23                      | ... | 12      | 11                    | ...     | 0 |
| MSB                     | LSB | MSB     | LSB                   |         |   |

LSB = 1°

Prog\_ID number of the selected interlace pattern (Interrogation Program)

$$\text{prevScanSize} = (\text{curScan} - 1) * 2 + 4 * \sum_{x=1}^{\text{curScan}-1} \text{SecN}(x)$$

$$\text{fullScanSize} = \text{ScanN} * 2 + 4 * \sum_{x=1}^{\text{ScanN}} \text{SecN}(x)$$

#### prevScanSize

Is the number of bytes, that are used by the scan definitions that are positioned before the current scan.

#### fullScanSize

Is the number of bytes, that are used by all scan definitions. It must not exceed 253 Bytes. So the maximum number of defined sectors in all scans must not exceed 59. The maximum number of sectors per scan is 16. The maximum number of scans is 8.

#### Notes:

- 1) The content of the Interlace Pattern (part of the interrogation programs) has to be defined offline and loaded into the interrogator before use.
- 2) If a not assigned Interlace Pattern is selected, the default Interlace Pattern will be used. If this default value is not available the command will be rejected.
- 3) The MSSR 2000 ID checks the plausibility of the parameter. If the parameters do not comply with the defined structure, the parameter will be rejected and alarm messages will be sent (E\_SMC\_INVALID\_PARAMETER\_VALUE).
- 4) A new definition will be active as soon as sent to the MSSR 2000 ID and the currently valid sequence of the former SCAN\_SEQ\_PROGRAM parameter has been completed (this time will last less than two times the antenna beam width).

#### 4.11.8.11 CRYPTO\_Control

For a detailed description see /MSSR ICD AI - Annex/.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.8.12 CHAIN\_SWITCH

This message commands a manual switchover from Master to Slave and vice versa.

This command is only processed if:

- it is transmitted to the master
- the AUTO/MANUAL switch is in position AUTO
- the slave is in redundant mode and is ready for operation (delivering of alive signals).
- 

This is a write only command.

If the command was processed successfully, it results in a delta status message from both Master and Slave, if no delta status message was received, the processing fails.

|                                                           |                                                                 |
|-----------------------------------------------------------|-----------------------------------------------------------------|
| I253/080                                                  |                                                                 |
| Number                                                    | 0x0010 for CHAIN_SWITCH                                         |
| Synchronise                                               | no                                                              |
| Access                                                    | write Level 1                                                   |
| Content of Item I253/100 - Transparent Application Data 1 |                                                                 |
| DataType                                                  | UINT_8                                                          |
| Value                                                     | 0x00: No Action                                                 |
| Value                                                     | 0x01: Switch over force switch MASTER ◇ SLAVE<br>SLAVE ◇ MASTER |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.8.13 MAINTENANCE\_SELECT

This message commands the Operational Modes of the MSSR 2000 ID.

This command is only available if the MSSR 2000 ID is a dual redundant variant (**not** for a Standalone System).

|             |                               |
|-------------|-------------------------------|
| I253/080    |                               |
| Number      | 0x0011 for MAINTENANCE_SELECT |
| Synchronise | no                            |
| Access      | write Level 1                 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                                                        |
|----------|----------------------------------------------------------------------------------------|
| DataType | UINT_8                                                                                 |
| Value    | 0x00: REDUNDANT Redundant Operate (if the AUTO/MAN switch is set to AUTO or own chain) |
| Value    | 0x01: MAINTENANCE Redundant Maintenance                                                |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.8.14 POWER\_MONITORING\_RESET

This message commands the MSSR 2000 ID to reset the POWER\_MONITORING.

The data containers for minimal, maximal and average power and for VSWR are cleared.

|             |                                   |
|-------------|-----------------------------------|
| I253/080    |                                   |
| Number      | 0x0414 for POWER_MONITORING_RESET |
| Synchronise | no                                |
| Access      | write Level 0                     |

Content of Item I253/100 - Transparent Application Data 1

|          |                           |
|----------|---------------------------|
| DataType | UINT_8                    |
| Value    | 0x00: RESET               |
| Value    | all other values not used |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.8.15 LOAD\_CFGSET

This commands the MSSR 2000 ID to load a certain configuration set (cfgset) to the current system parameter

|             |               |
|-------------|---------------|
| I253/080    |               |
| Number      | 0x0346        |
| Synchronise | yes           |
| Access      | write Level 1 |

Content of ITEM I253/100 - Transparent Application Data 1

|          |                                                                  |
|----------|------------------------------------------------------------------|
| DataType | UINT_8                                                           |
| Value    | 0: Default cfgset, cannot be overwritten                         |
| Values   | 1-9: User specific cfgsets, can be overwritten with save command |

Note:

The cfgset is checked during loading and the following Alarm Message from type WARNING is generated in case of a failure: E\_SMC\_INVALID\_PARAMETER\_VALUE

Content of ITEM I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.8.16 SAVE\_CFGSET

This commands the MSSR 2000 ID to save the current system parameter to a certain configuration set (cfgset).

|             |               |
|-------------|---------------|
| I253/080    |               |
| Number      | 0x0347        |
| Synchronise | yes           |
| Access      | write Level 1 |

Content of ITEM I253/100 - Transparent Application Data 1

|           |                                             |
|-----------|---------------------------------------------|
| Data Type | UINT_8                                      |
| Values    | 1-9: User specific cfgsets, can be reloaded |
| Value     | 10: Maintenance cfgset, always free to save |

Content of ITEM I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.8.17 Buzzer Control

This message commands the MSSR 2000 ID to:

- set the buzzer (a ringing buzzer is reset)
- enable the buzzer (the buzzer is enabled to ring)
- disable the buzzer (the buzzer is disabled, in no case the buzzer will ring)

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0002      |
| Synchronise | yes         |
| Access      | r/w Level 0 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                      |
|-----------|--------------------------------------|
| Data Type | UINT_8                               |
| Value     | 0x00: Buzzer enabled                 |
| Value     | 0x01: Buzzer disabled                |
| Value     | 0x02: Reset Buzzer / Buzzer enabled  |
| Value     | 0x03: Reset Buzzer / Buzzer disabled |

Note:

The Reset Bit (= Bit 1 of the value) is temporary only. After the action has been performed by the MSSR 2000 ID, this Bit will be reset automatically.

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.8.18 SUR\_PULSE\_SECTORS

This parameter defines the SUR pulse sectors which are north oriented. When this parameter is enabled (sectors are defined) then the system will perform a SUR pulse for all MkXII and Mode S all-call interrogations within these sectors - with those exclusions defined in the RS.

Note that either north-oriented (this parameter), or bow-oriented (parameter SUR\_PULSE\_SECTORS\_BOW) may be enabled, but not both.

For the sector definition, always the complete 360° are used.

The given sector boundaries are the stop values of the sector.

Sector one starts at 0° and ends with the given azimuth. The next sector starts with the stop azimuth of the previous sector and ends with the given azimuth. The last sector ends with 360°.

The sector stop values must be set that they are in ascending order of azimuth.

At least, there is always one sector defined.

Note:

- SUR Pulse = 0 indicates no SUR Pulse sector
- SUR Pulse = 1 indicates SUR Pulse sector - not for Directed Interrogations
- SUR Pulse = 2 indicates SUR Pulse sector - including Directed Interrogations
- In case of a change of the SUR PULS SECTOR (via SMC), an automatic Delta Status Transfer Message is sent as an acknowledgment under the SUR\_PULSE\_SECTORS format.

Rules to create a new sector:

$0^\circ < \text{current sector stop} \leq 360^\circ$

$\text{previous sector stop} < \text{current sector stop} < \text{next sector stop}$

the given "Stop of last Sector" is always 360°

value of SUR Pulse = as defined above

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x700       |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|                                  |                                       |               |
|----------------------------------|---------------------------------------|---------------|
| 1. Byte                          | Overall Length of Item                | $2 + (4 * S)$ |
| 2. Byte                          | Number of Sectors for SUR Pulse       | S             |
| 3. Byte until $(4*S + 2)$ . Byte | List of Definitions for the SUR Pulse |               |

S: Number of SUR Pulse sectors per 360°.

$1 \leq S \leq 16$ : 1 to 16 sectors are available.

|                                |                                   |  |
|--------------------------------|-----------------------------------|--|
| 1. Byte of a Sector definition | Stop of sector in deg (high byte) |  |
| 2. Byte of a Sector definition | Stop of sector in deg (low byte)  |  |
| 3. Byte of a Sector definition | SUR Pulse (high Byte)             |  |
| 4. Byte of a Sector definition | SUR Pulse (low Byte)              |  |

The LSB of "Stop of Sector" = 1°. Note, that the accuracy and resolution of the sector boundaries is 1°.

Example:

The following sectors shall be set:

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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Sector 1 from 0° to 30° with SUR Pulse (not for DIs)  
Sector 2 from stop of Sector 1 to 360° with No SUR Pulse

|                                     |  |                           |
|-------------------------------------|--|---------------------------|
| Set the SUR Pulse sector            |  | Contents of Item I253/110 |
| Overall length - 10 bytes           |  | 0A                        |
| Number of Sectors                   |  | 02                        |
| Stop of Sector 1 = 30° (high Byte)  |  | 00                        |
| Stop of Sector 1 = 30° (low Byte)   |  | 1D                        |
| set SUR Pulse (high Byte)           |  | 00                        |
| set SUR Pulse (low Byte)            |  | 01                        |
| Stop of Sector 2 = 360° (high Byte) |  | 01                        |
| Stop of Sector 2 = 360° (low Byte)  |  | 68                        |
| no SUR Pulse                        |  | 00                        |
| no SUR Pulse                        |  | 00                        |

#### 4.11.8.19 SUR\_PULSE\_SECTORS\_BOW

This parameter defines the SUR pulse sectors which are bow oriented. When this parameter is enabled (sectors are defined) then the system will perform a SUR pulse for all MkXII and Mode S all-call interrogations within these sectors - with those exclusions as defined in the RS.

Note that either north-oriented (parameter SUR\_PULSE\_SECTOR), or bow-oriented (this parameter) may be enabled, but not both.

For the sector definition, always the complete 360° are used.

The given sector boundaries are the stop values of the sector.

Sector one starts at 0° and ends with the given azimuth. The next sector starts with the stop azimuth of the previous sector and ends with the given azimuth. The last sector ends with 360°.

The sector stop values must be set that they are in ascending order of azimuth.  
At least, there is always one sector defined.

Note:

- SUR Pulse = 0 indicates no SUR Pulse sector
- SUR Pulse = 1 indicates SUR Pulse sector - not for Directed Interrogations
- SUR Pulse = 2 indicates SUR Pulse sector - including Directed Interrogations
- In case of a change of the SUR PULSE SECTOR (via SMC), an automatic Delta Status Transfer Message is sent as an acknowledgment under the SUR\_PULSE\_SECTORS format.

Rules to create a new sector:

0° < current sector stop ≤ 360°  
previous sector stop < current sector stop < next sector stop  
the given "Stop of last Sector" is always 360°  
value of SUR Pulse = as defined above

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x701       |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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This item is not used for this message.

#### Content of Item I253/110 - Transparent Application Data 2

|                                  |                                       |               |
|----------------------------------|---------------------------------------|---------------|
| 1. Byte                          | Overall Length of Item                | $2 + (4 * S)$ |
| 2. Byte                          | Number of Sectors for SUR Pulse       | S             |
| 3. Byte until $(4*S + 2)$ . Byte | List of Definitions for the SUR Pulse |               |

S: Number of SUR Pulse sectors per 360°.

$1 \leq S \leq 16$ : 1 to 16 sectors are available.

|                                |                                   |  |
|--------------------------------|-----------------------------------|--|
| 1. Byte of a Sector definition | Stop of sector in deg (high Byte) |  |
| 2. Byte of a Sector definition | Stop of sector in deg (low Byte)  |  |
| 3. Byte of a Sector definition | SUR Pulse (high Byte)             |  |
| 4. Byte of a Sector definition | SUR Pulse (low Byte)              |  |

The LSB of "Stop of Sector" = 1°. Note, that the accuracy and resolution of the sector boundaries is 1°.

Example:

The following sectors shall be set:

Sector 1 from 0° to 30° with SUR Pulse

Sector 2 from stop of Sector 1 to 360° with no SUR Pulse

| Set the SUR Pulse Bow sector        |  | Contents of Item I253/110 |
|-------------------------------------|--|---------------------------|
| Overall length - 10 bytes           |  | 0A                        |
| Number of Sectors                   |  | 02                        |
| Stop of Sector 1 = 30° (high Byte)  |  | 00                        |
| Stop of Sector 1 = 30° (low Byte)   |  | 1D                        |
| set SUR Pulse (high Byte)           |  | 00                        |
| set SUR Pulse (low Byte)            |  | 01                        |
| Stop of Sector 2 = 360° (high Byte) |  | 01                        |
| Stop of Sector 2 = 360° (low Byte)  |  | 68                        |
| no SUR Pulse (high Byte)            |  | 00                        |
| no SUR Pulse (low Byte)             |  | 00                        |

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.9 Description of Configuration Parameters

##### 4.11.9.1 PASSWORD

This parameter configures the system passwords. Note that read or write access requires the access level specified below.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0200      |
| Synchronise | yes         |
| Access      | r/w Level 3 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|                    |
|--------------------|
| Octet              |
| 1                  |
| 0x41               |
| Length of the data |

|                  |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |
|------------------|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|
| Octet            |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |
| 2                | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| x                | x | x | x | x | x | x | x | x  | x  | x  | x  | x  | x  | x  | x  |
| Password Level 0 |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |

|                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Octet            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 18               | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 |
| x                | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  |
| Password Level 1 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

|                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Octet            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 34               | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 |
| x                | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  |
| Password Level 2 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

|                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Octet            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 50               | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
| x                | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  | x  |
| Password Level 3 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

For password level 0 all 16 characters are field filled with "0x00" (= ASCII Value "NUL")

X = 0x00: not used fields for the password

X = ASCII- values for digits (0x30 - 0x39) as characters of the password

The passwords are left-aligned.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.9.2 SYSTEM TIME SOURCE

This parameter selects the external time source.

Note, it is not recommended to use GPS as an external time source. NTP is the preferred time source.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0205      |
| Synchronise | yes         |
| Access      | r/w Level 0 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                           |
| Value     | 0x00: not used<br>0x01: use GPS as external time source<br>0x02: use NTP as external time source |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.3 SYSTEM TIME

This parameter sets or reads the system date and time for the internal clock.

It is recommended to use this command only if no external time source is used. Otherwise the external time source overwrite the just configured time immediately and results leap in time.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0320      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|         |                        |             |
|---------|------------------------|-------------|
| 1. Byte | Overall Length of Item | = 8         |
| 2. Byte | Year (high byte)       |             |
| 3. Byte | Year (low byte)        |             |
| 4. Byte | Month                  | 0x01 - 0x0C |
| 5. Byte | Day                    | 0x01 - 0x1F |
| 6. Byte | Hour                   | 0x00 - 0x17 |
| 7. Byte | Minute                 | 0x00 - 0x3B |
| 8. Byte | Second                 | 0x00 - 0x3B |

The time zone used is UTC.

Time values which are not valid are ignored and the following Alarm Message from type WARNING is generated: E\_SMC\_INVALID\_PARAMETER\_VALUE

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.4 ANTENNA LONGITUDE

This parameter defines the longitude position (WGS 84) of the radar head.

Note: The configset parameter corresponding to this Cat253 message does NOT use the same unit, cf. /Parameter/

**Note:** If Bit 4 in parameter MIN\_CFG\_NAV\_MOVING\_PLATFORM is set, the CAT253 command will be ignored. Setting of longitude via Cat253 is not allowed in this case.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0209      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                            |
|----------|------------------------------------------------------------|
| DataType | REAL                                                       |
| Values   | $-\pi \leq N \leq \pi$                                     |
| Values   | A positive value is "east".<br>A negative value is "west". |
| Unit     | Radians                                                    |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.5 ANTENNA LATITUDE

This parameter defines the latitude position (WGS 84) of the radar head.

Note: The configset parameter corresponding to this Cat253 message does NOT use the same unit, cf. /Parameter/

**Note:** If Bit 4 in parameter MIN\_CFG\_NAV\_MOVING\_PLATFORM is set, the CAT253 command will be ignored. Setting of latitude via Cat253 is not allowed in this case.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x020A      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                              |
|----------|--------------------------------------------------------------|
| DataType | REAL                                                         |
| Values   | $-\frac{\pi}{2} \leq N \leq \frac{\pi}{2}$                   |
| Values   | A positive value is "north".<br>A negative value is "south". |
| Unit     | Radians                                                      |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.6 ANTENNA ALTITUDE

This parameter defines the altitude position of the radar head (phase centre SSR antenna). It is recommended to use the height relating to WGS 84 reference ellipsoid as defined for I034/120. To adapt the system to site specific constraints it is possible to use the altitude above mean sea level. The value is used to define the 3D-Position of Data Source as reported in I034/120, if not overwritten by connected navigation or GPS system.

Note:

When this altitude is added manually inside the config set, it should be the altitude above mean sea level of the phase centre of the SSR antenna connected to the interrogator.

This parameter is automatically updated with data from the GPS receiver when it changes by at least 5m, at the latest every 10 minutes if nav data processing is disabled and USE\_GPS\_POSITION is enabled.

**Note:** This parameter is used to set the stationary antenna position of a land based system once. The cycli setting of the position is not permitted and cannot be used as a substitute for a NAV interface.

Note: The maximal operating altitude is 5000 m, due to low pressure.

**Note:** If Bit 4 in parameter MIN\_CFG\_NAV\_MOVING\_PLATFORM is set, the CAT253 command will be ignored. Setting of altitude via Cat253 is not allowed in this case.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x020B      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

#### Content of Item I253/100 - Transparent Application Data 1

|          |         |
|----------|---------|
| DataType | REAL    |
| Min      | -1000.0 |
| Max      | 10000.0 |
| Unit     | meter   |

#### Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.7 NOMINAL ATT

This parameter defines the nominal Antenna Turn Time of the connected antenna.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x020F      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

#### Content of Item I253/100 - Transparent Application Data 1

|          |        |
|----------|--------|
| DataType | REAL   |
| Min      | 0.7    |
| Max      | 30.0   |
| Default  | 10.0   |
| Unit     | second |

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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Content of Item I253/110 - Transparent Application Data 2  
This item is not used for this message.

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|                                                |            |                  |          |           |
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#### 4.11.9.8 ANTENNA RANGE BIAS

This parameter defines the range bias of the system.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0213      |
| Synchronise | no          |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                   |
|----------|-----------------------------------|
| DataType | REAL                              |
| Values   | Range Bias in nautical miles [NM] |
| Min      | -10.0                             |
| Max      | 10.0                              |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.9 ANTENNA NORTH BIAS

This parameter defines the antenna north bias.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0214      |
| Synchronise | no          |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|          |                         |
|----------|-------------------------|
| DataType | REAL                    |
| Values   | azimuth bias in deg [°] |
| Min      | -10.0                   |
| Max      | 10.0                    |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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**4.11.9.10 ANTENNA TYPE MP**

This parameter defines the monopulse capability of the antenna. If the Antenna is not Monopulse capable, the default value must be changed to 0.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0217      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

**Content of Item I253/100 - Transparent Application Data 1**

|          |                                       |
|----------|---------------------------------------|
| DataType | UINT_8                                |
| Value    | 0x00: no monopulse<br>0x01: monopulse |

**Content of Item I253/110 - Transparent Application Data 2**

This item is not used for this message.

|                                                |            |                         |          |           |
|------------------------------------------------|------------|-------------------------|----------|-----------|
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#### 4.11.9.11 ASTERIX OUTPUT CONTROL: NORTH MARKER MESSAGES

Enables/disables the output of the following ASTERIX Messages at LAN interface (TCP Port 1, 2, 3 and UDP Port 1, 2, 3):

North Marker Message

Note, in case of moving platform, or if the ACP/ACR pulses are not north oriented: If the sector messages at the selected port are bow oriented, then the north message is also bow oriented (and vice versa). If no sector message is enabled, the north message is disabled!

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0322      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                         |          |           |
|------------------------------------------------|------------|-------------------------|----------|-----------|
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#### 4.11.9.12 ASTERIX OUTPUT CONTROL: SECTOR CROSSING MESSAGES

Enables/disables the output of the following ASTERIX Messages at LAN interface (TCP Port 1, 2, 3 and UDP Port 1, 2, 3):

Sector Crossing Message (north orientated)

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0323      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.13 ASTERIX OUTPUT CONTROL: RAW PLOTS

Enables/disables the output of the following ASTERIX Messages at LAN interface (TCP Port 1, 2, 3 and UDP Port 1, 2, 3):

Raw Plot Message (bow orientated)

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0324      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.14 ASTERIX OUTPUT CONTROL: PROCESSED PLOTS

Enables/disables the output of the following ASTERIX Messages at LAN interface (TCP Port 1, 2, 3 and UDP Port 1, 2, 3):

Processed Plot Message (north or bow orientated)

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0325      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.15 ASTERIX OUTPUT CONTROL: TRACKS

Enables/disables the output of the following ASTERIX Messages at LAN interface (TCP Port 1, 2, 3 and UDP Port 1, 2, 3):

Track Message (north orientated)

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0326      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.16 ASTERIX OUTPUT CONTROL: JAMMING MESSAGES

Enables/disables the output of the following ASTERIX Messages at LAN interface (TCP Port 1, 2, 3 and UDP Port 1, 2, 3):

Jamming Message

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0327      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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**4.11.9.17 ASTERIX OUTPUT CONTROL: ADS-B MESSAGES**

Enables/disables the output of the ADS-B plots over ASTERIX CAT 021.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0328      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

**Content of Item I253/100 - Transparent Application Data 1**

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

**Content of Item I253/110 - Transparent Application Data 2**

This item is not used for this message.

|                                                |            |                         |          |           |
|------------------------------------------------|------------|-------------------------|----------|-----------|
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#### 4.11.9.18 ASTERIX OUTPUT CONTROL: MODE 5 LEVEL 2 PLOTS

Enable and disable the output of the Mode 5 Level 2 raw plots over ASTERIX CAT 048.

Note, starting from config 9.01, the Mode 5 Level 2 tracker is implemented. When Mode 5 Level 2 raw plots (AST\_TX\_M5\_LVL2) are output at the same port as tracks (AST\_TX\_TRACK) or processed plots (AST\_TX\_PLOT), Mode 5 Level 2 targets will be output multiple times on the same port. To avoid that, configure the output of Mode 5 Level 2 raw plots only on ports which do not output processed plots or tracks.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0329      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                           |
| Value     | 00: No output<br>01: output on Port 1<br>02: output on Port 2<br>03: output on Port 1 and 2<br>04: output on Port 3<br>05: output on Port 1 and 3<br>06: output on Port 2 and 3<br>07: output on Port 1, 2 and 3 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.19 ASTERIX TX NO ALLCALL

The purpose of this message is to set or read the setting of the AST\_TX\_NO\_ALLCALL parameter. The output off Mode-S all calls can be disabled by setting this parameter to "1".

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0758      |
| Synchronise | yes         |
| Access      | r/w Level 3 |

Content of Item I253/100 - Transparent Application Data 1

|          |                               |
|----------|-------------------------------|
| DataType | UINT_16                       |
| Value    | 0x00: Transmit all targets    |
| Value    | 0x01: Transmit only Roll-Call |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.20 STT DISABLE

The purpose of this message is to set or read the setting of the STT\_ENABLE parameter. The generation of Test Targets can be enabled or disabled.

NOTE: The message STT\_DISABLE is logically inverted with respect to the parameter STT\_ENABLE

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x032C      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------|
| DataType | UINT_8                                                                                             |
| Value    | 0x00: STT not disabled (parameter STT_ENABLE = 1)<br>0x01: STT disabled (parameter STT_ENABLE = 0) |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.21 INTERR JITTER MODE

The purpose of this message is to set or read the setting of the Interrogation Jitter. The Jitter mode can be selected.

|             |               |
|-------------|---------------|
| Update      | Status Change |
| I253/080    |               |
| Number      | 0x021E        |
| Synchronise | yes           |
| Access      | r/w Level 0   |

##### Content of Item I253/100 - Transparent Application Data 1

|          |                                                                                              |
|----------|----------------------------------------------------------------------------------------------|
| DataType | UINT_8                                                                                       |
| Value    | 0x00: no jitter selected<br>0x01: sequential jitter selected<br>0x02: random jitter selected |

##### Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.9.22 INTR MODE ENABLE

This parameter enables or disables modes for interrogation.  
Only enabled modes can be used in the interrogation programs.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x02F2      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|              |                         |         |
|--------------|-------------------------|---------|
| 1. Byte      | Overall length of Item  | = 2 + N |
| 2. Byte      | Number of Modes Enabled |         |
| (2 + N) Byte | Enabled Modes           |         |

$$0 \leq N \leq 12$$

Codes for the Modes:

- 01 = 0x01 = Mode 1
- 02 = 0x02 = Mode 2
- 03 = 0x03 = Mode 3/A
- 04 = 0x04 = Mode 4
- 05 = 0x05 = Mode B (unused)
- 06 = 0x06 = Mode C
- 07 = 0x07 = Mode D (unused)
- 08 = 0x08 = Supermode
- 09 = 0x09 = Mode S
- 10 = 0x0a = Intermode A/C-only All-Call
- 11 = 0x0b = (unused)
- 12 = 0x0c = Mode 5

The "Mode 4" and "Supermode" must be enable/disable at the same time.

Note: The Mode 4 status (e.g., Enabled Crypto Modes and Alarm Status) changes due to enable/disable of "Mode 4 and Supermode".

Examples:

|                                       |                           |
|---------------------------------------|---------------------------|
|                                       | Contents of Item I253/110 |
| No mode is enabled                    | 02 00                     |
| Mode 1, Mode 2 and Mode C are enabled | 05 03 01 02 06            |

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.9.23 II/SI SECTORS

II\_SI\_SECTORS defines azimuth areas in which the II or SI code is used.

For the sector definition, always the complete 360° are used.

The given sector boundaries are the stop values of each sector.

Sector one starts at 0° and ends with the given azimuth. The next sector starts with the stop azimuth of the previous sector and ends with the given azimuth. The last sector ends with 360°. This means that with this parameter - 360° is always covered - even if the last defined sector does not explicitly stop at 360.

The sector stop values must be set that they are in ascending order of azimuth. At least, there is always one sector defined.

The 16 bit IISl code value is defined as follows:

IISl, bit 15..8: SI code

IISl, bit 7..0: II code

The IISl code value must be entered in hexadecimal. See the examples below.

Notes:

- If SI = 0, the II Code is used.
- If SI > 0, the SI code is used, regardless of the II code value.
- When this parameter is enabled (sectors are defined) then it has priority over the II\_CODE and SI\_CODE parameters.
- In case of a change of the II Code or SI Code (through the HMI or via SMC), an automatic Delta Status Transfer Message (cf ICD Asterix part 4.11.4.8) is sent as an acknowledgment under the II\_SI\_SECTORS format.

Rules to create a new sector:

0° < current sector stop ≤ 360°

previous sector stop < current sector stop < next sector stop

the given "Stop of last Sector" is always 360°

0 = SI if SI = 0 then the II Code is used, else the SI Code is used

1 ≤ SI ≤ 63

0 ≤ II ≤ 15

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x02D7      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|                               |                                           |             |
|-------------------------------|-------------------------------------------|-------------|
| 1. Byte                       | Overall Length of Item                    | 2 + (4 * S) |
| 2. Byte                       | Number of Sectors for SI/II Code          | S           |
| 3. Byte until (4*S + 2). Byte | List of Definitions for the SI/II Sectors |             |

S: Number of II or SI Codes sectors per 360°.

1 ≤ S ≤ 16: 1 to 16 sectors are available.

|                                |                                   |  |
|--------------------------------|-----------------------------------|--|
| 1. Byte of a Sector definition | Stop of sector in deg (high byte) |  |
| 2. Byte of a Sector definition | Stop of sector in deg (low byte)  |  |
| 3. Byte of a Sector definition | SI Code                           |  |
| 4. Byte of a Sector definition | II Code                           |  |

The LSB of "Stop of Sector" = 1°. Note, that the accuracy and resolution of the sector boundaries is 1°.

|                                                |            |                  |          |           |
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Example:

The following sectors shall be set:

Sector 1 from 0° to 30° with II Code 14

Sector 2 from stop of Sector 1 to 360° with SI Code 1

|                                     |  |                           |
|-------------------------------------|--|---------------------------|
| Set the II SI Code sectorised       |  | Contents of Item I253/110 |
| Overall length - 10 bytes           |  | 0A                        |
| Number of Sectors                   |  | 02                        |
| Stop of Sector 1 = 30° (high Byte)  |  | 00                        |
| Stop of Sector 1 = 30° (low Byte)   |  | 1D                        |
| SI Code (not used - must be 0)      |  | 00                        |
| II Code                             |  | 0D                        |
| Stop of Sector 2 = 360° (high Byte) |  | 01                        |
| Stop of Sector 2 = 360° (low Byte)  |  | 68                        |
| SI Code = 1                         |  | 01                        |
| II Code (don't care if SI is used)  |  | XX                        |

|                                                |                     |                                             |                |                   |
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#### 4.11.9.24 II CODE

This parameter defines the II (Interrogator Identifier) Code of the site.

This parameter is substituted by II\_SI\_SECTORS but still supported for compatibility.

It is recommended to use II\_SI\_SECTORS instead of this parameter.

Note:

- The usage of the II Code (between 0 and 15) is activated when SI Code is equal to 0.
- The parameter II\_CODE is used together with the parameter II\_SI\_SECTORS. If the II Code is defined sector-wise:
  - this parameter is not readable.
  - this parameter writes the II Code to one sector from 0° to 360° and the following sectors deleted. Therefore it is recommended to use II\_CODE and II\_SI\_SECTORS exclusively.
- If a coverage map is defined, this parameter is read only and not synchronised with the Slave chain. The II\_CODE is set automatically by the selected solution within the coverage map
- In case of a change of the II Code (through the HMI or via SMC), an automatic Delta Status Transfer Message is sent as an acknowledgment under the II\_SI\_SECTORS format.

|             |                            |
|-------------|----------------------------|
| Update      | Status Change              |
| I253/080    |                            |
| Number      | 0x02D2                     |
| Synchronise | no, if cluster map is used |
| Access      | r/w Level 1                |

Content of Item I253/100 - Transparent Application Data 1

|          |                                 |
|----------|---------------------------------|
| DataType | UINT_8                          |
| Values   | $0 \leq \text{II Code} \leq 15$ |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.25 SI CODE

This parameter defines the SI (Surveillance Identifier) Code of the site.

This parameter is substituted by II\_SI\_SECTORS but still used in the Asterix ICD for compatibility.

It is recommended to use II\_SI\_SECTORS instead of this parameter.

Note:

- SI = 0 indicates, that no SI Code has been assigned to the Interrogator. The II Code is used.
- The SI Code is defined from 1 to 63.
- The parameter SI\_CODE is used together with the parameter II\_SI\_SECTORS. If the SI Code is defined sector-wise:
  - this parameter is not readable.
  - this parameter writes the SI Code to one sector from 0° to 360° and the following sectors deleted. Therefore it is recommended to use SI\_CODE and II\_SI\_SECTORS exclusively.
- If a coverage map is defined, this parameter is read only and not synchronised with the Slave chain. The SI\_CODE is set automatically by the selected solution within the coverage map
- In case of a change of the SI Code (through the HMI or via SMC), an automatic Delta Status Transfer Message (cf part 4.11.4.8) is sent as an acknowledgment under the II\_SI\_SECTORS format.

|             |                            |
|-------------|----------------------------|
| Update      | Status Change              |
| I253/080    |                            |
| Number      | 0x02D3                     |
| Synchronise | no, if cluster map is used |
| Access      | r/w Level 1                |

Content of Item I253/100 - Transparent Application Data 1

|          |                  |
|----------|------------------|
| DataType | UINT_8           |
| Values   | 0 ≤ SI Code ≤ 63 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.26 OP MODE SYNCH

This parameter defines if the interrogation control operates autonomously or if it is synchronised with an external trigger (e.g. Primary Radar).

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x02F1      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------|
| DataType | UINT_8                                                                                           |
| Values   | 0x00: NO_SYNCH_OPERATION (autonomous operation)<br>0x01: SYNCH_OPERATION (pre trigger operation) |
| Default  | 0x00                                                                                             |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
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#### 4.11.9.27 MAX RANGE

This parameter defines the Maximum instrumented range of the sensor.

Detection with a larger slant range is impossible and will be discarded. It is also used in various polar maps within the Tracking SW.

This parameter is thus an upper limit on the range of the sensor but the instantaneous processed slant range may be less. It is the minimum of this parameter and the maximal slant range defined within the interrogation program. The resulting range value is used to define the power for All-Call interrogation.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x02BE      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

##### Content of Item I253/100 - Transparent Application Data 1

|          |       |
|----------|-------|
| DataType | REAL  |
| Min      | 0.12  |
| Max      | 331.0 |
| Default  | 256.0 |
| Unit     | NM    |

##### Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.28 MIN RANGE

This parameter defines the Minimum instrumented range of the sensor.

Detection with a smaller slant range is impossible and will be discarded.

The MIN\_RANGE must be smaller than the MAX\_RANGE.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x02C2      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

##### Content of Item I253/100 - Transparent Application Data 1

|          |       |
|----------|-------|
| DataType | REAL  |
| Min      | 0.08  |
| Max      | 256.0 |
| Default  | 0.5   |
| Unit     | NM    |

##### Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |                     |                                             |                |                   |
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#### 4.11.9.29 POWER LEVELS SECT

This defines range/azimuth areas in which the corresponding power level is used. These range areas can be different in different sectors.

The given range values are the outer boundaries of the range areas. These values are used to define the power level of an interrogation signal. For All-Call operations the power level which is defined for instrumented range, is used. For Roll-Call operations, the appropriate power level determined by the range and azimuth of the target is used. Re-interrogations are handled as All-Call operations.

For the sector definition it is assumed, that the complete 360° are used.

The given sector boundaries are the start values of the sector. Sector one starts from the given azimuth and ends with the start azimuth of the next sector. The last sector ends with the start azimuth of the first sector, even if this sector is crossing north.

The selectable power levels are predefined and cannot be changed.

##### Note:

This parameter substitutes the old parameter POWER\_LEVELS, which is due to compatibility reasons still supported by the SW. If POWER\_LEVELS is used it overwrites the full content of this parameter.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0343      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

##### Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

##### Content of Item I253/110 - Transparent Application Data 2

|                             |                            |       |
|-----------------------------|----------------------------|-------|
| 1. Byte                     | Overall Length of Item     | 2 + X |
| 2. Byte                     | Number of Sectors          | S     |
| 3. Byte until (X + 2). Byte | List of Sector definitions |       |

S: Number of Power Level Sectors per 360°.

0 ≤ S ≤ 15: Maximal 15 sectors are available.

S = 0: No Power Levels are defined. The maximum power is used over the full scan and the full range

S = 1: The given "Start of Sector" is not relevant.

For each sector, range areas for up to 7 Power Levels can be defined.

##### Sector definition:

|                    |                                                    |   |
|--------------------|----------------------------------------------------|---|
| 1. Byte            | Start of sector in deg (high byte)                 |   |
| 2. Byte            | Start of sector in deg (low byte)                  |   |
| 3. Byte            | number of entries                                  | N |
| (4 + (N-1)*3) Byte | Power Level for Range area                         | P |
| (5 + (N-1)*3) Byte | End of Range of this Power Level (high byte) in RU |   |
| (6 + (N-1)*3) Byte | End of Range of this Power Level (low byte) in RU  |   |

Start of Sector: LSB = 1° (accuracy and resolution of the sector boundaries is 1°)

1 ≤ N ≤ 7: Number of defined Range areas for a dedicated Power Level

0 ≤ P ≤ 6: Used Power Level for defined Range area.

The attenuation for the dedicated Power Levels is intern configured enable the following for P:

P = 0: MAX\_POWER

P = 1: MAX\_POWER - 2 dB

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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P = 2: MAX\_POWER - 4 dB  
P = 3: MAX\_POWER - 6 dB  
P = 4: MAX\_POWER - 8 dB  
P = 5: MAX\_POWER - 10 dB  
P = 6: MAX\_POWER - 12 dB

0 ≤ RU ≤ 33 000: A RU (Range Unit) is defined as = 0.0101142 NM = 18.7314984 m

If the instrumented range is greater than the range boundary of last range area, from the last given boundary to the instrumented range, Power Level 0 is used.

#### Note:

The user can define the range areas (with assigned power) in any order, as the software places them in the correct power order. However, an increase in the power should always increase the range.

Unused rings are set to 0 and ignored by the software.

#### Example 1:

The following range areas for the power levels shall be set:

Sector 1 from 0° to start of Sector 2 with 4 range areas

Power Level 4 from 0 NM to 41.43 NM

Power Level 3 from 41.43 NM to 82.87 NM

Power Level 2 from 82.87 NM to 124.31 NM

Power Level 1 from 124.31 NM to 165.75 NM

Sector 2 from 180° to start of Sector 1 with 3 range areas

Power Level 6 from 0 NM to 50 NM

Power Level 5 from 50 NM to 130 NM

Power Level 4 from 130 NM to 180 NM

|                                   |  |                           |
|-----------------------------------|--|---------------------------|
| Set the Power Levels              |  | Contents of Item I253/110 |
| Overall length                    |  | 1D                        |
| Number of Sectors                 |  | 02                        |
| Start of Sector 1 = 0°            |  | 00 00                     |
| Number of Power Levels (Sector 1) |  | 04                        |
| Power Level 4 until RU = 1000     |  | 04 10 00                  |
| Power Level 3 until RU = 2001     |  | 03 20 01                  |
| Power Level 2 until RU = 3002     |  | 02 30 02                  |
| Power Level 1 until RU = 4003     |  | 01 40 03                  |
| Start of Sector 2 = 180°          |  | 00 B4                     |
| Number of Power Levels (Sector 2) |  | 03                        |
| Power Level 6 until RU = 12EC     |  | 06 12 EC                  |
| Power Level 5 until RU = 31D2     |  | 05 31 D2                  |
| Power Level 4 until RU = 4584     |  | 04 45 84                  |

#### Example 2:

2 sectors should be defined with different maximum power for the instrumented range (=100 NM)

Sector 1: from 90° to start of Sector 2; Power Level 1

Sector 2: from 270° to start of Sector 1; Power Level 4

|                         |  |                           |
|-------------------------|--|---------------------------|
| Set the Power Levels    |  | Contents of Item I253/110 |
| Overall length          |  | 0E                        |
| Number of Sectors       |  | 02                        |
| Start of Sector 1 = 90° |  | 00 5A                     |

|                                                |                     |                                             |                |                   |
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|                                   |  |          |
|-----------------------------------|--|----------|
| Number of Power Levels (Sector 1) |  | 01       |
| Power Level 1 until RU = 269F     |  | 01 26 9F |
| Start of Sector 2 = 270°          |  | 01 0E    |
| Number of Power Levels (Sector 2) |  | 01       |
| Power Level 4 until RU = 269F     |  | 04 26 9F |

#### 4.11.9.30 FFM

This parameter is to set or read information about the FFM associated with the MSSR 2000 ID.

For the MSSR 2000 ID the maximum number of FFM is defined as 4.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0218      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|                     |                                     |                                                                             |
|---------------------|-------------------------------------|-----------------------------------------------------------------------------|
| 1. Byte             | Overall Length of Item              | = 2 + N * 15                                                                |
| 2. Byte             | Number of FFM entries (N)           | $0 \leq N \leq 4$                                                           |
| 3. + (N-1)*15 Byte  | azimuth in deg of FFM N (high byte) | $0 \leq \text{azimuth} \leq 360$ , LSB = $360^\circ/2^{16}$ = about 0,0055° |
| 4. + (N-1)*15 Byte  | azimuth in deg of FFM N (low byte)  |                                                                             |
| 5. + (N-1)*15 Byte  | range in NM of FFM N (high byte)    | $0 \leq \text{range} \leq 256$ NM, LSB = $256 \text{ NM}/2^{16}$ = 1/256 NM |
| 6. + (N-1)*15 Byte  | range in NM of FFM N (low byte)     |                                                                             |
| 7. + (N-1)*15 Byte  | Status of FFM N                     | See a)                                                                      |
| 8. + (N-1)*15 Byte  | Mode S Address of FFM N             | See b)                                                                      |
| 9. + (N-1)*15 Byte  | Mode S Address of FFM N             |                                                                             |
| 10. + (N-1)*15 Byte | Mode S Address of FFM N             |                                                                             |
| 11. + (N-1)*15 Byte | Mode S Address of FFM N             |                                                                             |
| 12. + (N-1)*15 Byte | Mode 3/A Code of FFM N              | See c)                                                                      |
| 13. + (N-1)*15 Byte | Mode 3/A Code of FFM N              |                                                                             |
| 14. + (N-1)*15 Byte | Mode 2 Code of FFM N                | See c)                                                                      |
| 15. + (N-1)*15 Byte | Mode 2 Code of FFM N                |                                                                             |
| 16. + (N-1)*15 Byte | Mode C Code of FFM N                | See d)                                                                      |
| 17. + (N-1)*15 Byte | Mode C Code of FFM N                |                                                                             |

For N = 0: all FFM information will be deleted. The overall length of the Item = 2.

For N > 0 the following is applied:

a) Status of the FFM

|         |   |   |   |    |    |    |    |
|---------|---|---|---|----|----|----|----|
| Octet 1 |   |   |   |    |    |    |    |
| 8       | 7 | 6 | 5 | 4  | 3  | 2  | 1  |
| 0       | 0 | 0 | 0 | MC | M2 | M3 | MS |

MS: 0: FFM holds no Mode S Address 1: FFM holds a Mode S Address

|                                                |            |                  |          |           |
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M3: 0: FFM holds no Mode 3/A Code 1: FFM holds a Mode 3/A Code  
 M2: 0: FFM holds no Mode 2 Code 1: FFM holds a Mode 2 Code  
 MC: 0: FFM holds no Mode C Code 1: FFM holds a Mode C Code

b) Mode S Address as 24 Bit value

| Octet 1 |    |    |    |    |    |    |    | Octet 2 |    |    |    |    |    |    |     |
|---------|----|----|----|----|----|----|----|---------|----|----|----|----|----|----|-----|
| 32      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24      | 23 | 22 | 21 | 20 | 19 | 18 | 17  |
| 0       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | MSB     |    |    |    |    |    |    |     |
| Octet 3 |    |    |    |    |    |    |    | Octet 4 |    |    |    |    |    |    |     |
| 16      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8       | 7  | 6  | 5  | 4  | 3  | 2  | 1   |
|         |    |    |    |    |    |    |    |         |    |    |    |    |    |    | LSB |

c) Mode 3/A Code, Mode 2 Code and Mode C code

| Octet 1 |    |    |    |    |    |    |    | Octet 2 |    |    |    |    |    |    |    |
|---------|----|----|----|----|----|----|----|---------|----|----|----|----|----|----|----|
| 16      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8       | 7  | 6  | 5  | 4  | 3  | 2  | 1  |
| 0       | 0  | 0  | 0  | A4 | A2 | A1 | B4 | B2      | B1 | C4 | C2 | C1 | D4 | D2 | D1 |

Note:

Incorrect parameters of this command will cause the MSSR 2000 ID not to detect the defined FFM. If a FFM is defined but not detected the Alarm Message from type WARNING " FFM\_PLOT\_MISSING" will occur.

The Mode C here has to be entered in an octal representation, as the Mode A and Mode 2, unlike the format used for Mode C within Asterix Cat.048 data item I048/100.

#### 4.11.9.31 NTP ADDRESS

Network Time Protocol (NTP) Server Address of the first NTP server in host notation. The system will synchronise the internal clock with this NTP server, or the second NTP server (see NTP\_2\_ADDRESS) if the connection to the first NTP server is not possible or interrupted.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x033A      |
| Synchronise | no          |
| Access      | r/w Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                      |
|----------|--------------------------------------|
| DataType | UINT_32                              |
| Values   | IP- Address in 32 Bit representation |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.32 NTP2 ADDRESS

Network Time Protocol (NTP) Server Address of the second NTP server in host notation. The system will synchronise the internal clock with this NTP server, if the connection to the first NTP server is not possible or interrupted.

|             |                                                           |
|-------------|-----------------------------------------------------------|
| I253/080    |                                                           |
| Number      | 0x033B                                                    |
| Synchronize | no                                                        |
| Access      | r/w Level 2                                               |
|             | Content of Item I253/100 - Transparent Application Data 1 |

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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|          |                                     |
|----------|-------------------------------------|
| DataType | UINT_32                             |
| Values   | IP-ADDRESS in 32 Bit representation |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message

#### 4.11.9.33 NTP\_TIME\_OFFSET

The MSSR reports the time-source as accurate, when offset of the MSSR internal time to the NTP server time is less than this configured value.

DR System:

Additionally, the MSSR reports an alarm message in case the time difference between the MSSR-times of the two chains exceeds this value.

|             |                                                           |
|-------------|-----------------------------------------------------------|
| I253/080    |                                                           |
| Number      | 0x033D                                                    |
| Synchronize | yes                                                       |
| Access      | r/w Level 3                                               |
|             | Content of Item I253/100 - Transparent Application Data 1 |
| DataType    | UINT_16                                                   |
| Values      | Time offset in milliseconds                               |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.34 GTC (Gain Time Control)

These parameters define the starting values of the GTC threshold. The GTC threshold is configurable for each reply processor independently.

|             |                                              |
|-------------|----------------------------------------------|
| I253/080    |                                              |
| Number      | 0x0287 MX_GTC for the MK XA reply processing |
| Number      | 0x0232 M4_GTC for Mode 4 reply processing    |
| Number      | 0x0274 MS_GTC for Mode S reply processing    |
| Number      | 0x0275 M5_GTC for Mode 5 reply processing    |
| Synchronise | yes                                          |
| Access      | r/w                                          |

#### Content of Item I253/100 - Transparent Application Data 1

|                         |                                       |
|-------------------------|---------------------------------------|
| Data Type               | UINT_32                               |
| Values                  | as defined below                      |
| Octet 1                 | Octet 2                               |
| 32 31 30 29 28 27 26 25 | 24 23 22 21 20 19 18 17               |
|                         | Time Varying Threshold Start Offset   |
| Octet 3                 | Octet 4                               |
| 16 15 14 13 12 11 10 9  | 8 7 6 5 4 3 2 1                       |
|                         | Time Varying Threshold Constant $C_N$ |

empty fields = not used

Time Varying Threshold Start Offset (LSB = 0.005058585 NM)

|     |                                          |
|-----|------------------------------------------|
| Min | 0 (internal 0H) corresponds to 1NM       |
| Max | 1.7 (internal 14FH) corresponds to 2.7NM |

Time Varying Threshold Constant  $C_N$  (LSB = 0.023443 dB)

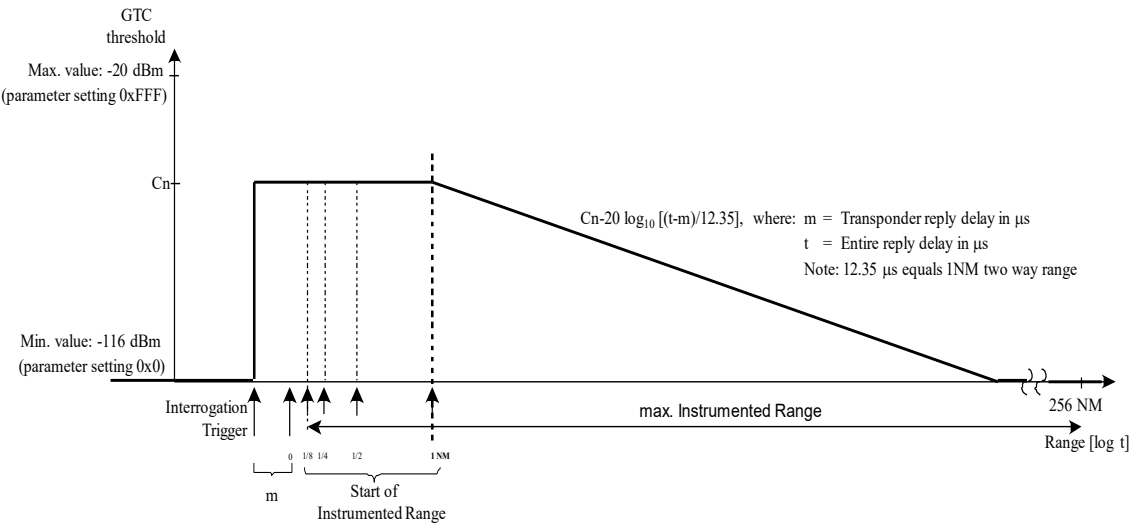
|     |                                                              |
|-----|--------------------------------------------------------------|
| Min | 0 (internal 0H, corresponding to a threshold of -116 dBm)    |
| Max | 96 (internal 0FFFH, corresponding to a threshold of -20 dBm) |

#### Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

*Figure 4-14. GTC Time Behaviour with DigRX*

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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Cn is adjustable via parameters (MX/MS/M4)\_THRES\_VARY\_CONST. Each parameter holds an integer Cn\_int in the limits of 0 to 4095 (hex: 0xFFFF).  
Cn = -116 dBm + (Cn\_int\*96/4095) dB  
In case of reaching the end of the instrumented range, the gtc threshold shall be set to the minimum value.

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|                                                |            |                  |          |           |
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#### 4.11.9.35 SURVEILLANCE CO-ORDINATION FUNCTION

This parameter enables or disables the Surveillance Co-ordination Function (SCF) .

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0420      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                      |
| Value     | 0x00: OFF: SCF disabled, no cluster functionality,<br>local solution list or default solution used<br>0x01: ON: SCF enabled |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.36 SURVEILLANCE CO-ORDINATION NETWORK

This parameter enables or disables the connection of the system's SCF to the Surveillance Co-ordination Network (SCN).

See SCF\_ENABLE.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0421      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                             |
| Value     | 0x00: OFF: network disabled,<br>SCF stand-alone states possible (e. g. STANDBY)<br>0x01: ON: network enabled,<br>SCF network aided states possible |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |                     |                                             |                |                   |
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#### 4.11.9.37 SELECT SOLUTION SET

If the SCF state is DISABLED, this parameter select a solution set (coverage map) from the local solution list or from the default solution. This message has no operational effect in other SCF states but is stored and kept as fall back solution if the SCF enters the state DISABLED.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0422      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                                                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                                                                                                                                                                                                |
| Values    | <p>Number of solution set selected from the local solution list.</p> <p>0 : default solution set ("hard-coded")</p> <p>1...63 : solution set, used by distributed cluster</p> <p>64...127 : solution set, reserved for central cluster controller</p> |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.38 SOLUTION LIST VERSION

This parameter defines the current solution list version. A solution list with this version has to be present at the file system, otherwise the parameter is rejected.

Note: This parameter selects the Coverage Map directory to be used.

The coverage map directory is created out of the main and sub version indexes.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0423      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

Structure:

|          |                                                 |        |
|----------|-------------------------------------------------|--------|
| 01. Byte | overall length of Item                          | 0x06   |
| 02. Byte | type = 6 (Unsigned integer 32 bit)              | 0x06   |
| 03. Byte | current solution list major version (high byte) | see a) |
| 04. Byte | current solution list major version (low byte)  |        |
| 05. Byte | current solution list minor version (high byte) | see b) |
| 06. Byte | current solution list minor version (low byte)  |        |

|           |                                                                                                            |  |
|-----------|------------------------------------------------------------------------------------------------------------|--|
| Data Type | UINT_16[2] (Array of 2 UINT_16)<br>array[1] : see a)<br>array[0] : see b)                                  |  |
| Values    | a) current solution list major version<br>1 to 999: version numbers<br>0: no solution list version is used |  |
| Values    | b) current solution list minor version<br>1 to 999: version numbers<br>0: no solution list version is used |  |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
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#### 4.11.9.39 SOLUTION LIST NEXT VERSION

This parameter defines the next solution list version.

The parameter is rejected if no solution list with the selected version is present at the file system.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0424      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

Structure:

|           |                                                                                                         |        |
|-----------|---------------------------------------------------------------------------------------------------------|--------|
| 01. Byte  | overall length of Item                                                                                  | 0x06   |
| 02. Byte  | type = 6 (Unsigned integer 32 bit)                                                                      | 0x06   |
| 03. Byte  | next solution list major version (high byte)                                                            | see a) |
| 04. Byte  | next solution list major version (low byte)                                                             |        |
| 05. Byte  | next solution list minor version (high byte)                                                            | see b) |
| 06. Byte  | next solution list minor version (low byte)                                                             |        |
| Data Type | UINT_16[2] (Array of 2 UINT_16)<br>array[1] : see a)<br>array[0] : see b)                               |        |
| Values    | a) next solution list major version<br>1 to 999: version numbers<br>0: no solution list version is used |        |
| Values    | b) next solution list minor version<br>1 to 999: version numbers<br>0: no solution list version is used |        |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.40 SOLUTION LIST NEXT TIME

This parameter defines the time (in future), when the next solution list version shall be activated.

The message shall be rejected if the next solution list version is not present or the time is not within reliable limits. (e.g. time in the past).

The next solution list substitutes the current solution list, if the time expires because time = now.

After expiration the time is reset to zero.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0425      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                              |
|-----------|----------------------------------------------------------------------------------------------|
| Data Type | time_t (UINT_32, seconds since 00:00:00UTC on 1-1-1970)                                      |
| Value     | value = 0: timer not used<br>value > 0: actual time (UTC) seconds since 00:00:00 on 1.1.1970 |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.9.41 MATCH II/SI ENABLE

If the interrogator is operating with a SI code, this operational parameter enables the matching II-Code processing.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0430      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                       |
|-----------|-------------------------------------------------------|
| Data Type | UINT_8                                                |
| Value     | 0 = disable<br>1 = enable matching II-Code processing |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.42 MATCH II/SI LOCKOUT

If the interrogator is operating with a SI code and if the matching II-Code processing is enabled, this operational parameter selects the type of lockout to be used (intermittent lockout or no lockout) for targets which operates with matching II-Code independently from the lockout map. If the matching II-Code processing is not enabled, this parameter has no effect.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0431      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                            |
|-----------|--------------------------------------------|
| Data Type | UINT_8                                     |
| Value     | 0 = no lockout<br>1 = intermittent lockout |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.43 NONSICAP PROCEDURE ENABLE

If the interrogator is operating with an II code this operational parameter enables the special processing of targets which indicates no SI-Code capability

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0432      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|           |                                                                         |
|-----------|-------------------------------------------------------------------------|
| Data Type | UINT_8                                                                  |
| Value     | 0 = disable<br>1 = enable special processing for non SI-Code capability |

|                                                |                     |                                             |                |                   |
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Content of Item I253/110 - Transparent Application Data 2  
This item is not used for this message.

#### 4.11.9.44 NONSICAP PROCEDURE LOCKOUT

If the interrogator is operating with an II code and if the NON-SI\_Capability processing is enabled, this operational parameter selects the type of lockout to be used (intermittent lockout or no lockout) for targets which indicates no SI-Code Capability independently from the lockout map. If the NON-SI\_Capability processing is not enabled, this parameter has no effect.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x0433      |
| Synchronise | yes         |
| Access      | r/w Level 1 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                            |
|----------|--------------------------------------------|
| DataType | UINT_8                                     |
| Value    | 0 = no lockout<br>1 = intermittent lockout |

Content of Item I253/110 - Transparent Application Data 2  
This item is not used for this message.

#### 4.11.9.45 BDS REGISTER NUMBER

This parameter defines the number of the (first) BDS register, which shall be extracted periodically. Input is in decimal numbers (e.g. BDS1.0 = 16, BDS5.0 = 80 and so on)

Exceptions:

Registers for air-initiated Comm-B:

BDS0.0, BDS0.1, BDS0.2, BDS0.3 and BDS0.4

ACAS resolution advisory:

BDS3.0

|             |                         |
|-------------|-------------------------|
| I253/080    |                         |
| Number      | 0x0434 for BDS_1_REG_NO |
| Number      | 0x0437 for BDS_2_REG_NO |
| Number      | 0x043A for BDS_3_REG_NO |
| Number      | 0x043D for BDS_4_REG_NO |
| Number      | 0x0440 for BDS_5_REG_NO |
| Synchronise | yes                     |
| Access      | r/w Level 2             |

Content of Item I253/100 - Transparent Application Data 1

|          |                               |
|----------|-------------------------------|
| DataType | UINT_8                        |
| Values   | 0..255 (exclusive exceptions) |

Content of Item I253/110 - Transparent Application Data 2  
This item is not used for this message.

#### Recommendations for the use of the internal GICB application

The possibility of periodical GICB extraction depends on the used **Mode Interlace Pattern (MIP)**, the **Antenna Turn Time**, the number of Mode S Roll-Call Periods in Beam, the time available for Mode S Roll-Call and the use of the Split GICB capability (config parameter).

|                                                |            |                  |          |           |
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In most of the configured interrogation programs (MIPs) there are only three Roll-Call Periods in Beam and in case of GICB split enable it may be that only two Roll-Call periods are available for GICB extraction in one scan for some targets.

To take into consideration the possibility of air initiated Comm-B and ACAS-Comm-B the values for the periodicity shall not be below the following limits

- one periodical GICB active periodicity  $\geq 1 * ATT$
- two periodical GICB active periodicity  $\geq 1.5 * ATT$
- three periodical GICB active periodicity  $\geq 2.0 * ATT$
- four periodical GICB active periodicity  $\geq 3.0 * ATT$
- five periodical GICB active periodicity  $\geq 4.0 * ATT$

It is not recommended to use the internal GICB application and the special periodical extraction of BDS1.7 at the same time. If the special periodical BDS1.7 extraction is activated, a maximum of three additional periodical GICB extractions may be used.

It is recommended to use only low priorities for BDS extractions with the internal GICB application, to suppress conflicts with higher priority air initiated Comm-B and ACAS Comm-B messages.

The recommended priority values for use with the internal GICB extraction are from 0 up to 4.

Additional recommendation for the use of the internal GICB application in conjunction with a MSSR 2000 ID 1500W

To take into consideration the limited Mode S capability (additional TX delay between interrogations) the values for the periodicity shall not be below the following limits:

- one periodical GICB active periodicity  $\geq 1.5 * ATT$
- two periodical GICB active
  - one per. GICB with periodicity  $\geq 1.5 * ATT$
  - one per. GICB with periodicity  $\geq 2.5 * ATT$
- three periodical GICB active
  - one per. GICB with periodicity  $\geq 1.5 * ATT$
  - two per. GICB with periodicity  $\geq 2.5 * ATT$
- four periodical GICB active
  - one per. GICB with periodicity  $\geq 1.5 * ATT$
  - one per. GICB with periodicity  $\geq 2.5 * ATT$
  - two per. GICB with periodicity  $\geq 3.5 * ATT$

#### 4.11.9.46 BDS REGISTER PERIODICITY

The periodicity for the extraction of the first BDS register in seconds.

All values without 0,

0 means functionality disabled

Note:

For AIMS 17-1000 compliant systems the minimum value is 180s when BDS 0,8 /1,0/1,7/2,0 or 2,5 is configured.

|             |                           |
|-------------|---------------------------|
| I253/080    |                           |
| Number      | 0x0435 for BDS_1_REG_PERI |
| Number      | 0x0438 for BDS_2_REG_PERI |
| Number      | 0x043B for BDS_3_REG_PERI |
| Number      | 0x043E for BDS_4_REG_PERI |
| Number      | 0x0441 for BDS_5_REG_PERI |
| Synchronise | yes                       |
| Access      | r/w Level 2               |

|                                                |            |                  |          |           |
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## Content of Item I253/100 - Transparent Application Data 1

|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| Data Type | UINT_8                                                                         |
| Value     | value = 0: functionality disabled<br>value > 0: periodicity for the extraction |

## Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

## 4.11.9.47 BDS REGISTER PRIORITY

The priority of the extraction of the first periodical BDS register.

To more fully understand the effect of the priority parameter, it is useful to understand more about the scheduling in the system.

When scheduling interrogations for targets in the beam, the SW uses a so-called transaction-list for each target. The transaction-list for each target is generated before the target enters the beam, and defines all the data required from the target in this beam-dwell. It more or less corresponds to an ideal sequence of interrogations for this target.

(Note that the transaction list may be adapted as the target passes thru the beam - depending on the replies received from the target!)

The sequence of transactions within the transaction-list depends on the priorities or the data-requests making-up the list.

Requests are:

- the surveillance transaction (FL) - always
- A-code request (see parameter **3A\_REINTERROGATION\_PERIOD**)
- the maximum 5 periodical BDS registers (priorities given by user 0 - 31)
- DBDS 17 (see parameter **BDS17\_REINTERROGATION\_PERIOD**: priority 6)
- Capability request( BDS 10: priority 1 - 6 depending on situation )
- Flight-ID request (Aquisition Process: priority 6)
- a maximum of 3 DI BDS registers (priority 30)
- requests generated by CAT18 (priorities given by user: 0 - 31)

The surveillance transaction is always the first transaction for the target.

The A-code request has the next highest priority, and is the second element in the transaction-list when it is requested.

The remaining requests are achieved via BDS requests and these are first ordered (hi to low priority) into a list of BDS-message requests for the target. The messages are then entered into the transaction list in the order of hi priority (first) to low priority(last) - **but something rather odd happens here. If the priority of the BDS-message has bit 0x08 set, then these transactions are considered highest priority, and moved into the transaction-list ahead of the other BDS-requests.**

**For example, this means that a BDS request with priority 8, is actually higher priority than BDS request with priority 16!!!!**

Note that when adding the BDs-requests to the transaction list, these can be combined with the surveillance transaction (UF 4 changes to UF 20) or with the A-code request (UF 21).

|             |                           |
|-------------|---------------------------|
| I253/080    |                           |
| Number      | 0x0436 for BDS_1_REG_PRIO |
| Number      | 0x0439 for BDS_2_REG_PRIO |
| Number      | 0x043C for BDS_3_REG_PRIO |
| Number      | 0x043F for BDS_4_REG_PRIO |
| Number      | 0x0442 for BDS_5_REG_PRIO |
| Synchronise | yes                       |

|                                                |            |                         |          |           |
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|        |             |
|--------|-------------|
| Access | r/w Level 2 |
|--------|-------------|

## Content of Item I253/100 - Transparent Application Data 1

|          |              |
|----------|--------------|
| DataType | UINT_8       |
| Values   | Priority     |
| Min      | 0 = lowest   |
| Max      | 31 = highest |

## Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

## Recommendation

When more than one BDS Register extraction is defined the priority should be different.

**4.11.9.48 OPTION A CRYPTO PROFILE A/B**

For a detailed description see /MSSR ICD AI - Annex/.

**4.11.9.49 CRYPTO LEGACY MODE**

For a detailed description see /MSSR ICD AI - Annex/.

**4.11.9.50 CRYPTO MODE4 A/B WINDOW**

For a detailed description see /MSSR ICD AI - Annex/.

**4.11.9.51 BDS REINTERROGATION PERIOD**

Mode S parameters used by SUR.

The parameter defines the timer for periodic BDS 1.7 request. The request is conducted only if BDS 1.0 is present.

Set 0 to disable periodic request.

## Note:

For AIMS 17-1000 compliant systems the minimum value is 180s.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x06DC      |
| Synchronise | yes         |
| Access      | r/w Level 3 |

## Content of Item I253/100 - Transparent Application Data 1

|          |               |
|----------|---------------|
| DataType | UINT_32       |
| Min      | 0             |
| Max      | E10H (=3600s) |
| Default  | 0             |
| Unit     | s             |

## Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |                     |                                             |                |                   |
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#### 4.11.9.52 INITIAL PR

Define the initial PR value used in UF11 by stochastic acquisition at start-up.

After INITIAL\_SCAN\_COUNT scans the PR value is decreased by one step until Nominal PR is reached.

Note: When the PR value in UF11 is decreased, then the probability of reply is increased.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x06E7      |
| Synchronise | yes         |
| Access      | r/w Level 3 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------|
| DataType | UINT_8                                                                                                         |
| Values   | 0 = Probability 1<br>1 = Probability 1/2<br>2 = Probability 1/4<br>3 = Probability 1/8<br>4 = Probability 1/16 |
| Default  | 4                                                                                                              |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.53 MAX Z VEL

Max. allowable vertical speed (HMI name MAX VERT SPEED):

Determines the size of the vertical association gate.

This limits the possible number of association candidates wrt. their height difference and the time between measurements. If an aircraft climbs or descends with a higher vertical rate, the Tracking SW will in general not be able to follow it consistently.

[knots]=[NM/h]

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x07AC      |
| Synchronise | yes         |
| Access      | r/w Level 3 |

Content of Item I253/100 - Transparent Application Data 1

|          |       |
|----------|-------|
| DataType | REAL  |
| Min      | 1.0   |
| Max      | 800.0 |
| Default  | 800.0 |
| Step     | 0.1   |
| Unit     | knots |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
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#### 4.11.9.54 MAX MISSES

This parameter defines the maximum number of radar scans during which an existing track may be coasted.

This is equivalent to the maximum number of "plot misses" for a track.

If the number of scans during which a track is coasted exceeds this parameter value, the track will be deleted. Note: If the track is in the cone of silence, it can be coasted more often; if the track is of doubtful quality or leaving the radar coverage, it may be deleted earlier.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x07A6      |
| Synchronise | yes         |
| Access      | r/w Level 2 |

##### Content of Item I253/100 - Transparent Application Data 1

|          |         |
|----------|---------|
| DataType | UINT_16 |
| Min      | 2       |
| Max      | 8       |
| Step     | 1       |
| Default  | 4       |

##### Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.9.55 GHOST AZIMUTH GATE

Azimuth gate for ghost suppression.

Note:

A large gate increases the processing load.

The gate (range and azimuth) determines the area around the given target where the SW checks for a possible reflection.

Note:

For testing purposes, the value shall be set to 1.0. This value is also defined in the standard config set.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x078B      |
| Synchronise | yes         |
| Access      | r/w Level 3 |

##### Content of Item I253/100 - Transparent Application Data 1

|          |       |
|----------|-------|
| DataType | REAL  |
| Min      | 1.0   |
| Max      | 180.0 |
| Default  | 180.0 |
| Unit     | deg   |

##### Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
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#### 4.11.9.56 GHOST RANGE GATE

Range gate for ghost suppression.

Note: A large gate increases the processing load.

Note: For testing purposes, the value shall be set to 1.0. This value is also defined in the standard config set.

|             |             |
|-------------|-------------|
| I253/080    |             |
| Number      | 0x078C      |
| Synchronise | yes         |
| Access      | r/w Level 3 |

Content of Item I253/100 - Transparent Application Data 1

|          |       |
|----------|-------|
| DataType | REAL  |
| Min      | 0.1   |
| Max      | 256.0 |
| Default  | 2.0   |
| Unit     | NM    |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                         |          |           |
|------------------------------------------------|------------|-------------------------|----------|-----------|
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#### 4.11.10 Description of Diagnostic Parameters

These parameters are read only parameters.

##### 4.11.10.1 TEMPERATURE

The purpose of this message is to read the temperature of the:

|             |                                       |
|-------------|---------------------------------------|
| I253/080    |                                       |
| Number      | 0x032E PSU_TEMP for Power Supply      |
| Number      | 0x032F ADAP_TEMP for Adaptation Board |
| Synchronise | no                                    |
| Access      | read Level 0                          |

Content of Item I253/100 - Transparent Application Data 1

|           |            |
|-----------|------------|
| Data Type | UINT_16    |
| Values    | See below: |

| Temperature | Binary Representation | Hexadecimal Representation |
|-------------|-----------------------|----------------------------|
| +125 °C     | 01111101 00000000     | 7D 00                      |
| +25 °C      | 00011001 00000000     | 19 00                      |
| +0.5 °C     | 00000000 10000000     | 00 80                      |
| 0 °C        | 00000000 00000000     | 00 00                      |
| -0.5 °C     | 11111111 10000000     | FF 80                      |
| -25 °C      | 11100111 00000000     | E7 00                      |
| -55 °C      | 11001001 00000000     | C9 00                      |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.10.2 PSU STATUS

The purpose of this message is to read the status of the PSU.

|             |                       |
|-------------|-----------------------|
| I253/080    |                       |
| Number      | 0x032D for PSU_STATUS |
| Synchronise | no                    |
| Access      | read Level 0          |

Content of Item I253/100 - Transparent Application Data 1

|           |            |
|-----------|------------|
| Data Type | UINT_16    |
| Values    | See below: |

| BIT      | Description |
|----------|-------------|
| 00       | P50V        |
| 01       | N75V        |
| 02       | P3V3        |
| 03       | P5V         |
| 04       | N5V         |
| 05       | P15V        |
| 06       | N15V        |
| 07       | P2V5        |
| 08       | N15VP       |
| 09       | not used    |
| 10       | P24V        |
| 11 .. 15 | not used    |

0 = not OK,

1 = OK

unused Bits shall be ignored

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.10.3 MSSR 2000 ID - INTERROGATOR HW-STATUS

This parameter is to read the current state of the Interrogator HW. This parameter is sent automatically if one value of the parameter has changed.

|             |              |
|-------------|--------------|
| I253/080    |              |
| Number      | 0x0384       |
| Synchronise | no           |
| Access      | read Level 0 |

Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|          |                                        |      |
|----------|----------------------------------------|------|
| 01. Byte | Overall length of Item                 | 0x14 |
| 02. Byte | HW-Status of Data Processor A          |      |
| 03. Byte | HW-Status of Data Processor B          |      |
| 04. Byte | HW-Status of SPARE 1                   |      |
| 05. Byte | HW-Status of SPARE 2                   |      |
| 06. Byte | HW-Status of RTCC                      |      |
| 07. Byte | HW-Status of Adaptation Board          |      |
| 08. Byte | HW-Status of Receiver                  |      |
| 09. Byte | HW-Status of Modulator                 |      |
| 10. Byte | HW-Status of Frontend                  |      |
| 11. Byte | HW-Status of Transmitter Module 1      |      |
| 12. Byte | HW-Status of Transmitter Module 2      |      |
| 13. Byte | HW-Status of Transmitter Module 3      |      |
| 14. Byte | HW-Status of Transmitter Module 4      |      |
| 15. Byte | HW-Status of Transmitter Module 5      |      |
| 16. Byte | HW-Status of Capacity Extension Module |      |
| 17. Byte | HW-Status of Front Panel               |      |
| 18. Byte | SPARE (reserved for future upgrade)    |      |
| 19. Byte | SPARE (reserved for future upgrade)    |      |
| 20. Byte | HW-Status of PSU                       |      |
| Value    | 0: HW Status = OK                      |      |
| Values   | all other values HW Status = not OK    |      |

Not present modules are indicated with 0xFF.

If a LRU is present which should not be present, the HW status is indicated as not OK.

If a LRU is not present which should be present, the HW status is indicated as not OK.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.10.4 MSSR 2000 ID - BOOSTER HW-STATUS

This parameter is to read the current state of the Booster HW. This parameter is sent automatically if one value of the parameter has changed. This parameter is available only, if a booster is used.

|             |              |
|-------------|--------------|
| I253/080    |              |
| Number      | 0x0385       |
| Synchronise | no           |
| Access      | read Level 0 |

Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|          |                                          |      |
|----------|------------------------------------------|------|
| 01. Byte | Overall length of Item                   | 0x14 |
| 02. Byte | HW-Status of Interface Booster           |      |
| 03. Byte | HW-Status of Transmitter Module 1        |      |
| 04. Byte | HW-Status of Transmitter Module 2        |      |
| 05. Byte | HW-Status of Transmitter Module 3        |      |
| 06. Byte | HW-Status of Transmitter Module 4        |      |
| 07. Byte | HW-Status of Transmitter Module 5        |      |
| 08. Byte | HW-Status of Transmitter Module 6        |      |
| 09. Byte | HW-Status of Capacity Extension Module 1 |      |
| 10. Byte | HW-Status of Capacity Extension Module 2 |      |
| 11. Byte | SPARE (reserved for future upgrade)      |      |
| 12. Byte | SPARE (reserved for future upgrade)      |      |
| 13. Byte | SPARE (reserved for future upgrade)      |      |
| 14. Byte | SPARE (reserved for future upgrade)      |      |
| 15. Byte | SPARE (reserved for future upgrade)      |      |
| 16. Byte | SPARE (reserved for future upgrade)      |      |
| 17. Byte | SPARE (reserved for future upgrade)      |      |
| 18. Byte | SPARE (reserved for future upgrade)      |      |
| 19. Byte | SPARE (reserved for future upgrade)      |      |
| 20. Byte | HW-Status of PSU                         |      |
| Value    | 0: HW Status = OK                        |      |
| Values   | all other values HW Status = not OK      |      |

Not present modules are indicated as 0xFF.

If a LRU is present which should not be present, the HW status is indicated as not OK.

If a LRU is not present which should be present, the HW status is indicated as not OK.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.10.5 MSSR 2000 ID - SOU HW-STATUS

This parameter is to read the current state of the SOU HW. This parameter is sent automatically if one value of the parameter is changed

|             |              |
|-------------|--------------|
| I253/080    |              |
| Number      | 0x0386       |
| Synchronise | no           |
| Access      | read Level 0 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|          |                                       |      |
|----------|---------------------------------------|------|
| 01. Byte | Overall length of Item                | 0x14 |
| 02. Byte | SPARE (reserved for future upgrade)   |      |
| 03. Byte | SPARE (reserved for future upgrade)   |      |
| 04. Byte | SPARE (reserved for future upgrade)   |      |
| 05. Byte | SPARE (reserved for future upgrade)   |      |
| 06. Byte | SPARE (reserved for future upgrade)   |      |
| 07. Byte | SPARE (reserved for future upgrade)   |      |
| 08. Byte | SPARE (reserved for future upgrade)   |      |
| 09. Byte | SPARE (reserved for future upgrade)   |      |
| 10. Byte | SPARE (reserved for future upgrade)   |      |
| 11. Byte | SPARE (reserved for future upgrade)   |      |
| 12. Byte | HW-Status of SOU                      |      |
| 13. Byte | SPARE (reserved for future upgrade)   |      |
| 14. Byte | SPARE (reserved for future upgrade)   |      |
| 15. Byte | SPARE (reserved for future upgrade)   |      |
| 16. Byte | HW-Status of COAX Relay 1 (not a LRU) |      |
| 17. Byte | HW-Status of COAX Relay 2 (not a LRU) |      |
| 18. Byte | HW-Status of COAX Relay 3 (not a LRU) |      |
| 19. Byte | SPARE (reserved for future upgrade)   |      |
| 20. Byte | SPARE (reserved for future upgrade)   |      |

|        |                                     |
|--------|-------------------------------------|
| Value  | 0: HW Status = OK                   |
| Values | all other values HW Status = not OK |

Not present modules are indicated as 0xFF.

If a LRU is present which should not be present, the HW status is indicated as not OK.

If a LRU is not present which should be present, the HW status is indicated as not OK.

The contents of spare bytes are not defined up to now, each value between 0x00 and 0xFF is possible.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.10.6 MSSR 2000 ID ONLINE STATUS

This parameter is to read the current state of the Interrogator functions. This parameter is sent automatically if one value of the parameter has changed.

|             |                          |
|-------------|--------------------------|
| I253/080    |                          |
| Number      | 0x0387 for ONLINE_STATUS |
| Synchronise | no                       |
| Access      | read Level 0             |

Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|          |                                     |        |
|----------|-------------------------------------|--------|
| 01. Byte | Overall length of Item              | 0x14   |
| 02. Byte | Antenna - State                     | see a) |
| 03. Byte | Input Control Signals State         | see b) |
| 04. Byte | SPARE (reserved for future upgrade) |        |
| 05. Byte | SPARE (reserved for future upgrade) |        |
| 06. Byte | Temperature DP4/5 Board             |        |
| 07. Byte | Temperature DP4/5 CPU               |        |
| 08. Byte | Temperature DigRx Sum               | see c) |
| 09. Byte | Temperature DigRx Omega             | see c) |
| 10. Byte | Temperature Interrogator TX1        | see c) |
| 11. Byte | Temperature Interrogator TX2        | see c) |
| 12. Byte | Temperature Interrogator TX3        | see c) |
| 13. Byte | Temperature Interrogator TX4        | see c) |
| 14. Byte | Temperature Interrogator TX5        | see c) |
| 15. Byte | Temperature Booster TX1             | see c) |
| 16. Byte | Temperature Booster TX2             | see c) |
| 17. Byte | Temperature Booster TX3             | see c) |
| 18. Byte | Temperature Booster TX4             | see c) |
| 19. Byte | Temperature Booster TX5             | see c) |
| 20. Byte | Temperature Booster TX6             | see c) |

a) Antenna State:

|   |   |   |   |   |   |   |     |
|---|---|---|---|---|---|---|-----|
| 7 | 8 | 5 | 4 | 3 | 2 | 1 | 0   |
|   |   |   |   |   |   |   | ASY |

ASY = 1: Antenna synchronised

ASY = 0: Antenna not synchronised

b) Input Control Signals State

|   |   |   |   |   |   |   |     |
|---|---|---|---|---|---|---|-----|
| 7 | 8 | 5 | 4 | 3 | 2 | 1 | 0   |
|   |   |   |   |   |   |   | ASS |

ASS = 0: Antenna Safety switch closed

ASS = 1: Antenna Safety switch open (safety Switch activated, system not operable)

c) Temperature Values:

For not used positions the temperature value is set to -128 to mark it as invalid.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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Note for versions later than 1x.05.02 the parameter AST\_TX\_CONTROL can be set such that invalid temperature are set to 0, and bytes 05, 06, and 07 are set to 0.  
The value of the temperature is in °C in 2ers complement representation.

#### 4.11.10.7 ANTENNA ATT

This parameter is used to read the actual antenna turn time, calculated from the north pulses.

|             |              |
|-------------|--------------|
| I253/080    |              |
| Number      | 0x020C       |
| Synchronise | no           |
| Access      | read Level 0 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                     |
|----------|-------------------------------------|
| DataType | REAL                                |
| Values   | Actual Antenna Turn Time in seconds |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.10.8 LOCAL REMOTE

This message indicates if the MSSR 2000 ID is switched to local or to remote control.

|             |              |
|-------------|--------------|
| I253/080    |              |
| Number      | 0x0201       |
| Synchronise | yes          |
| Access      | read Level 0 |

Content of Item I253/100 - Transparent Application Data 1

|          |                       |
|----------|-----------------------|
| DataType | UINT_8                |
| Value    | 0: local<br>1: remote |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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#### 4.11.10.9 ASTERIX RDPC

The purpose of this message is to read the ASTERIX RDPC (Radar Data Processing Chain) of the MSSR 2000 ID.

In a Standalone configuration this value is set to 1 by default. In a dual redundant configuration the value is calculated from HW- Lines from the SOU (Switch Over Unit).

This parameter is read only.

|             |              |
|-------------|--------------|
| I253/080    |              |
| Number      | 0x0208       |
| Synchronise | no           |
| Access      | read Level 2 |

Content of Item I253/100 - Transparent Application Data 1

|          |                                                 |
|----------|-------------------------------------------------|
| DataType | UINT_8                                          |
| Value    | 0x01: Radar Data Processing Chain 1 is selected |
| Value    | 0x02: Radar Data Processing Chain 2 is selected |

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

#### 4.11.10.10 CRYPTO STATUS

For a detailed description see /MSSR ICD AI - Annex/.

|                                                |            |                         |          |           |
|------------------------------------------------|------------|-------------------------|----------|-----------|
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#### 4.11.10.11 CURRENT PROGRAM ID

This parameter shows the PROGRAM\_ID of the currently executed interrogation sector. This parameter is transmitted automatically, if one of the values of the content has changed. The parameter is read only.

|             |                               |
|-------------|-------------------------------|
| I253/080    |                               |
| Number      | 0x0344 for CURRENT_PROGRAM_ID |
| Synchronise | no                            |
| Access      | read Level 0                  |

Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|          |                                      |                           |
|----------|--------------------------------------|---------------------------|
| 1. Byte  | LEN                                  | 0x0A                      |
| 2. Byte  | PROGAM_ID_TYPE                       | 0x00 - 0x0F               |
| 3. Byte  | PROGRAM_ID                           | 0x00 - 0xFF               |
| 4. Byte  | current_SCAN                         | 0x00 - 0x08               |
| 5. Byte  | start of current Program (high byte) | 0x0000 - 0x0167, LSB = 1° |
| 6. Byte  | start of current Program (low byte)  |                           |
| 7. Byte  | TIME-OF-DAY                          | LSB = 1/128 s             |
| 8. Byte  |                                      |                           |
| 9. Byte  |                                      |                           |
| 10. Byte | FLAGS                                |                           |

##### PROGRAM\_ID\_TYPE:

The PROGRAM\_ID\_TYPE indicates if the PROGRAM\_ID references to a normal PROGRAM\_ID or an exceptional PROGRAM\_ID. A normal PROGRAM\_ID is one of the interrogation programs defined in the Interlace Pattern Reference Table. An exceptional PROGRAM\_ID is an interrogation program executed during calibration processes or default interrogation programs executed if the normal program does not exist.

0x00 normal PROGRAM\_ID  
0x01 default PROGRAM\_ID  
0x02 calibration PROGRAM\_ID  
0x03 - 0x07 reserved for future extension

The PROGRAM\_ID number used for default programs or calibration programs is not related at all to the number used for a normal PROGRAM\_ID. They are processed completely separate within the MSSR 2000 ID.

##### PROGRAM\_ID:

This parameter shows the ID of the current active interrogation program of the program type specified by PROGRAM\_ID\_TYPE.

##### current\_SCAN:

If the MSSR 2000 ID is controlled using the SCAN\_SEQ\_PROGRAM parameter, this parameter shows the currently processed scan.

If the MSSR 2000 ID is controlled using the SECTOR\_MODE parameter, current\_SCAN can have the values 1 and 2. The azimuth value is with respect to a 360° scan

If the MSSR 2000 ID is controlled using the INTERROGATION\_SEQUENCE parameter, current\_SCAN is set to 1 and the azimuth value is set to 0x0000. In this case the message will be transmitted with every north crossing even if no value has changed.

NOTE: "current\_SCAN=0" indicates that the MSSR 2000 ID is in an operational state other than TRANSMIT or CALIBRATION. In this case, all values are set to 0x00.

If current\_SCAN is not equal to zero, the MSSR 2000 ID is in operational state TRANSMIT or CALIBRATION and executes the selected Scan Mode.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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start of current Program:

This parameter shows the beginning of the described interrogation sector in the current scan in deg.

TIME-OF-DAY:

This parameter shows the time when the currently described interrogation sector was started. The time of day value is reset to zero each day at midnight.

LSB =  $2^{-7}s = 1/128 s$

FLAGS:

|   |   |   |   |   |   |   |        |
|---|---|---|---|---|---|---|--------|
| 7 | 8 | 5 | 4 | 3 | 2 | 1 | 0      |
|   |   |   |   |   |   |   | SILENT |

SILENT: Flag for Silent Sector

1 SILENT SECTOR is set, the system does not transmit

0 SILENT SECTOR not set, normal operation

The Silent Sector flag has priority over the interrogation type and program ID. If it is set, the system will not transmit in the current sector. The given interrogation type and program ID are the configured ones, that would be processed, if the silent sector would be deleted.

The MSSR 2000 ID will automatically transmit this parameter when one of the values of the parameter has changed.

The start of a Directed Interrogation Sector will trigger the transmission of this parameter if the program ID of the Directed Interrogation is different from the actual executed program ID.

#### 4.11.10.12 MSR STATE

This message indicates the functional state of the MSSR 2000 ID.

This message is read only.

|             |                      |
|-------------|----------------------|
| I253/080    |                      |
| Number      | 0x0400 for MSR_STATE |
| Synchronise | no                   |
| Access      | read Level 0         |

Content of Item I253/100 - Transparent Application Data 1

|          |                     |
|----------|---------------------|
| DataType | UINT_16             |
| Value    | 0x0004: NOGO        |
| Value    | 0x0005: STANDBY     |
| Value    | 0x0006: TRANSMIT    |
| Value    | 0x0007: CALIBRATION |
| Value    | 0x0008: BITE        |

All other values are not defined and indicate an UNKNOWN state of the MSSR 2000 I

Content of Item I253/110 - Transparent Application Data 2

This item is not used for this message.

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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#### 4.11.10.13 MSSR 2000 ID EXTENDED SYSTEM STATE

This parameter is to read the current state of the Interrogator functions. This parameter is sent automatically if one value of the parameter is changed.

|             |                               |
|-------------|-------------------------------|
| I253/080    |                               |
| Number      | 0x0410 for EXTENDED_SYS_STATE |
| Synchronise | no                            |
| Access      | read Level 0                  |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|          |                                                         |        |
|----------|---------------------------------------------------------|--------|
| 01. Byte | overall length of Item                                  | 0x12   |
| 02. Byte | general system state                                    | see a) |
| 03. Byte | master-slave state                                      | see b) |
| 04. Byte | proxy state                                             | see c) |
| 05. Byte | chain identification                                    | see d) |
| 06. Byte | auto master slave switch enabled                        | see e) |
| 07. Byte | antenna turning                                         | see f) |
| 08. Byte | system mode                                             | see g) |
| 09. Byte | time source                                             | see h) |
| 10. Byte | indicate used azimuth encoder                           | see i) |
| 11. Byte | time accuracy                                           | see j) |
| 12. Byte | navigation status                                       | see k) |
| 13. Byte | time synchronization status (for 10.04/11.04 or higher) | see l) |
| 14. Byte | SPARE (reserved for future upgrade)                     | see x) |
| 15. Byte | SPARE (reserved for future upgrade)                     | see x) |
| 16. Byte | SPARE (reserved for future upgrade)                     | see x) |
| 17. Byte | SPARE (reserved for future upgrade)                     | see x) |
| 18. Byte | SPARE (reserved for future upgrade)                     | see x) |

Values:

a) general system state

|    |                  |                                                      |
|----|------------------|------------------------------------------------------|
| 00 | Unknown          | undetermined status                                  |
| 01 | Wait             | Power Up: waiting for synchronisation (SW processes) |
| 02 | Synchronised     | Power Up: system synchronised (SW processes)         |
| 03 | Status ready     | Power Up: system status ready                        |
| 04 | Nogo             | system status NOGO                                   |
| 05 | Standby          | system status STANDBY                                |
| 06 | Transmit         | system status TRANSMIT                               |
| 07 | Calibration      | calibration is running                               |
| 08 | Bite             | BITE is running                                      |
|    | all other values | not defined                                          |

b) master-slave state

|    |                  |                          |
|----|------------------|--------------------------|
| 00 | unknown          | undetermined status      |
| 01 | master           | this is the master chain |
| 02 | slave            | this is the slave chain  |
| 03 | alone            | Standalone Interrogator  |
|    | all other values | not defined              |

**Note:**

|                                                |            |                  |          |           |
|------------------------------------------------|------------|------------------|----------|-----------|
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**The slave chain doesn't deliver CAT-48 messages (plots / tracks) if the proxy state (see c) is "disconnected" !**

c) proxy state (connection between master and slave chain)

00 PX-disconnected connection is not established  
 01 PX-connected connection is established  
 all other values not defined

**Note:**

**The slave chain (see b) doesn't deliver CAT-48 messages (plots / tracks) if the proxy state is disconnected" !**

d) chain identification

00 CHAIN-unknown chain cannot be determined (this state indicates a hardware error)  
 01 CHAIN\_1 this is Chain 1  
 02 CHAIN\_2 this is Chain 2  
 03 CHAIN\_LONE Standalone to indicate a Single Chain HW  
 all other values not defined

e) auto master slave switch enabled

00 DISABLED auto master-slave switch disabled  
 01 ENABLED auto master-slave switch enabled  
 all other values not defined

f) antenna turning

00 FALSE antenna not turning  
 all other values TRUE antenna turning

g) Operational Modes

00 REDUNDANT Redundant Operate  
 01 MAINTENANCE Redundant Maintenance  
 02 STD\_ALONE Standalone System  
 all other values not defined

h) time source

00 INTERNAL no external time source found  
 01 GPS synchronised to GPS as external time source  
 02 NTP synchronised to NTP as external time source  
 03 OTHER CHAIN synchronised to 'other chain' in a dual redundant configuration  
 04 EXTERNAL an external HW at J61 or X25 (ECU) is used as time source  
 05 NTP2 synchronised to NTP2 as external time source  
 all other values not defined

i) azimuth encoder identification

00 unknown encoder cannot be determined (this state indicates a hardware error)  
 01 ENCODER\_1 encoder 1 is used  
 02 ENCODER\_2 encoder 2 is used  
 03 ENCODER\_LONE Standalone, SOU not fitted, only encoder 1 in use.  
 all other values not defined

j) time accuracy

00 inaccurate inaccurate when the time offset between NTP server and system time is not within the configured threshold  
 01 accurate accurate when the time offset between NTP server and system time is within the configured threshold

For Single Chain (SC) systems the threshold is set by the NTP\_TIME\_OFFSET parameter.

For Dual Redundant (DR) systems the threshold is set by the

\* NTP\_TIME\_OFFSET parameter, in case the chain is synchronized to its own NTP server.

\* NTP\_TIME\_OFFSET of the other chain, in case the chain is synchronized to the NTP server of the other chain.

**Notes:**

- These bits are set independent of the time source indicator, i.e. the MSSR can be synchronised to NTP although time accuracy is set to "inaccurate".
- This value is always set to accurate when synchronised to GPS.
- For all other time sources defined above, this field is not defined.
- for the DR system each chain has a time accuracy parameter.

|                                                |            |                  |          |           |
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## k) navigation status

- 01 NAV\_POS\_VALID inform when latitude, longitude information are valid
- 02 NAV\_ALT\_VALID inform when altitude information is valid
- 04 NAV\_HEADING\_VALID inform when heading information is valid
- 08 NAV\_SPEED\_VALID inform when speed information is valid
- 10 NAV\_ATT\_VALID inform when pitch and roll information are valid
- 20 NAV\_HEADING\_REQ inform when heading information is required
- 40 NAV\_ATT\_REQ inform when pitch and roll information are required
- all other values not defined

## l) time synchronization status

## Single Channel (SC) systems:

- Bit 0 TIME\_SYNC\_STATUS\_VALID set to 1 when time synchronization status is valid
- Bit 1 TIME\_UPDATE\_AVAILABLE set to 1 when external time source is available.
- Bit 2 TIME\_UPDATE\_ACCEPTED set to 1 when the last external time update is accepted (due to drift threshold verification)
- Bit 3 TOD\_WILL\_BECOME\_STALE set to 1 when Time Of Day (TOD) will become stale within 12h (0 when TOD is stale)
- Bit 4 TOD\_IS\_STALE set to 1 when TOD is stale
- all other values not defined

## Dual Redundant (DR) systems:

For DR systems this bit has to be set for each chain:

- Bit 0 TIME\_SYNC\_STATUS\_VALID set to 1 when time synchronization status is valid (For DR systems this bit is set to 1 in case the chain synchronizes to NTP, ECU or the other chain, where the other chain synchronizes to NTP or ECU).
- Bit 1 TIME\_UPDATE\_AVAILABLE set to 1 when external time source is available at this chain.  
(set to 0 when an external time source is available only via the other chain).
- Bit 2 TIME\_UPDATE\_ACCEPTED set to 1 when the last external time update is accepted (due to drift threshold verification)  
(set to 0 when an external time source is accepted only via the other chain)
- Bit 3 TOD\_WILL\_BECOME\_STALE set to 1 when Time Of Day (TOD) will become stale within 12h (0 when TOD is stale)  
(refers to the last load time of the ECU of this chain)
- Bit 4 TOD\_IS\_STALE set to 1 when TOD is stale  
(refers to the last load time of the ECU of this chain)
- all other values not defined

## Notes:

- Bit 0 is set to 1 only set when a Mode 5 capable ECU is connected and NTP is selected as external time source
- Bits 1-4 are only valid when bit 0 is set to 1

x) SPARE contents of spare bytes are not defined each value between 0x00 and 0xFF is possible

#### 4.11.10.14 MSSR 2000 ID CLUSTER STATE

This parameter is to read the current cluster state of the cluster functions. This parameter is sent automatically if one value of the parameter is changed.

|             |                          |
|-------------|--------------------------|
| I253/080    |                          |
| Number      | 0x0411 for CLUSTER_STATE |
| Synchronise | no                       |
| Access      | read Level 0             |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|                                                |            |                  |          |           |
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|          |                                        |        |
|----------|----------------------------------------|--------|
| 01. Byte | overall length of Item                 | 0x12   |
| 02. Byte | used cluster-file and system-map-files | see a) |
| 03. Byte | SCF state                              | see b) |
| 04. Byte | node                                   | see c) |
| 05. Byte | current topology node 1                | see d) |
| 06. Byte | current topology node 2                | see d) |
| 07. Byte | current topology node 3                | see d) |
| 08. Byte | current topology node 4                | see d) |
| 09. Byte | current topology node 5                | see d) |
| 10. Byte | current topology node 6                | see d) |
| 11. Byte | current topology cluster controller    | see e) |
| 12. Byte | current Solution Set                   | see f) |
| 13. Byte | Cluster Sstate                         | see g) |
| 14. Byte | SPARE (reserved for future upgrade)    | see x) |
| 15. Byte | SPARE (reserved for future upgrade)    | see x) |
| 16. Byte | SPARE (reserved for future upgrade)    | see x) |
| 17. Byte | SPARE (reserved for future upgrade)    | see x) |
| 18. Byte | SPARE (reserved for future upgrade)    | see x) |

Values:

a) used cluster-file and system-map-files (=solution list)

00 not used

cluster- and system-map-files not found

01 default

the default solution is used, because no solution set within the local solution list was found.

02 local

the local solution list from local surveillance coordinator is used because the SCF is not connected.

03 SCF

solution list from the SCF is used all other values are not defined

b) SCF state

stand-alone states

00 DISABLED

SCF disabled or not connected

01 NOGO

SCF error

02 INIT

SCF is in initialisation phase (wait for interrogator)

03 STANDBY

SCF is ready but SCN is off or interrogator not in TX

04 SLAVE

SCF is ready but SCN is off because MSSR is slave

05 STANDALONE

not used

06 WARMUP

SCF is warming up

Note:

Wait for 10 scans if the MSSR 2000 ID changes state to TX after power up or

wait for 2 scans if the MSSR 2000 ID changes state to TX and the system was in distributed before last power up

|                                                |                     |                                             |                |                   |
|------------------------------------------------|---------------------|---------------------------------------------|----------------|-------------------|
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network aided states (SCN on)  
 07 AUTONOMOUS SCF network aided autonomous  
 08 DISTRIBUTED SCF network aided distributed  
 09 CENTRAL SCF network aided central controlled  
 all other values are not defined

**Notes:**

1) The States 0 to 2 indicate that the SCF is either not connected, not enabled or not in an operational state. Cluster operation is then not possible.  
 2) States 03 to 09 are taken when cluster operation is possible - given that the MSSR is also operational (ie. not in NOGO, BIT or CALIBRATION). Note that Cluster operation is considered as possible even if the SCN\_ENABLE switch is set to disable.

**c) node (own node reference number)**

0..5 own cluster node, determined by own SIC/SAC code and cluster-file.  
 6 reserved for cluster controller  
 255 (hex 0xFF) no node found or no solution list loaded.  
 all other values are not defined

**d) current topology (matrix with the view of node 0..5)**

0 SCF disabled or not connected  
 1..127 The first six bits indicates which nodes can be seen from the dedicated node in the cluster network. The seventh bit indicate if the cluster controller can be seen from the dedicated node in the cluster network. The slave only delivers it's own node when SCF is enabled.

**e) current topology (view of cluster controller)**

0 CC disabled or not connected  
 1..127 The first six bits indicates which nodes can be seen from the cluster controller in the cluster network. The seventh bit indicate if the cluster controller can see oneself.

**f) current solution set**

This number selects a solution from the solution list.  
 The current solution set corresponds with the current topology of the own node in distributed mode.  
 0 no solution set  
 1 ... 63 solution set, used by distributed cluster  
 65 ... 127 solution set, used from central cluster controller

**g) Cluster State**

Bit 0:

0 = not operational - cluster operation is not possible,  
 1 = operational - cluster operation possible.

This bit is the logical and of the state SCF operational **AND** MSSR is operational (Transmit or STANDBY).

x) SPARE contents of spare bytes are not defined each value between 0x00 and 0xFF is possible

#### 4.11.10.15 MSSR 2000 ID VERSIONS

This parameter is to read the current version information from various SW and HW modules.  
 This parameter is sent after request.

**Note:**

The MSSR ignores the version request message during sytem state "BITE".

|             |                          |
|-------------|--------------------------|
| I253/080    |                          |
| Number      | 0x0412 for MSSR_VERSIONS |
| Synchronise | no                       |
| Access      | read Level 0             |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

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## Content of Item I253/110 - Transparent Application Data 2

|                          |                          |               |
|--------------------------|--------------------------|---------------|
| 1. Byte                  | Overall Length of Item   | 3 + VersN *28 |
| 2. Byte                  | index of version message | IndexN        |
| 3. Byte                  | number of versions       | VersN         |
| (4 + (Vidx -1)*28) Byte  | 1. string character      | ASCII         |
| (5 + (Vidx -1)*28) Byte  | 2. string character      | ASCII         |
| (6 + (Vidx -1)*28) Byte  | 3. string character      | ASCII         |
| :                        |                          | :             |
| :                        |                          | :             |
| (29 + (Vidx -1)*28) Byte | 26. string character     | ASCII         |
| (30 + (Vidx -1)*28) Byte | 27. string character     | ASCII         |
| (31 + (Vidx -1)*28) Byte | string termination       | 0             |

$1 \leq \text{IndexN} \leq 16$  = index for version message. It possible to carry a maximum of 9 versions per message, therefore it is necessary to send this message multiple times to transmit all versions. The index of the first message is one, for each further message the index is incremented.

Maximum number of versions: 16 messages \* 9 versions = 144 versions

$1 \leq \text{VersN} \leq 9$  = number of versions per message.

Up to 9 version strings per message are possible. ( $9 * 28\text{Byte} = 252\text{ Byte}$ ). Maximum length for Item110 = 255 Byte.

$1 \leq \text{Vidx} \leq \text{VersN}$  = index of each version in the message.

## Example

|                   |                           |                        |
|-------------------|---------------------------|------------------------|
| 1. Byte           | Overall Length of Item    | 255                    |
| 2. Byte           | index of version message  | 1                      |
| 3. Byte           | number of versions        | 9                      |
| 4. to 31. Byte    | version string (Vidx = 1) | "Global Version V6.07" |
| 32. to 59. Byte   | version string (Vidx = 2) | "SCP V6.59"            |
| 60. to 87. Byte   | version string (Vidx = 3) | "BASE V6.79"           |
| 88. to 115. Byte  | version string (Vidx = 4) | "SCP WATCHDOG V6.59"   |
| 116. to 143. Byte | version string (Vidx = 5) | "SUP WATCHDOG V6.59"   |
| 144. to 171. Byte | version string (Vidx = 6) | "PAP WATCHDOG V6.59"   |
| 172. to 199. Byte | version string (Vidx = 7) | "LCP WATCHDOG V6.59"   |
| 200. to 227. Byte | version string (Vidx = 8) | "SC LOADMON V6.59"     |
| 228. to 255. Byte | version string (Vidx = 9) | "SUP LOADMON V6.59"    |

## 4.11.10.16 POWER MONITORING

This parameter is used to estimate output powers and VSWR (average, min and max). The values are reported for each used power level and for SUM and OMNI channel.

The command POWER\_MONITORING\_RESET clears the data container for minimal, maximal and average power and VSWR.

Note: The power measurement must be calibrated to deliver valid data and the accuracy of the power sensor is very low (about  $\pm 0.5\text{ dB}$ ).

This parameter is read only.

This parameter is sent after request.

|                                                |            |                  |          |           |
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|             |                             |
|-------------|-----------------------------|
| I253/080    |                             |
| Number      | 0x0413 for POWER_MONITORING |
| Synchronise | no                          |
| Access      | read Level 0                |

Content of Item I253/100 - Transparent Application Data 1  
This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

Definition of data container for each Power Level:

|                   |                        |                        |
|-------------------|------------------------|------------------------|
| 1. Byte           | Overall Length of Item | 170                    |
| 2. Byte           | Number of Power Level  | 7                      |
| 3. to 26. Byte    | Data for Power Level 1 | see Table below, N=3   |
| 27. to 50. Byte   | Data for Power Level 2 | see Table below, N=27  |
| 51. to 74. Byte   | Data for Power Level 3 | see Table below, N=51  |
| 75. to 98. Byte   | Data for Power Level 4 | see Table below, N=75  |
| 99. to 122. Byte  | Data for Power Level 5 | see Table below, N=99  |
| 123. to 146. Byte | Data for Power Level 6 | see Table below, N=123 |
| 147. to 170. Byte | Data for Power Level 7 | see Table below, N=147 |

Definition of Data for Power Level:

|                       |                              |              |
|-----------------------|------------------------------|--------------|
| (N+0) to (N+1) Byte   | average power SUM channel    | 0..65536 [W] |
| (N+2) to (N+3) Byte   | average power OMNI channel   | 0..65536 [W] |
| (N+4) to (N+5) Byte   | minimum power SUM channel    | 0..65536 [W] |
| (N+6) to (N+7) Byte   | minimum power OMNI channel   | 0..65536 [W] |
| (N+8) to (N+9) Byte   | maximum power SUM channel    | 0..65536 [W] |
| (N+10) to (N+11) Byte | maximum power OMNI channel   | 0..65536 [W] |
| (N+12) to (N+13) Byte | average VSWR of SUM channel  | 0..65536     |
| (N+14) to (N+15) Byte | average VSWR of OMNI channel | 0..65536     |
| (N+16) to (N+17) Byte | minimum VSWR of SUM channel  | 0..65536     |
| (N+18) to (N+19) Byte | minimum VSWR of OMNI channel | 0..65536     |
| (N+20) to (N+21) Byte | maximum VSWR of SUM channel  | 0..65536     |
| (N+22) to (N+23) Byte | maximum VSWR of OMNI channel | 0..65536     |

Definition of Power:

|            |                   |          |
|------------|-------------------|----------|
| N Byte     | power (high byte) |          |
| (N+1) Byte | power (low byte)  | LSB = 1W |

0 indicates the level is not used, or no measured data received since last reset or power up.  
1 .. 0xFFFF Power in Watt

Definition of VSWR:

|            |                  |                    |
|------------|------------------|--------------------|
| N Byte     | VSWR (high byte) |                    |
| (N+1) Byte | VSWR (low byte)  | LSB = (1/100) VSWR |

0 indicates the level is not used, or no measured data received since last reset or power up.  
1 .. 0xFFFF equals to a VSWR of 0 .. 655.36 (max. useful VSWR values up to 8.0)

|                                                |            |                  |          |           |
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VSWR values below 1.5 recommended

Example:

For the ratio forward to reflected power, expressed by a VSWR of 1.5:1 the value 150 is transmitted.

#### 4.11.10.17 CRYPTO TIME INFORMATION

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.18 OPTION A CRYPTO - STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.19 OPTION A CRYPTO - PROFILE A KEY STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.20 OPTION A CRYPTO - PROFILE B TEK STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.21 OPTION A CRYPTO - PROFILE B KEK STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.22 OPTION B CRYPTO - STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.23 OPTION B CRYPTO - KEY AND FILL STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.24 QRTK4 NG - ETI AND STATE

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.25 QRTK4 NG - CRYPTO STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.26 QRTK4 NG - LOADED KEYS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.27 QRTK4 NG - TEK STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.28 QRTK4 NG - KEK STATUS

For a detailed description see /MSSR ICD AI - Annex/.

#### 4.11.10.29 MIP LIST

This parameter is to read the current defined MIP list of the MSSR 2000 ID.

This parameter is sent after request.

Note: The MSSR ignores the MIP LIST request message during sytem state "BITE".

|             |              |
|-------------|--------------|
| I253/080    |              |
| Number      | 0x0415       |
| Synchronise | no           |
| Access      | read Level 0 |

Content of Item I253/100 - Transparent Application Data 1

This item is not used for this message.

Content of Item I253/110 - Transparent Application Data 2

|         |                        |               |
|---------|------------------------|---------------|
| 1. Byte | Overall Length of Item | 4 + MIP-N *33 |
|---------|------------------------|---------------|

|                                                |            |                  |          |           |
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|                            |                                      |         |
|----------------------------|--------------------------------------|---------|
| 2. Byte                    | index of MIP message (start by 1)    | IndexN  |
| 3. Byte                    | number of configured pages           | Page no |
| 4. Byte                    | number of MIPs (inside this message) | MIP-N   |
| (5 + (MIP-N - 1)*33) Byte  | MIP ID                               | MIP-ID  |
| (6 + (MIP-N - 1)*33) Byte  | 1. string character                  | ASCII   |
| (7 + (MIP-N - 1)*33) Byte  | 2. string character                  | ASCII   |
| :                          |                                      | :       |
| :                          |                                      | :       |
| (36 + (MIP-N - 1)*33) Byte | 31. string character                 | ASCII   |
| (37 + (MIP-N - 1)*33) Byte | 32. string termination               | 0       |

**Note:**

The MIP ID is the number of the interrogation program and corresponds to the Program ID (used for the SCAN\_SEQ\_PROGRAM parameter).

It possible to carry maximal 6 MIP names per message. Typically, the number of MIPs defined in the system will exceed 6. In this case, multiple messages will be generated to transmit all MIPs. The index IndexN of the first message is one, for each further message the index is incremented.

**Note:**

The system supports a maximum of 256 MIPs. If 256 MIPs are defined, they will be transmitted as 43 messages, with 42 messages containing 6 MIPs and one message containing 4 MIPs.

$1 \leq \text{MIP-N} \leq 6 = \text{number of MIPs per message.}$

Maximal 6 MIP strings per message are possible. Max length in Bytes for Item110 =  $4 + 6 * 33 = 202$

**Example:**

|                   |                            |                   |
|-------------------|----------------------------|-------------------|
| 1. Byte           | Overall Length of Item     | 202               |
| 2. Byte           | index of mip               | 1                 |
| 3. Byte           | number of configured pages | 2                 |
| 4. Byte           | number of MIPs             | 6                 |
| 5. Byte           | MIP-ID                     | x00               |
| 6. to 37. Byte    | MIP string                 | "000: 4s:M3.58Hz" |
| 38. Byte          | MIP-ID                     | x01               |
| 39. to 70. Byte   | MIP string                 | "4s:MC.58Hz"      |
| 71. Byte          | MIP-ID                     | x02               |
| 72. to 103. Byte  | MIP string                 | "4s:M2.58Hz"      |
| 104. Byte         | MIP-ID                     | x03               |
| 105. to 136. Byte | MIP string                 | "4s:M4.58Hz"      |
| 137. Byte         | MIP-ID                     | x15 = 21          |
| 138. to 169. Byte | MIP string                 | "4s:M3.MC.116Hz"  |
| 170. Byte         | MIP-ID                     | x16 = 22          |
| 171. to 202. Byte | MIP string                 | "4s:CMC.M3.116Hz" |

|                                                |                     |                                             |                |                   |
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#### 4.11.11 Alarm Messages handled with CAT253

The MSSR 2000 ID uses an error handling system to send Alarm Messages to the connected stations.

##### 4.11.11.1 Structure of an Alarm Message

Alarm Messages are messages with I253/040 Message Type 17 = Alarm.

The information of an Alarm Message is transferred in the Items I253/100 and I253/110 according to the Specific Content of the Data Items for CAT 253.

##### 4.11.11.1.1 Content of Item I253/100

I253/100 Data Type: TEXT

The 'text' payload in ASCII characters of this item is built the following way:

- 4 digits of the code (alarm number) as defined in ASCII representation
- SPACE in ASCII representation
- Classification (Severity) in ASCII representation
  - '0' MESSAGE for information only
  - '1' WARNING
  - '2' ERROR
  - '3' FATAL cause the system to go to NOGO
- SPACE in ASCII representation
- Symbolic name of the message in ASCII representation.

Example:

The message E\_LCP\_STARTED with the alarm number: 0x4000 is coded as:

*Table 4-38. Example of an Alarm Message*

| OCTET    | CONTENT | ASCII | HEX  |                                                 |
|----------|---------|-------|------|-------------------------------------------------|
| Octet 1  | LEN     | --    | 0x14 | Calculated from the actual length, maximum: 255 |
| Octet 2  | Type    | --    | 0x08 | Type = Text string                              |
| Octet 3  | value   | 4     | 0x34 | first ASCII digit to the error number           |
| Octet 4  | value   | 0     | 0x30 | second ASCII digit to the error number          |
| Octet 5  | value   | 0     | 0x30 | third ASCII digit to the error number           |
| Octet 6  | value   | 0     | 0x30 | forth ASCII digit to the error number           |
| Octet 7  | value   |       | 0x20 | always the ASCII representation of SPACE        |
| Octet 8  | value   | 2     | 0x32 | ASCII representation of the Classification      |
| Octet 9  | value   |       | 0x20 | always the ASCII representation of SPACE        |
| Octet 10 | value   | L     | 0x4c |                                                 |
| Octet 11 | value   | C     | 0x43 |                                                 |
| Octet 12 | value   | P     | 0x50 |                                                 |
| Octet 13 | value   | _     | 0x5f |                                                 |
| Octet 14 | value   | S     | 0x53 |                                                 |
| Octet 15 | value   | T     | 0x54 |                                                 |
| Octet 16 | value   | A     | 0x41 |                                                 |
| Octet 17 | value   | R     | 0x52 |                                                 |
| Octet 18 | value   | T     | 0x54 |                                                 |
| Octet 19 | value   | E     | 0x45 |                                                 |
| Octet 20 | value   | D     | 0x44 |                                                 |

|                                                |                     |                                             |                |                   |
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#### 4.11.11.1.2 Content of Item I253/110

This item contains additional information about the alarm message.

The content is free text in ASCII code.

Octet 1 contains the explicit length (the LEN field + the number of ASCII characters).

Octet 2 till Octet LEN contains the free text.

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